Topological Energy Condensate Theory (TECT): 26

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I. MINIMAL THEOREM/LEMMA CHECKLIST FOR AN SM+GR-TYPE INFRARED LIMIT IN TECT

II. 1. MAIN INFRARED TARGET

The correct infrared goal is not to assume the Standard Model plus General Relativity as a fundamental axiom, but to show that TECT admits an effective long-distance description of SM+GR type.

Accordingly, the appropriate main target theorem is:

$$\boxed{\textbf{Main IR Theorem (target form).}\quad \text{If conditions } (C1)\text{–}(C8) \text{ hold, then the infrared effective theory of TECT is of SM+GR}} \tag{1}$$

Here the conditions $(C1)$–$(C8)$ should be separated into theorem/lemma-level subproblems.

III. 2. MINIMAL CHECKLIST

A. Lemma C1: Continuum infrared sector

A controlled long-wavelength continuum limit must exist, with sufficiently suppressed residual anisotropy and an effectively Lorentzian propagation structure.

B. Lemma C2: Einstein-type gravitational sector

The infrared theory must contain a metric or massless spin-2 sector with universal coupling, and non-Einstein extra gravitational polarizations must decouple or be sufficiently suppressed.

C. Lemma C3: Gauge sector emergence

The infrared theory must contain the observed Standard Model gauge content, or an infrared-equivalent gauge structure reproducing the same tested gauge phenomenology.

D. Lemma C4: Chiral matter and family sector

The infrared theory must contain a chiral matter sector with the observed family structure, while unwanted mirror sectors must be removed, lifted, or screened.

E. Lemma C5: Canonical kinetic normalization

After projection to the infrared degrees of freedom, the kinetic terms must be reducible to canonical form so that masses and mixings are physically well-defined.

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F. Lemma C6: Decoupling of extra states

All non-Standard-Model light modes must either become sufficiently heavy, sufficiently weakly coupled, or otherwise invisible in tested infrared regimes.

G. Lemma C7: Suppression of dangerous higher operators

Potentially dangerous Lorentz-violating, flavor-violating, baryon/lepton-violating, or non-Einstein operators must be sufficiently suppressed in the infrared effective action.

H. Lemma C8: Consistency conditions

The infrared theory must satisfy anomaly consistency, stability, unitarity, and positivity requirements.

IV. 3. STATUS LOGIC

The present goal is not yet to prove the full main theorem, but to organize the proof structure:

$$\boxed{\text{(i) isolate the minimal conditions,}\quad\text{(ii) tag each as closed / conditional / open,}\quad\text{(iii) determine the dependency order for futur}} \tag{2}$$

This converts an otherwise overwhelming infrared-emergence problem into a structured research program.

V. TECT INFRARED PROGRAM: STATUS-TAGGED THEOREM/LEMMA CHECKLIST FOR AN SM+GR-TYPE IR LIMIT

VI. 1. MAIN TARGET

The appropriate infrared target is not to assume the Standard Model plus General Relativity as fundamental input, but to establish that TECT admits an effective infrared description of SM+GR type.

The target theorem is therefore:

$$\boxed{\textbf{Main IR Theorem (target form).}\quad \text{If conditions } (C1)\text{–}(C8) \text{ hold, then the infrared effective theory of TECT is of SM+GR}} \tag{3}$$

At the present stage, the correct task is not the full proof of the Main IR Theorem, but the status classification of the minimal conditions $(C1)$–$(C8)$.

VII. 2. STATUS LABELS

We use the following proof-status labels:

$$\boxed{\text{Closed}: \text{ rigorously established within the present TECT framework.}} \tag{4}$$

$$\boxed{\text{Conditional}: \text{ strongly supported by explicit constructions or partial derivations, but still dependent on unclosed assumptions o}} \tag{5}$$

$$\boxed{\text{Open}: \text{ not yet derived at the level required for an infrared theorem.}} \tag{6}$$

VIII. 3. THE C1–C8 CHECKLIST WITH CURRENT STATUS

A. C1. Continuum infrared sector

Statement. There exists a controlled long-wavelength continuum limit with sufficiently suppressed residual anisotropy, together with an effectively Lorentzian propagation structure.
Status.

$$\boxed{\text{Conditional}} \tag{7}$$

Reason. TECT has already developed a substantial body of work suggesting that the BCC condensate can yield an approximately isotropic infrared sector, and several structural arguments indicate that the long-distance theory should admit continuum effective fields. However, a full theorem establishing the precise Lorentzian infrared limit, including all required suppression estimates for anisotropic corrections, has not yet been completed.

What is already strong. The BCC-based infrared isotropy program and the continuum-effective-field perspective provide a credible partial foundation.

What is still missing. A theorem-level derivation of the effective infrared kinetic sector with explicit control of Lorentz-violating and cubic-anisotropy corrections.

B. C2. Einstein-type gravitational sector

Statement. The infrared theory contains a metric or massless spin-2 sector with universal coupling of Einstein type, while unwanted extra gravitational polarizations are absent or sufficiently suppressed.
Status.

$$\boxed{\text{Open}} \tag{8}$$

Reason. TECT contains a conceptual and structural gravity-emergence program, but the theorem-level derivation of the Einstein-Hilbert sector in the precise infrared sense is not yet closed. In particular, the universal coupling property, the decoupling of non-Einstein gravitational polarizations, and the precise infrared effective action remain unproved.

Why this is not merely conditional. The missing steps are not just small technical refinements; they are structurally essential for calling the IR limit “GR-type” in a theorem-level sense.

C. C3. Gauge sector emergence

Statement. The infrared theory contains the observed Standard Model gauge sector, or an infrared-equivalent gauge structure reproducing the same tested phenomenology.
Status.

$$\boxed{\text{Open}} \tag{9}$$

Reason. There are meaningful partial advances, especially in the direction of internal-geometry and bundle-based emergence mechanisms, but a full derivation of the complete $SU(3) \times SU(2) \times U(1)$ gauge content with the correct infrared couplings and matter representations is not yet established.

What is already present. Several internal-geometry constructions and partial emergence routes suggest that gauge structure is not being inserted arbitrarily by hand.

What is still missing. A theorem-level derivation of the complete observed gauge content and its low-energy coupling structure.

D. C4. Chiral matter and family sector

Statement. The infrared theory contains a chiral matter sector with the observed family structure, and unwanted mirror sectors are absent, lifted, or otherwise screened.

Status.

$$\boxed{\text{Conditional}} \tag{10}$$

Reason. TECT has already progressed significantly farther here than in several other sectors. In particular, the flavor problem has been reduced to a finite family Hamiltonian problem once a common family subspace exists:

$$(H_f^{\text{fam}})_{AB} = \langle \phi_A, \mathcal{Y}_f \phi_B \rangle. \tag{11}$$

Moreover, a substantial Dirac-emergence and family-carrier program has been developed. However, the actual microscopic construction of the family subspace, the elimination of unwanted mirror sectors, and the complete chiral low-energy matter content are not yet fully proven.

What is already strong. The reduction of flavor to a finite Hermitian family Hamiltonian is a genuine structural achievement.

What is still missing. Microscopic derivation of the actual common family subspace and a theorem-level mirror-lifting/removal mechanism.

E. C5. Canonical kinetic normalization

Statement. After projection to the infrared degrees of freedom, the kinetic form can be reduced to canonical form so that masses, couplings, and mixings are physically well-defined.

Status.

$$\boxed{\text{Conditional}} \tag{12}$$

Reason. At the general effective-field-theory level, this requirement is standard and its necessity is completely clear. Within the TECT flavor program, the need for common family-space kinetic normalization is already sharply understood. However, the actual microscopic computation of the projected kinetic form and its explicit canonical reduction in the intended Class II setting has not yet been completed.

Why conditional rather than open. This is not conceptually mysterious; it is a concrete missing derivation step. The mathematical task is clear, but the TECT-specific calculation is unfinished.

F. C6. Decoupling of extra states

Statement. All non-SM light modes become sufficiently heavy, sufficiently weakly coupled, or otherwise invisible in experimentally tested infrared regimes.

Status.

$$\boxed{\text{Open}} \tag{13}$$

Reason. TECT naturally risks generating extra collective modes, residual defect excitations, mirror sectors, or other low-energy remnants. At present there is no closed theorem proving that all such unwanted degrees of freedom decouple in the relevant IR regime.

Why this matters. Even if the desired SM-like sector emerges, the infrared limit would still fail phenomenologically if extra light states remain observable.

G. C7. Suppression of dangerous higher operators

Statement. Potentially dangerous higher-dimensional operators, including Lorentz-violating, flavor-violating, baryon/lepton-violating, and non-Einstein gravitational corrections, are sufficiently suppressed in the IR effective action.

Status.

$$\boxed{\text{Open}} \tag{14}$$

Reason. This is one of the most demanding conditions for any deeper microscopic theory. TECT has not yet produced the needed theorem-level bounds or scaling laws ensuring that all dangerous irrelevant operators are suppressed below experimental limits.

Why this is essential. Without C7, even a qualitatively correct emergent low-energy sector can be ruled out by precision tests.

H. C8. Consistency conditions

Statement. The infrared theory satisfies anomaly consistency, stability, unitarity, and positivity requirements.
Status.

$$\boxed{\text{Open}} \tag{15}$$

Reason. Although the framework is being developed with these requirements in mind, a theorem-level consistency analysis for the full intended infrared sector has not yet been carried out.

Why this is fundamentally necessary. Even a phenomenologically attractive effective theory is unacceptable if it fails anomaly consistency, unitarity, or stability.

IX. 4. SUMMARY TABLE

The current TECT infrared status may therefore be summarized as follows:

Condition	Current Status
$C1$ Continuum IR sector	Conditional
$C2$ Einstein-type gravity sector	Open
$C3$ Gauge sector emergence	Open
$C4$ Chiral matter and family sector	Conditional
$C5$ Canonical kinetic normalization	Conditional
$C6$ Decoupling of extra states	Open
$C7$ Suppression of dangerous higher operators	Open
$C8$ Consistency conditions	Open

(16)

In particular:

$$\boxed{\text{Closed: none at the full SM+GR-type IR level.}} \tag{17}$$

$$\boxed{\text{Conditional: } C1,\ C4,\ C5.} \tag{18}$$

$$\boxed{\text{Open: } C2,\ C3,\ C6,\ C7,\ C8.} \tag{19}$$

X. 5. DEPENDENCY ORDER

The proof dependencies are not symmetric. The correct order is:

$$\boxed{C1 \longrightarrow (C2, C3, C4) \longrightarrow C5 \longrightarrow (C6, C7, C8).} \tag{20}$$

This order means:

1. Without a controlled continuum infrared sector (C1), one cannot even state the intended low-energy Einstein/gauge/matter content cleanly.
2. Once C1 is sufficiently controlled, the three main emergence sectors must be addressed:
$$C2 \text{ (gravity)}, \qquad C3 \text{ (gauge)}, \qquad C4 \text{ (matter/family)}.$$
3. After the low-energy degrees of freedom are isolated, canonical kinetic normalization (C5) becomes mandatory for physical interpretation.
4. Only then can one meaningfully attack the phenomenological viability conditions:
$$C6 \text{ (decoupling)}, \qquad C7 \text{ (dangerous operators)}, \qquad C8 \text{ (consistency)}.$$

XI. 6. IMMEDIATE STRATEGIC CONSEQUENCE

At the current stage, the most productive next step is not to attempt the full infrared theorem at once. Instead, the next mathematically rational step is to sharpen the strongest conditional block:

$$\boxed{C4 \text{ (chiral matter/family sector)}} \tag{21}$$

while simultaneously clarifying the precise continuum control required for

$$\boxed{C1 \text{ (continuum infrared sector).}} \tag{22}$$

This is because the present TECT program is already structurally strongest on the matter/flavor side, whereas the gravity and full gauge sectors remain substantially more open.

XII. 7. RELATION TO THE CKM PROGRAM

The CKM program lies strictly inside $C4$ and $C5$. More precisely, a physically meaningful CKM extraction additionally requires the following subconditions:

$$\boxed{\text{(i) common left-handed family subspace,} \quad \text{(ii) canonical kinetic normalization,} \quad \text{(iii) nondegenerate splittings,} \quad \text{(iv) hierarchy s}} \tag{23}$$

Therefore the CKM program should not be treated as separate from the main IR-emergence checklist. It is a refined subproblem of the matter/family sector plus kinetic control.

XIII. 8. HONEST FINAL STATUS

The strongest honest conclusion is:

$$\boxed{\text{TECT already has a nontrivial and promising structural matter/flavor program,}} \tag{24}$$

$$\boxed{\text{but an SM+GR-type infrared theorem is still far from closed.}} \tag{25}$$

Nevertheless, the checklist above converts the problem from an amorphous ambition into a structured proof program.

XIV. FINAL SUMMARY

The minimal theorem/lemma checklist for an SM+GR-type infrared limit in TECT consists of eight conditions:

$$C1: \text{ continuum infrared sector,} \qquad C2: \text{ Einstein-type gravity sector,} \qquad C3: \text{ gauge sector emergence,} \tag{26}$$

$$C4: \text{ chiral matter and family sector,} \qquad C5: \text{ canonical kinetic normalization,} \qquad C6: \text{ decoupling of extra states,} \tag{27}$$

$$C7: \text{ suppression of dangerous higher operators,} \qquad C8: \text{ consistency conditions.} \tag{28}$$

The honest present status is:

$$\boxed{\text{Closed: none.}} \tag{29}$$

$$\boxed{\text{Conditional: } C1,\ C4,\ C5.} \tag{30}$$

$$\boxed{\text{Open: } C2,\ C3,\ C6,\ C7,\ C8.} \tag{31}$$

The correct proof dependency order is:

$$\boxed{C1\ \rightarrow\ (C2, C3, C4)\ \rightarrow\ C5\ \rightarrow\ (C6, C7, C8).} \tag{32}$$

Thus the present TECT program is strongest on the matter/flavor side, especially in the reduction of flavor to a finite family Hamiltonian, but the full SM+GR-type infrared theorem remains unclosed. The immediate rational next step is to sharpen the strongest conditional sector, namely the matter/family program, while clarifying the continuum control required for the infrared limit itself.

XV. STATUS-TAGGED CKM ENTRY CHECKLIST INSIDE THE TECT MATTER/FAMILY PROGRAM

XVI. 1. PURPOSE

Before attempting a full CKM extraction, one should isolate the minimal conditions under which the projected family Hamiltonians can be interpreted as physically meaningful quark flavor data.

We denote these minimal CKM-entry conditions by $K1$–$K5$.

XVII. 2. THE FIVE MINIMAL CKM-ENTRY CONDITIONS

A. K1. Common left-handed family subspace

There must exist a common three-dimensional left-handed family subspace

$$\mathcal{H}_{\text{fam}} = \text{span}\{\phi_A\}_{A=1}^{3} \tag{33}$$

on which both the up-sector and down-sector operators act.

Status:

$$\boxed{\text{Conditional}} \tag{34}$$

Reason. The flavor problem has already been reduced structurally to a finite family Hamiltonian problem,

$$(H_f^{\text{fam}})_{AB} = \langle \phi_A, \mathcal{Y}_f \phi_B \rangle, \tag{35}$$

which is a genuine achievement. However, the actual microscopic derivation of a common left-handed family subspace in the intended Class II setting is not yet fully closed.

B. K2. Canonical kinetic normalization

After projection to the family sector, the kinetic matrix must be reducible to canonical form:

$$Z_{AB} \rightarrow \delta_{AB}. \tag{36}$$

Status:

$$\boxed{\text{Conditional}} \tag{37}$$

Reason. The necessity of this condition is fully understood, but the TECT-specific microscopic computation of the projected kinetic form and its explicit canonical reduction has not yet been completed.

C. K3. Nondegenerate diagonal splittings

The projected family Hamiltonians must have nondegenerate diagonal splittings:

$$\Delta_{21} \neq 0, \qquad \Delta_{32} \neq 0, \qquad \Delta_{31} \neq 0. \tag{38}$$

Status:

$$\boxed{\text{Conditional}} \tag{39}$$

Reason. Toy-level projected family matrices already exhibit nontrivial diagonal entries, so this requirement is not empty or implausible. However, nondegenerate splittings have not yet been derived robustly in the actual microscopic Class II setting.

D. K4. Hierarchy seed in the off-diagonal structure

The projected family Hamiltonian must structurally favor the observed CKM hierarchy pattern, at least at the level of a microscopic seed:

$$|h_{12}|,\ |h_{23}| \gg |h_{13}|, \tag{40}$$

or more sharply in the desired regime,

$$|h_{12}| > |h_{23}| > |h_{13}|. \tag{41}$$

Status:

$$\boxed{\text{Conditional}} \tag{42}$$

Reason. This is presently the strongest conditional block. The earlier toy potential-like operator produced the wrong hierarchy, but the Hermitian corrected operator has been shown to possess an exact link-local derivative structure:

$$\Pi_d'(x) = \frac{1}{R^2} \sum_{A<B} (t_A - t_B)(X_A - X_B) p_A(x) p_B(x), \tag{43}$$

which structurally suppresses the long direct 1–3 channel. Moreover, the degenerate spurion

$$T_d \propto \mathrm{diag}(2, -1, -1) \tag{44}$$

was correctly excluded as a first CKM toy spurion, and the minimal viable nondegenerate choice

$$T_d = \mathrm{diag}(1, 0, -1) \tag{45}$$

was identified. Nevertheless, the hierarchy seed is still not derived from the actual microscopic Class II sector, and therefore it remains conditional rather than closed.

E. K5. Irreducible CP seed

There must exist a physically non-removable phase source such that

$$\Im(h_{12} h_{23} h_{31}) \neq 0 \tag{46}$$

is possible after all family rephasings.
Status:

$$\boxed{\text{Open}} \tag{47}$$

Reason. A one-dimensional toy model with a single commuting diagonal spurion may generate complex-looking matrix entries without guaranteeing a genuinely physical rephasing-invariant phase. At present, no theorem-level mechanism has yet been established ensuring an irreducible CP-violating seed in the microscopic TECT flavor construction.

XVIII. 3. SUMMARY TABLE

The current CKM-entry status is therefore:

Condition	Current Status
$K1$ Common family subspace	Conditional
$K2$ Canonical kinetic normalization	Conditional
$K3$ Nondegenerate splittings	Conditional
$K4$ Hierarchy seed	Conditional
$K5$ Irreducible CP seed	Open

(48)

In particular:

$$\boxed{\text{Closed: none.}} \tag{49}$$

$$\boxed{\text{Conditional: } K1,\ K2,\ K3,\ K4.} \tag{50}$$

$$\boxed{\text{Open: } K5.} \tag{51}$$

XIX. 4. STRATEGIC CONSEQUENCE

This status split implies that the TECT CKM program should currently be divided into two phases.

A. Phase A: hierarchy-only CKM program

Use the strongest presently supported ingredients $K1$–$K4$ to test whether the microscopic mechanism can reproduce the observed hierarchy pattern in the magnitudes:

$$|V_{us}| > |V_{cb}| > |V_{ub}|. \tag{52}$$

B. Phase B: full CP-violating CKM program

Only after Phase A is stabilized should one attempt to close $K5$, namely the existence of an irreducible CP-violating seed. This likely requires at least one of the following:

$$\text{(i) a noncommuting second spurion,} \qquad \text{(ii) genuinely three-dimensional geometric misalignment,} \qquad \text{or (iii) intrinsically comple} \tag{53}$$

XX. 5. HONEST FINAL DIAGNOSIS

The present TECT CKM program is already meaningful at the level of hierarchy generation, but it is not yet a closed theory of physical CKM CP violation. The honest present statement is therefore:

$$\boxed{\text{TECT has a conditionally supported CKM-hierarchy program, but not yet a closed full CKM-phase program.}} \tag{54}$$

XXI. FINAL SUMMARY

Inside the TECT matter/family program, the minimal CKM-entry conditions are:

$$K1: \text{ common left-handed family subspace}, \qquad K2: \text{ canonical kinetic normalization}, \tag{55}$$

$$K3: \text{ nondegenerate diagonal splittings}, \qquad K4: \text{ hierarchy seed in the off-diagonal structure}, \qquad K5: \text{ irreducible CP seed}. \tag{56}$$

The honest current status is:

$$\boxed{\text{Closed: none.}} \tag{57}$$

$$\boxed{\text{Conditional: } K1,\ K2,\ K3,\ K4.} \tag{58}$$

$$\boxed{\text{Open: } K5.} \tag{59}$$

Thus the present TECT CKM program is strong enough to justify a hierarchy-focused next phase, but not yet strong enough to claim a fully closed theory of physical CKM CP violation. The mathematically rational next step is therefore to push $K1$–$K4$ as far as possible in the actual microscopic setting, while separately identifying the minimal structure required to close $K5$.

XXII. DEPENDENCY ORDER AND WORK ORDER FOR THE CKM-ENTRY CONDITIONS $K1$–$K4$

XXIII. 1. THE FOUR PRE-CKM CONDITIONS

Before a physically meaningful CKM extraction, the following four conditions must be addressed:

$$K1: \text{ common left-handed family subspace}, \qquad K2: \text{ canonical kinetic normalization}, \tag{60}$$

$$K3: \text{ nondegenerate diagonal splittings}, \qquad K4: \text{ hierarchy seed in the off-diagonal structure}. \tag{61}$$

Here $K5$ (irreducible CP seed) is deliberately postponed, since the present goal is to organize the hierarchy-only CKM program first.

XXIV. 2. LOGICAL DEPENDENCY ORDER

The logical dependency structure is

$$\boxed{K1 \longrightarrow K2 \longrightarrow (K3,\ K4).} \tag{62}$$

A. 2.1. Why $K1$ comes first

The CKM matrix is defined as the relative misalignment of the up-sector and down-sector operators acting on the same left-handed family space. Therefore, without a common three-dimensional family subspace

$$\mathcal{H}_{\text{fam}} = \operatorname{span}\{\phi_A\}_{A=1}^{3}, \tag{63}$$

the CKM problem is not yet even well-defined.

Hence:

$$\boxed{K1 \text{ is the first logical prerequisite.}} \tag{64}$$

B. 2.2. Why $K2$ comes before the physical interpretation of $K3$ and $K4$

After projection to the family sector, the kinetic form is generically

$$Z_{AB} \neq \delta_{AB}. \tag{65}$$

Then the projected Hamiltonian entries

$$d_A, \qquad h_{AB} \tag{66}$$

are not yet physical observables, since they still depend on a non-canonical family metric. Only after canonical normalization,

$$Z \to I, \tag{67}$$

does one obtain the physically meaningful normalized family Hamiltonian

$$H_f^{\text{phys}} = Z^{-1/2} H_f^{\text{fam}} Z^{-1/2}. \tag{68}$$

Therefore:

$$\boxed{K2 \text{ must precede the physical reading of both } K3 \text{ and } K4.} \tag{69}$$

C. 2.3. Why $K3$ and $K4$ come only after $K1$ and $K2$

The nondegenerate splittings

$$\Delta_{21},\ \Delta_{32},\ \Delta_{31} \tag{70}$$

and the hierarchy seed

$$|h_{12}| > |h_{23}| > |h_{13}| \tag{71}$$

must both be read from the normalized physical family Hamiltonian. Otherwise they remain basis-dependent artifacts.

Hence the logical order is exactly:

$$\boxed{K1 \to K2 \to (K3, K4).} \tag{72}$$

XXV. 3. PRACTICAL WORK ORDER

Although the logical dependency order is $K1 \to K2 \to (K3, K4)$, the most effective practical work order is slightly different:

$$\boxed{K4^{\text{toy}} \longrightarrow K1 \longrightarrow K2 \longrightarrow K3 \longrightarrow K4^{\text{micro}}.} \tag{73}$$

A. 3.1. Why $K4^{\text{toy}}$ comes first in practice

At present, the strongest conditional component of the TECT CKM program is the hierarchy-seed mechanism in the corrected toy operator. In particular:

1. the earlier potential-like toy operator was explicitly shown to produce the wrong hierarchy,
2. the Hermitian corrected operator has an exact link-local derivative structure,
3. the long direct 1–3 channel is structurally suppressed,
4. and the minimal viable nondegenerate spurion

   $$T_d = \text{diag}(1, 0, -1)$$

   has already been identified.

Thus $K4^{\text{toy}}$ already provides a strong guiding pattern for the microscopic program, even though it is not yet a closed microscopic theorem.

B. 3.2. Why $K1$ and $K2$ are the first true closure targets

The next actual closure targets must be $K1$ and $K2$, because they turn the CKM program from a toy matrix exercise into a basis-independent physical problem.

Concretely, one must:

$$\boxed{\text{(i) construct the actual common family subspace,} \quad \text{(ii) compute the projected kinetic matrix,} \quad \text{(iii) canonically normalize}} \tag{74}$$

C. 3.3. Why $K3$ follows

Only after $K1$ and $K2$ are in place can the diagonal splittings

$$\Delta_{21},\ \Delta_{32},\ \Delta_{31} \tag{75}$$

be meaningfully interpreted as physical input for CKM extraction.

Thus $K3$ is not conceptually prior to $K1$ and $K2$, even if it may look numerically easier in toy runs.

D. 3.4. Why $K4^{\text{micro}}$ comes last

The final step of the hierarchy-only program is to verify that the hierarchy seed survives in the actual microscopic setting after the family space and kinetic normalization are fixed.

This is the true microscopic closure of $K4$, as opposed to the presently available toy-level version.

XXVI. 4. RECOMMENDED IMMEDIATE RESEARCH ORDER

The immediate recommended sequence is therefore:

$$\boxed{\text{Priority 1: close } K1, \quad \text{Priority 2: close } K2, \quad \text{Priority 3: close } K3, \quad \text{Priority 4: close } K4^{\text{micro}}.} \tag{76}$$

Equivalently:

$$\boxed{\text{actual family space} \longrightarrow \text{projected kinetic form} \longrightarrow \text{normalized family Hamiltonian} \longrightarrow \text{splittings and hierarchy.}} \tag{77}$$

XXVII. 5. IMMEDIATE COMPUTATIONAL CONSEQUENCE

The next concrete task is therefore not yet the final CKM fit. Instead, one should first compute, in the actual Class II setting,

$$S^{\text{fam}}_{AB}, \qquad Z^{\text{fam}}_{AB}, \qquad H^{\text{fam}}_{u,d}, \tag{78}$$

and then pass to the normalized physical Hamiltonians

$$H^{\text{phys}}_{u,d} = (Z^{\text{fam}})^{-1/2} H^{\text{fam}}_{u,d} (Z^{\text{fam}})^{-1/2}. \tag{79}$$

Only after this step should one extract

$$\Delta_{ab}, \qquad h_{12},\ h_{23},\ h_{13}. \tag{80}$$

XXVIII. 6. FINAL DIAGNOSIS

The honest conclusion is:

$$\boxed{\text{The logical gate order is } K1 \to K2 \to (K3, K4),} \tag{81}$$

but

$$\boxed{\text{the practical research order should keep the toy hierarchy seed as a guide while closing } K1 \text{ and } K2 \text{ first.}} \tag{82}$$

XXIX. FINAL SUMMARY

For the hierarchy-only CKM program inside TECT, the four relevant entry conditions are:

$$K1: \text{ common left-handed family subspace}, \qquad K2: \text{ canonical kinetic normalization}, \tag{83}$$

$$K3: \text{ nondegenerate diagonal splittings}, \qquad K4: \text{ hierarchy seed}. \tag{84}$$

Their logical dependency order is

$$\boxed{K1 \to K2 \to (K3, K4).} \tag{85}$$

However, the most effective practical work order is

$$\boxed{K4^{\text{toy}} \to K1 \to K2 \to K3 \to K4^{\text{micro}}.} \tag{86}$$

This is because the toy hierarchy seed is already the strongest conditional part of the present TECT CKM program and should be retained as a guide, while the true closure targets are the construction of the actual family space and the canonical normalization of the projected kinetic sector.

Therefore the immediate next step is to compute the actual projected family overlap/kinetic data and pass to the normalized physical family Hamiltonians before reading off splittings and hierarchy.

XXX. DECOMPOSITION OF $K1$: MINIMAL LEMMAS FOR A COMMON LEFT-HANDED FAMILY SUBSPACE

XXXI. 1. PURPOSE

The condition

$$K1: \text{ there exists a common left-handed family subspace for the up and down sectors} \tag{87}$$

is not a single statement. It should be decomposed into a sequence of smaller lemmas, each with a separate proof status.

We write:

$$\boxed{K1A \longrightarrow K1B \longrightarrow K1C \longrightarrow K1D.} \tag{88}$$

XXXII. 2. THE FOUR SUBCONDITIONS

A. K1A. Common carrier sector exists

There exists a common low-energy defect-localized carrier sector for the unperturbed background P_0, determined by an operator of the form

$$H_0[P_0] \quad \text{or} \quad \mathcal{L}_{\text{loc}}[P_0]. \tag{89}$$

This sector must be meaningful before introducing the flavor-sector perturbations δH_u and δH_d.

Status:

$$\boxed{\text{Conditional}} \tag{90}$$

Reason. A substantial low-lying carrier-mode program already exists, and toy/local Hessian constructions produce defect-localized low modes. However, the actual microscopic Class II Hessian has not yet been derived and analyzed at theorem level.

B. K1B. A three-dimensional family subspace is isolated

Inside the common carrier sector, there exists an isolated three-dimensional subspace

$$\mathcal{H}_{\mathrm{fam}} = \mathrm{span}\{\phi_1, \phi_2, \phi_3\}, \tag{91}$$

separated from the remainder of the spectrum by a sufficient gap or effective Schur-complement control.

Status:

$$\boxed{\text{Conditional}} \tag{92}$$

Reason. The toy multi-defect constructions naturally suggest a triplet of family-like modes, and the generalized-eigenproblem pipeline can isolate three low-lying modes. But the robust microscopic proof of a genuinely isolated three-dimensional family subspace is not yet complete.

C. K1C. The up/down sector perturbations preserve the family sector effectively

For $f \in \{u, d\}$, the perturbations δH_f must act in such a way that the family subspace remains an effective closed sector to leading order:

$$P_{\mathrm{fam}} \delta H_f P_{\mathrm{fam}} \quad \text{dominates}, \qquad P_{\mathrm{fam}} \delta H_f Q_{\mathrm{fam}} \quad \text{is controllably subleading.} \tag{93}$$

This is the structural precondition for the effective family Hamiltonian

$$\mathcal{Y}_f = P_{\mathrm{fam}} \delta H_f P_{\mathrm{fam}} - P_{\mathrm{fam}} \delta H_f Q_{\mathrm{fam}} (Q_{\mathrm{fam}} H_0 Q_{\mathrm{fam}} - E_*)^{-1} Q_{\mathrm{fam}} \delta H_f P_{\mathrm{fam}} + \cdots . \tag{94}$$

Status:

$$\boxed{\text{Conditional}} \tag{95}$$

Reason. The reduction formula is already structurally established, but its actual microscopic validity in the intended Class II setting still depends on the missing derivation of the true sector operators and their couplings to the carrier sector.

D. K1D. Physical left-handed interpretation

The common family subspace must be interpretable as the physical left-handed quark family space. This requires more than linear algebra: it demands compatibility with chirality, gauge-coupling structure, and the removal or lifting of unwanted mirror sectors.

Status:

$$\boxed{\text{Open}} \tag{96}$$

Reason. Although there is substantial progress toward family-carrier and Dirac-type emergence, the theorem-level identification of the common family space with the actual physical left-handed quark family sector has not yet been achieved.

XXXIII. 3. STATUS SUMMARY

The honest present classification is:

$$\boxed{\text{Closed: none}} \tag{97}$$

$$\boxed{\text{Conditional: } K1A,\ K1B,\ K1C} \tag{98}$$

$$\boxed{\text{Open: } K1D} \tag{99}$$

XXXIV. 4. DEPENDENCY ORDER

The correct internal proof order is:

$$\boxed{K1A \to K1B \to K1C \to K1D.} \tag{100}$$

This means:

1. one must first establish the existence of a common carrier sector;
2. then isolate a three-dimensional family subspace within it;
3. then show that the relevant sector perturbations preserve this family subspace effectively;
4. and only then can one attempt the full physical left-handed interpretation.

XXXV. 5. IMMEDIATE NEXT TASKS

The most productive immediate tasks are:

$$\boxed{\text{(i) isolate the low-lying triplet in an actual or quasi-actual background,}} \tag{101}$$

$$\boxed{\text{(ii) test the relative size of } P_{\mathrm{fam}}\delta H_f P_{\mathrm{fam}} \text{ versus } P_{\mathrm{fam}}\delta H_f Q_{\mathrm{fam}}.} \tag{102}$$

These two tasks advance $K1A$, $K1B$, and $K1C$ simultaneously, and are therefore the correct next frontiers for the $K1$ program.

XXXVI. 6. FINAL DIAGNOSIS

The common-family-space problem is not yet closed. However, the main unresolved part is not the final physical interpretation $K1D$ by itself; rather, it is the actual microscopic realization of the earlier linear-algebraic stages $K1A$–$K1C$. Thus the correct immediate strategy is to close the existence, isolation, and preservation of the family subspace before attempting a full physical chirality/gauge interpretation.

XXXVII. FINAL SUMMARY

The condition $K1$, namely the existence of a common left-handed family subspace for the up and down sectors, should be decomposed into four subconditions:

$$K1A: \text{ common carrier sector exists,} \qquad K1B: \text{ a three-dimensional family subspace is isolated,} \tag{103}$$

$$K1C: \text{ sector perturbations preserve the family space effectively,} \qquad K1D: \text{ physical left-handed interpretation.} \tag{104}$$

The honest present status is:

$$\boxed{\text{Closed: none}} \tag{105}$$

$$\boxed{\text{Conditional: } K1A,\ K1B,\ K1C} \tag{106}$$

$$\boxed{\text{Open: } K1D} \tag{107}$$

The correct internal proof order is

$$\boxed{K1A \to K1B \to K1C \to K1D.} \tag{108}$$

Therefore the immediate next steps are not yet a full chirality/gauge interpretation, but rather:

$$\text{(i) isolate the low-lying triplet in an actual or quasi-actual background,} \tag{109}$$

$$\text{(ii) verify that the sector perturbations preserve this triplet effectively.} \tag{110}$$

These are the mathematically correct next targets for closing $K1$.

XXXVIII. LOW-LYING TRIPLET ISOLATION CRITERIA FOR $K1A$ AND $K1B$

XXXIX. 1. PURPOSE

To close the first part of the CKM-entry program, one must establish not merely the existence of some low-energy carrier sector, but the existence of an isolated three-dimensional defect-anchored triplet that can serve as the common family subspace.

Thus the correct task is to refine

$$K1A: \text{ common carrier sector exists,} \qquad K1B: \text{ a three-dimensional family subspace is isolated,} \tag{111}$$

into explicit low-lying triplet criteria.

XL. 2. REFINED SUBCONDITIONS

We refine $K1A$ and $K1B$ into the following minimal components:

$$\boxed{K1A1 \to K1A2 \to K1B1 \to K1B2 \to K1B3.} \tag{112}$$

A. K1A1. Localized low-energy carrier sector exists

There exists a low-energy carrier sector of the unperturbed/localized operator

$$H_0[P_0] \quad \text{or} \quad \mathcal{L}_{\rm loc}[P_0] = -Z\nabla^2 + U[P_0](x), \tag{113}$$

whose states are localized near the multi-defect background rather than delocalized continuum modes.

Status:

$$\boxed{\text{Conditional}} \tag{114}$$

Reason. Toy and quasi-local Hessian models already produce defect-localized low modes, but the full Class II microscopic operator has not yet been derived and proven to possess such a sector.

B. K1A2. Defect-anchored quasi-local basis exists

There exist trial states

$$\chi_A, \qquad A = 1, 2, 3, \tag{115}$$

each anchored near defect center X_A, such that the low-energy carrier sector has large overlap with

$$\mathrm{span}\{\chi_1, \chi_2, \chi_3\}. \tag{116}$$

Status:

$$\boxed{\text{Conditional}} \tag{117}$$

Reason. This is already realized operationally in the Gaussian-basis pipeline and in the multi-defect toy constructions. However, the theorem-level justification that these quasi-local states converge to the actual microscopic low modes remains unfinished.

C. K1B1. Three low modes are spectrally separated

Let

$$\varepsilon_1 \le \varepsilon_2 \le \varepsilon_3 \le \varepsilon_4 \le \cdots \tag{118}$$

be the ordered eigenvalues (or generalized eigenvalues) of the localized carrier problem. Then there must exist a positive spectral gap

$$\boxed{\Delta_{\mathrm{gap}} := \varepsilon_4 - \varepsilon_3 > 0} \tag{119}$$

such that the first three modes are isolated from the remainder of the spectrum.

Status:

$$\boxed{\text{Conditional}} \tag{120}$$

Reason. The current numerical/tactical program already searches for exactly such a triplet structure, but the robust microscopic proof of a nonvanishing triplet gap is not yet complete.

D. K1B2. Stability of the family projector

Let

$$P_{\mathrm{fam}} = \sum_{a=1}^{3} |\phi_a\rangle\langle\phi_a| \tag{121}$$

be the spectral projector onto the three isolated low modes. Then the projector must be stable under the intended background deformations and discretization/trial-space refinements, in the sense that

$$\|P_{\mathrm{fam}}^{(N)} - P_{\mathrm{fam}}^{(N')}\| \to 0 \qquad \text{as the approximation is refined,} \tag{122}$$

and similarly under small background deformations within the same branch.

Status:

$$\boxed{\text{Conditional}} \tag{123}$$

Reason. The program already assumes such stability implicitly when passing from a numerical low-lying triplet to an effective family Hamiltonian, but explicit convergence/stability estimates have not yet been proven.

E. K1B3. Robust triplet isolation theorem in the actual Class II setting

There exists a theorem-level statement in the actual microscopic Class II theory that the relevant multi-defect background supports a robustly isolated three-dimensional low-energy triplet.

Status:

$$\boxed{\text{Open}} \tag{124}$$

Reason. This is the actual frontier. The toy and quasi-actual constructions strongly suggest such a structure, but the theorem has not yet been established in the intended microscopic theory.

XLI. 3. SINGLE-DEFECT PRECURSOR CRITERION

A practical precursor to triplet isolation is the single-defect localization problem. Let

$$\mathcal{L}_A = -Z\nabla^2 + U_A(x) \tag{125}$$

be the operator for a single defect centered at X_A. The minimal precursor condition is:

$$\boxed{\text{Each single-defect operator } \mathcal{L}_A \text{ admits at least one localized low mode } \chi_A.} \tag{126}$$

This provides the seed states for the multi-defect family construction.

Interpretation. If the single-defect problem fails, then the multi-defect family triplet has no natural microscopic seed. If it succeeds and the inter-defect overlaps remain small, then the multi-defect problem naturally enters a tight-binding-like three-state regime.

XLII. 4. MULTI-DEFECT WEAK-OVERLAP CRITERION

Assume now that single-defect localized states χ_A exist. Define

$$S_{AB} := \langle \chi_A, \chi_B \rangle, \qquad L_{AB} := \langle \chi_A, \mathcal{L}_{\text{loc}} \chi_B \rangle. \tag{127}$$

The weak-overlap criterion is:

$$\boxed{|S_{AB}| \ll 1 \quad (A \neq B), \qquad |L_{AB}| \ll |L_{AA}| \quad \text{except for controlled nearest-neighbor mixing.}} \tag{128}$$

In this regime, the low-energy sector is approximately governed by the 3×3 generalized eigenproblem

$$Lc = \varepsilon Sc, \tag{129}$$

and the resulting low modes form a defect-anchored family triplet.

XLIII. 5. DEFECT ANCHORING CRITERION

The three low modes ϕ_a must not merely exist abstractly; they must be interpretable as family carriers attached to the three defect centers. A practical criterion is the existence of a unitary matrix U_{aA} such that the rotated basis

$$\psi_A := \sum_{a=1}^{3} U_{Aa} \phi_a \tag{130}$$

satisfies

$$\boxed{\langle \psi_A, \, \Theta_B \, \psi_A \rangle \approx \delta_{AB}} \tag{131}$$

for suitable localization projectors Θ_B onto neighborhoods of the defect centers X_B.

Interpretation. This means that, after a unitary change of basis inside the triplet, one can identify one mode per defect center. This is the precise defect-anchoring property required before speaking of a family triplet.

XLIV. 6. GAP CRITERION AND SCHUR REDUCTION

Once the first three modes are isolated, the Schur-complement reduction used in the flavor program becomes justified only if the gap to the complement sector is sufficiently controlled. Concretely, one needs

$$\boxed{\|(Q_{\rm fam} H_0 Q_{\rm fam} - E_*)^{-1}\| \leq \frac{1}{\Delta_{\rm gap}}} \tag{132}$$

up to controlled constants, where E_* is the low-energy reference scale inside the triplet.

Thus the triplet gap is not merely an aesthetic feature; it is the operator-theoretic prerequisite for the effective family Hamiltonian program.

XLV. 7. SUMMARY STATUS OF $K1A$–$K1B$

The honest present classification is:

$$\boxed{\text{Closed: none}} \tag{133}$$

$$\boxed{\text{Conditional: } K1A1,\ K1A2,\ K1B1,\ K1B2} \tag{134}$$

$$\boxed{\text{Open: } K1B3} \tag{135}$$

XLVI. 8. IMMEDIATE COMPUTATIONAL TARGETS

The next mathematically rational targets are:

$$\boxed{\text{(i) solve the single-defect localization problem,}} \tag{136}$$

$$\boxed{\text{(ii) build the three-defect Gram/Hessian matrices } S_{AB}, L_{AB},} \tag{137}$$

$$\boxed{\text{(iii) verify the triplet gap } \Delta_{\rm gap} > 0,} \tag{138}$$

$$\boxed{\text{(iv) verify defect anchoring of the resulting low triplet.}} \tag{139}$$

These four tasks are the correct next frontiers for closing $K1A$ and $K1B$.

XLVII. FINAL SUMMARY

The first part of the common-family-space problem should be refined into the low-lying triplet isolation chain

$$K1A1 \to K1A2 \to K1B1 \to K1B2 \to K1B3, \tag{140}$$

where:

$$K1A1: \ \text{a localized low-energy carrier sector exists,} \tag{141}$$

$$K1A2: \ \text{a defect-anchored quasi-local basis exists,} \tag{142}$$

$$K1B1: \text{ three low modes are spectrally separated,} \tag{143}$$

$$K1B2: \text{ the family projector is stable,} \tag{144}$$

$$K1B3: \text{ a robust triplet-isolation theorem holds in the actual Class II theory.} \tag{145}$$

The honest status is:

$$\boxed{\text{Closed: none}} \tag{146}$$

$$\boxed{\text{Conditional: } K1A1,\ K1A2,\ K1B1,\ K1B2} \tag{147}$$

$$\boxed{\text{Open: } K1B3} \tag{148}$$

The practical criterion for triplet isolation is:

$$\text{localized single-defect seed modes } + \text{ weak inter-defect overlap } + \text{ positive triplet gap } + \text{ defect anchoring.} \tag{149}$$

Thus the immediate next tasks are:

$$\text{(i) solve the single-defect localization problem,} \tag{150}$$

$$\text{(ii) build the multi-defect Gram/Hessian matrices,} \tag{151}$$

$$\text{(iii) verify the triplet gap,} \tag{152}$$

$$\text{(iv) verify defect anchoring of the low triplet.} \tag{153}$$

These are the correct next steps for closing $K1A$ and $K1B$.

XLVIII. SINGLE-DEFECT LOCALIZATION PROBLEM AS THE SEED OF THE FAMILY PROGRAM

XLIX. 1. PURPOSE

Before one can justify a multi-defect family triplet, one must establish the existence of a localized seed mode for a single defect. This is the minimal precursor of the carrier-sector program.

Accordingly, the single-defect localization problem should be decomposed into the chain

$$\boxed{S1 \longrightarrow S2 \longrightarrow S3 \longrightarrow S4.} \tag{154}$$

L. 2. THE SINGLE-DEFECT BACKGROUND

Let $X_0 \in \mathbb{R}^3$ be a defect center, and let the corresponding background be denoted schematically by

$$P_{X_0}(x) \quad \text{or} \quad \rho_{X_0}(x). \tag{155}$$

The single-defect carrier problem is governed by an operator of the form

$$\boxed{\mathcal{L}_{X_0} = -Z\nabla^2 + U_{X_0}(x),} \tag{156}$$

where $U_{X_0}(x)$ is the effective defect-induced trapping profile extracted from the background.

At the toy/Class-II-surrogate level, one may take

$$U_{X_0}(x) = \mu^2 + 2\lambda\rho_{X_0}(x), \tag{157}$$

or more generally any effective local operator obtained from the second variation around the defect background.

LI. 3. REFINED SUBCONDITIONS

A. S1. Admissibility of the single-defect background

The single-defect background P_{X_0} (or ρ_{X_0}) must belong to the admissible branch of backgrounds under consideration, with sufficient regularity and localization to define a meaningful carrier operator.
Status:

$$\boxed{\text{Conditional}} \tag{158}$$

Reason. The project already uses well-motivated defect-centered toy and surrogate backgrounds, but the theorem-level existence and admissibility of the actual Class II single-defect background remains unclosed.

B. S2. Operator control: self-adjointness and ellipticity

The carrier operator $\mathcal{L}_{X_0}$ must define a well-posed spectral problem. At minimum, one needs:

1. symmetry/self-adjointness on an appropriate domain,
2. ellipticity or analogous coercive control,
3. lower-semiboundedness, so that the low-energy sector is meaningful.

Status:

$$\boxed{\text{Conditional}} \tag{159}$$

Reason. For the toy/local operators currently used, these properties are standard or straightforward. However, the actual Class II microscopic second-variation operator has not yet been fully derived and checked at theorem level.

C. S3. Existence of a localized low-energy seed mode

There exists at least one normalized mode χ_{X_0} satisfying

$$\mathcal{L}_{X_0}\chi_{X_0} = \varepsilon_0\chi_{X_0}, \qquad \|\chi_{X_0}\|_{L^2} = 1, \tag{160}$$

or a sufficiently controlled quasi-bound analogue, with the localization property

$$\boxed{\int_{|x-X_0|>R_{\mathrm{loc}}} |\chi_{X_0}(x)|^2\, dx \ll 1} \tag{161}$$

for some finite localization radius R_{loc}.
Status:

$$\boxed{\text{Conditional}} \tag{162}$$

Reason. This is already strongly suggested by the current surrogate/local-Hessian program. However, an actual theorem proving the existence of such a localized seed mode in the intended microscopic Class II setting is still missing.

D. S4. Robustness and defect anchoring

The localized seed mode must be robust under small admissible deformations of the defect profile and must remain anchored to the defect center X_0. Concretely, one needs:

$$\boxed{\langle \chi_{X_0}, \Theta_{X_0}\chi_{X_0}\rangle \approx 1} \tag{163}$$

for a localization projector Θ_{X_0} supported near the defect core, and one requires continuity/stability of χ_{X_0} under small background deformations within the same branch.

Status:

$$\boxed{\text{Open}} \tag{164}$$

Reason. This is the first genuinely delicate theorem-level step. It is not enough to have some low-energy mode; one needs a mode that remains defect-centered in a stable, branch-consistent sense. This has not yet been proven in the actual Class II theory.

LII. 4. VARIATIONAL LOCALIZATION CRITERION

A practical sufficient criterion for $S3$ is variational. Let $\psi \in H^1(\mathbb{R}^3)$ be a normalized trial function centered at X_0. Define the Rayleigh quotient

$$\mathcal{R}[\psi] = \frac{\langle \psi, \mathcal{L}_{X_0}\psi \rangle}{\langle \psi, \psi \rangle}. \tag{165}$$

If one can construct a defect-centered trial state ψ_{X_0} such that

$$\boxed{\mathcal{R}[\psi_{X_0}] < E_{\text{cont}},} \tag{166}$$

where E_{cont} denotes the bottom of the extended/continuum sector, then the operator admits at least one localized bound mode below the continuum threshold.

Interpretation. This is the most useful surrogate-level criterion: one does not need the full exact eigenfunction first; it suffices to exhibit a defect-centered trial state whose energy lies below the extended-state threshold.

LIII. 5. TOY GAUSSIAN SEED CRITERION

At the present surrogate level, a natural single-defect trial family is

$$\psi_\sigma(x) = \exp\left[-\frac{|x - X_0|^2}{2\sigma^2}\right]. \tag{167}$$

For this family one computes

$$\mathcal{R}(\sigma) = \frac{\langle \psi_\sigma, \mathcal{L}_{X_0}\psi_\sigma \rangle}{\langle \psi_\sigma, \psi_\sigma \rangle}. \tag{168}$$

Then a practical seed-localization test is:

$$\boxed{\exists\, \sigma_* > 0 \text{ such that } \mathcal{R}(\sigma_*) < E_{\text{cont}}.} \tag{169}$$

This does not yet prove the full microscopic theorem, but it provides the correct toy/quasi-actual diagnostic for the existence of a localized carrier seed.

LIV. 6. SPECTRAL MEANING OF THE CONTINUUM THRESHOLD

For a Schrödinger-type local operator

$$\mathcal{L}_{X_0} = -Z\nabla^2 + U_{X_0}(x), \tag{170}$$

with asymptotic value

$$U_{X_0}(x) \to U_\infty \qquad \text{as } |x| \to \infty, \tag{171}$$

the bottom of the extended sector is heuristically

$$E_{\text{cont}} \approx U_\infty \tag{172}$$

up to model-dependent corrections. Thus a single-defect seed mode requires an effective local well relative to the asymptotic background.

Key physical point. The defect must act as a trapping deformation of the local carrier operator, not merely as a weak perturbation of a featureless bulk.

LV. 7. WHY THIS MATTERS FOR THE FAMILY PROGRAM

If $S3$ fails, there is no natural defect-centered seed mode χ_{X_0}, and hence no natural building block for the three-defect family triplet. Conversely, if each single defect supports one localized seed mode and the inter-defect overlaps are small, the multi-defect problem naturally enters the triplet-isolation regime required for $K1A$–$K1B$.

Thus:

$$\boxed{\text{single-defect localization is the microscopic precursor of the family-space construction.}} \tag{173}$$

LVI. 8. HONEST STATUS SUMMARY

The present status is:

$$\boxed{\text{Closed: none}} \tag{174}$$

$$\boxed{\text{Conditional: } S1,\ S2,\ S3} \tag{175}$$

$$\boxed{\text{Open: } S4} \tag{176}$$

LVII. 9. IMMEDIATE NEXT TASKS

The correct immediate tasks are:

$$\boxed{\text{(i) formulate the actual or quasi-actual single-defect surrogate operator } \mathcal{L}_{X_0},} \tag{177}$$

$$\boxed{\text{(ii) compute a variational Rayleigh-quotient test using defect-centered trial states,}} \tag{178}$$

$$\boxed{\text{(iii) identify the continuum threshold } E_{\text{cont}},} \tag{179}$$

$$\boxed{\text{(iv) verify whether a localized seed mode exists below that threshold.}} \tag{180}$$

These are the mathematically correct next steps for closing the single-defect seed problem.

LVIII. FINAL SUMMARY

The single-defect localization problem, which provides the microscopic seed of the family program, should be decomposed into the chain

$$S1 \to S2 \to S3 \to S4, \tag{181}$$

where

$$S1: \text{ admissibility of the single-defect background,} \tag{182}$$

$$S2: \text{ self-adjoint and elliptic control of the carrier operator,} \tag{183}$$

$$S3: \text{ existence of at least one localized low-energy seed mode,} \tag{184}$$

$$S4: \text{ robustness and defect anchoring of that seed mode.} \tag{185}$$

The honest present status is:

$$\boxed{\text{Closed: none}} \tag{186}$$

$$\boxed{\text{Conditional: } S1,\ S2,\ S3} \tag{187}$$

$$\boxed{\text{Open: } S4} \tag{188}$$

A practical sufficient criterion for the seed mode is variational: if there exists a defect-centered trial state ψ_{X_0} such that its Rayleigh quotient satisfies

$$\mathcal{R}[\psi_{X_0}] < E_{\text{cont}}, \tag{189}$$

where E_{cont} is the bottom of the extended sector, then the single-defect operator admits a localized carrier seed mode below the continuum threshold.

Thus the correct next step is to formulate the single-defect surrogate operator explicitly, compute the Rayleigh-quotient test, identify the continuum threshold, and check whether a defect-centered localized seed mode exists.

LIX. VARIATIONAL TEST FOR SINGLE-DEFECT LOCALIZATION IN THE TOY/CLASS-II-SURROGATE CARRIER PROBLEM

LX. 1. MINIMAL SINGLE-DEFECT SURROGATE

To test whether a single defect can support a localized carrier seed mode, we consider the radial single-defect surrogate operator centered at X_0:

$$\boxed{\mathcal{L}_{X_0} = -Z\nabla^2 + U_{X_0}(x), \qquad Z > 0.} \tag{190}$$

The simplest Gaussian defect profile is

$$\rho_{X_0}(x) = \exp\left[-\frac{|x - X_0|^2}{R^2}\right]. \tag{191}$$

The most general minimal trapping form compatible with the present surrogate program is

$$\boxed{U_{X_0}(x) = U_\infty - V_0\,\rho_{X_0}(x), \qquad V_0 > 0.} \tag{192}$$

The previously used sign convention

$$U_{X_0}(x) = \mu^2 + 2\lambda\rho_{X_0}(x) \tag{193}$$

is recovered by

$$U_\infty = \mu^2, \qquad V_0 = -2\lambda. \tag{194}$$

Therefore:

$$\boxed{\text{local trapping requires } V_0 > 0 \iff \lambda < 0.} \tag{195}$$

LXI. 2. CONTINUUM THRESHOLD

Assume

$$U_{X_0}(x) \to U_\infty \qquad \text{as } |x| \to \infty. \tag{196}$$

Then for the present Schrödinger-type surrogate the bottom of the extended sector is

$$\boxed{E_{\text{cont}} = U_\infty.} \tag{197}$$

Thus a localized seed mode requires an eigenvalue below U_∞.

LXII. 3. GAUSSIAN TRIAL STATE

Take the normalized defect-centered Gaussian trial state

$$\boxed{\psi_\sigma(x) = (\pi\sigma^2)^{-3/4}\exp\left[-\frac{|x-X_0|^2}{2\sigma^2}\right], \qquad \sigma > 0.} \tag{198}$$

By translation invariance of the radial defect profile, we may set

$$r := |x - X_0| \tag{199}$$

and compute around the origin.

LXIII. 4. KINETIC EXPECTATION VALUE

For the normalized Gaussian,

$$\nabla\psi_\sigma = -\frac{x}{\sigma^2}\psi_\sigma, \tag{200}$$

hence

$$|\nabla\psi_\sigma|^2 = \frac{r^2}{\sigma^4}|\psi_\sigma|^2. \tag{201}$$

Using

$$|\psi_\sigma|^2 = (\pi\sigma^2)^{-3/2}e^{-r^2/\sigma^2}, \tag{202}$$

one obtains

$$\langle r^2\rangle_{\psi_\sigma} = \frac{3}{2}\sigma^2. \tag{203}$$

Therefore

$$\boxed{\langle\psi_\sigma, -Z\nabla^2\,\psi_\sigma\rangle = Z\int|\nabla\psi_\sigma|^2\,d^3x = \frac{3Z}{2\sigma^2}.} \tag{204}$$

LXIV. 5. DEFECT-OVERLAP EXPECTATION VALUE

We next compute

$$\langle\rho_{X_0}\rangle_{\psi_\sigma} = \int d^3x\,|\psi_\sigma(x)|^2 e^{-r^2/R^2}. \tag{205}$$

Since

$$|\psi_\sigma|^2 e^{-r^2/R^2} = (\pi\sigma^2)^{-3/2}\exp\left[-r^2\left(\frac{1}{\sigma^2}+\frac{1}{R^2}\right)\right], \tag{206}$$

the Gaussian integral gives

$$\boxed{\langle\rho_{X_0}\rangle_{\psi_\sigma} = \left(\frac{R^2}{R^2+\sigma^2}\right)^{3/2}.} \tag{207}$$

LXV. 6. RAYLEIGH QUOTIENT

The Rayleigh quotient of the trial family is

$$\mathcal{R}(\sigma) = \langle \psi_\sigma, \mathcal{L}_{X_0} \psi_\sigma \rangle. \tag{208}$$

Combining the previous results,

$$\boxed{\mathcal{R}(\sigma) = U_\infty + \frac{3Z}{2\sigma^2} - V_0 \left(\frac{R^2}{R^2 + \sigma^2} \right)^{3/2}.} \tag{209}$$

Hence the variational localization criterion is

$$\boxed{\exists\, \sigma > 0 \quad \text{such that} \quad \mathcal{R}(\sigma) < U_\infty.} \tag{210}$$

Equivalently,

$$\boxed{\frac{3Z}{2\sigma^2} < V_0 \left(\frac{R^2}{R^2 + \sigma^2} \right)^{3/2}.} \tag{211}$$

LXVI. 7. IMMEDIATE NO-GO STATEMENT FOR THE OLD POSITIVE-SIGN TOY POTENTIAL

If one uses

$$U_{X_0}(x) = \mu^2 + 2\lambda \rho_{X_0}(x) \tag{212}$$

with $\lambda \geq 0$, then

$$\mathcal{R}(\sigma) = \mu^2 + \frac{3Z}{2\sigma^2} + 2\lambda \left(\frac{R^2}{R^2 + \sigma^2} \right)^{3/2} \geq \mu^2 = E_{\text{cont}}. \tag{213}$$

Therefore:

$$\boxed{\lambda \geq 0 \quad \Longrightarrow \quad \text{no Gaussian variational bound seed mode exists below the continuum threshold.}} \tag{214}$$

This is an exact conclusion within the present surrogate family.

LXVII. 8. DIMENSIONLESS FORM OF THE TRAPPING CRITERION

Introduce the dimensionless ratio

$$a := \frac{\sigma^2}{R^2} > 0. \tag{215}$$

Then the bound-state criterion becomes

$$\frac{3Z}{2R^2} \frac{1}{a} < V_0 (1 + a)^{-3/2}. \tag{216}$$

Equivalently,

$$\boxed{\frac{2V_0 R^2}{3Z} \, \frac{a}{(1 + a)^{3/2}} > 1.} \tag{217}$$

Define

$$f(a) := \frac{a}{(1 + a)^{3/2}}. \tag{218}$$

Then

$$f'(a) = (1+a)^{-5/2}\left(1 - \frac{a}{2}\right), \tag{219}$$

so the unique maximum occurs at

$$\boxed{a_* = 2.} \tag{220}$$

Its maximal value is

$$\boxed{f_{\max} = f(2) = \frac{2}{3\sqrt{3}}.} \tag{221}$$

Therefore the Gaussian trial family admits a localized seed mode if and only if

$$\boxed{\frac{2V_0R^2}{3Z}\,\frac{2}{3\sqrt{3}} > 1,} \tag{222}$$

that is,

$$\boxed{\frac{V_0R^2}{Z} > \frac{9\sqrt{3}}{4}.} \tag{223}$$

Returning to the old notation $V_0 = -2\lambda$, this becomes

$$\boxed{\lambda < 0 \qquad \text{and} \qquad \frac{|\lambda|R^2}{Z} > \frac{9\sqrt{3}}{8}.} \tag{224}$$

Thus, within the present Gaussian variational test, this is the sharp threshold for the existence of a single-defect localized carrier seed.

LXVIII. 9. NEAR-THRESHOLD OPTIMAL WIDTH

At threshold, the optimal Gaussian width is

$$\boxed{\sigma_* = \sqrt{2}\,R.} \tag{225}$$

Hence the seed mode, when it first appears in the Gaussian family, has a width of order the defect radius itself.

This is physically reasonable: the seed mode is neither pointlike nor delocalized, but locked to the defect core scale.

LXIX. 10. INTERPRETATION FOR THE TECT PROGRAM

The single-defect seed criterion has an immediate interpretation:

$$\boxed{\text{A viable carrier seed requires the defect to lower the effective local carrier energy relative to the bulk.}} \tag{226}$$

In particular, the old sign choice

$$+2\lambda\rho_{X_0} \quad \text{with} \quad \lambda > 0 \tag{227}$$

does not trap a low mode. Thus, if localization is to emerge in the actual microscopic program, it must come from at least one of the following:

1. a negative local mass shift ($\lambda < 0$ in the surrogate language),
2. a more complicated matrix-valued or sectoral defect coupling,
3. a Schur-complement-induced effective well after integrating out heavy modes,
4. or a nonlocal carrier operator whose localization mechanism is not visible in the simplest local positive-sign surrogate.

LXX. 11. HONEST STATUS

What is now genuinely established at the surogate level is:

$$\boxed{\text{the Gaussian variational single-defect test is fully explicit.}} \tag{228}$$

$$\boxed{\text{the old positive-sign local toy operator does \textit{not} localize a carrier seed.}} \tag{229}$$

$$\boxed{\text{a trapping surrogate does localize a seed if } \frac{V_0 R^2}{Z} > \frac{9\sqrt{3}}{4}.} \tag{230}$$

What is not yet established is that the actual Class II microscopic Hessian produces such an effective well. That remains the next open bridge.

LXXI. FINAL SUMMARY

For the single-defect surrogate operator

$$\mathcal{L}_{X_0} = -Z\nabla^2 + U_\infty - V_0 e^{-|x-X_0|^2/R^2}, \qquad V_0 > 0, \tag{231}$$

and the normalized Gaussian trial state

$$\psi_\sigma(x) = (\pi\sigma^2)^{-3/4} \exp\left[-\frac{|x-X_0|^2}{2\sigma^2}\right], \tag{232}$$

the Rayleigh quotient is

$$\boxed{\mathcal{R}(\sigma) = U_\infty + \frac{3Z}{2\sigma^2} - V_0 \left(\frac{R^2}{R^2+\sigma^2}\right)^{3/2}.} \tag{233}$$

Since the continuum threshold is

$$E_{\text{cont}} = U_\infty, \tag{234}$$

a localized single-defect carrier seed exists within the Gaussian variational family if and only if

$$\exists\, \sigma > 0 \quad \text{such that} \quad \mathcal{R}(\sigma) < U_\infty. \tag{235}$$

This is equivalent to

$$\frac{V_0 R^2}{Z} > \frac{9\sqrt{3}}{4}, \tag{236}$$

with the threshold attained at

$$\sigma_* = \sqrt{2}\, R. \tag{237}$$

In the earlier sign convention

$$U_{X_0}(x) = \mu^2 + 2\lambda e^{-|x-X_0|^2/R^2}, \tag{238}$$

one has $V_0 = -2\lambda$. Therefore the localization condition becomes

$$\boxed{\lambda < 0 \qquad \text{and} \qquad \frac{|\lambda| R^2}{Z} > \frac{9\sqrt{3}}{8}.} \tag{239}$$

Hence the previous positive-sign local toy operator with $\lambda > 0$ cannot support a Gaussian variational single-defect carrier seed below the continuum threshold. A viable carrier seed requires an effective defect well relative to the bulk.

LXXII. SCHUR-COMPLEMENT MECHANISM FOR AN INDUCED CARRIER-LOWERING WELL

LXXIII. 1. GOAL

The previous single-defect variational analysis showed that the naive positive-sign local surrogate

$$\mathcal{L}_{X_0} = -Z\nabla^2 + \mu^2 + 2\lambda\rho_{X_0}(x), \qquad \lambda > 0, \tag{240}$$

cannot support a Gaussian variational localized carrier seed below the continuum threshold.

Therefore, if a single-defect carrier seed is to exist in the actual microscopic/Class-II program, a new mechanism must lower the effective carrier energy near the defect core.

The minimal candidate mechanism is the Schur-complement contribution obtained by integrating out a heavy sector.

LXXIV. 2. MINIMAL TWO-SECTOR SINGLE-DEFECT HESSIAN

Consider a light carrier field ψ and a heavy auxiliary/longitudinal sector X, expanded around a single-defect background P_{X_0}. The quadratic Hessian takes the block form

$$\boxed{\mathbb{H} = \begin{pmatrix} L_\ell & C \\ C^\dagger & L_h \end{pmatrix},} \tag{241}$$

where:

$$L_\ell = -Z_\ell\nabla^2 + U_\ell(x), \qquad L_h = -Z_h\nabla^2 + M_h^2 + U_h(x), \tag{242}$$

and C is the light-heavy mixing operator induced by the defect background.

We assume:

$$Z_\ell > 0, \qquad Z_h > 0, \qquad M_h^2 > 0, \tag{243}$$

and we study the low-energy light-sector problem near an energy E below the heavy-sector threshold.

LXXV. 3. SCHUR-COMPLEMENT REDUCTION

The block eigenvalue equation

$$\mathbb{H}\begin{pmatrix} \psi \\ X \end{pmatrix} = E\begin{pmatrix} \psi \\ X \end{pmatrix} \tag{244}$$

gives

$$(L_\ell - E)\psi + CX = 0, \tag{245}$$

$$C^\dagger\psi + (L_h - E)X = 0. \tag{246}$$

Assuming $L_h - E$ is invertible on the relevant domain, we solve for X:

$$X = -(L_h - E)^{-1}C^\dagger\psi. \tag{247}$$

Substituting back into the light equation yields the exact reduced light-sector problem

$$\boxed{L_{\text{eff}}(E)\psi = \Big(L_\ell - C(L_h - E)^{-1}C^\dagger\Big)\psi = E\psi.} \tag{248}$$

LXXVI. 4. SIGN THEOREM FOR THE INDUCED TERM

Assume that

$$L_h - E > 0 \tag{249}$$

as an operator on the heavy sector. Then $(L_h - E)^{-1}$ is positive definite, and for every light-sector test state φ,

$$\langle \varphi, C(L_h - E)^{-1}C^\dagger \varphi \rangle = \left\langle (L_h - E)^{-1/2}C^\dagger \varphi, (L_h - E)^{-1/2}C^\dagger \varphi \right\rangle \geq 0. \tag{250}$$

Therefore

$$\boxed{-C(L_h - E)^{-1}C^\dagger \text{ is negative semidefinite.}} \tag{251}$$

This is the key structural fact:

$$\boxed{\text{integrating out a positive heavy sector lowers the effective light-sector energy.}} \tag{252}$$

LXXVII. 5. LOCAL HEAVY-MASS EXPANSION

To make the effect explicit, suppose the heavy sector is gapped:

$$L_h \approx M_h^2 - Z_h \nabla^2 + U_h(x), \qquad M_h^2 \text{ dominant.} \tag{253}$$

Then for low energies $E \ll M_h^2$,

$$(L_h - E)^{-1} = \frac{1}{M_h^2 - E}\left[1 + \frac{Z_h \nabla^2 - U_h(x)}{M_h^2 - E} + \cdots\right]. \tag{254}$$

Hence the effective operator becomes

$$\boxed{L_{\text{eff}}(E) = L_\ell - \frac{1}{M_h^2 - E} CC^\dagger + O\left(\frac{\partial^2}{M_h^4}, \frac{U_h}{M_h^4}\right).} \tag{255}$$

Thus the leading induced effect is a negative local or quasi-local potential proportional to $CC^\dagger$.

LXXVIII. 6. DEFECT-LOCAL MIXING AND INDUCED GAUSSIAN WELL

Suppose the defect induces a local mixing profile

$$C = g\, f_{X_0}(x), \tag{256}$$

where g is a coupling constant and $f_{X_0}(x)$ is localized near the defect core. Then

$$CC^\dagger = g^2 |f_{X_0}(x)|^2, \tag{257}$$

and the effective light operator becomes

$$\boxed{L_{\text{eff}}(E) \approx -Z_\ell \nabla^2 + U_\ell(x) - \frac{g^2}{M_h^2 - E} |f_{X_0}(x)|^2.} \tag{258}$$

If we choose

$$f_{X_0}(x) = \exp\left[-\frac{|x - X_0|^2}{2R^2}\right], \tag{259}$$

then

$$|f_{X_0}(x)|^2 = \exp\left[-\frac{|x - X_0|^2}{R^2}\right]. \tag{260}$$

Therefore the Schur-complement-induced effective well is

$$\boxed{V_{\rm ind}(x) = \frac{g^2}{M_h^2 - E}\exp\left[-\frac{|x - X_0|^2}{R^2}\right].} \tag{261}$$

This is exactly of the same Gaussian-well form used in the single-defect variational test.

LXXIX. 7. EFFECTIVE TRAPPING CRITERION

Suppose the bare light-sector potential is

$$U_\ell(x) = U_\infty + U_{\rm rep}(x), \tag{262}$$

where $U_{\rm rep}(x)$ denotes any defect-induced repulsive shift relative to the asymptotic bulk value U_∞. Then

$$L_{\rm eff}(E) \approx -Z_\ell \nabla^2 + U_\infty + U_{\rm rep}(x) - V_{\rm ind}(x). \tag{263}$$

If $U_{\rm rep}(x)$ is also approximately Gaussian,

$$U_{\rm rep}(x) \approx V_{\rm rep}\exp\left[-\frac{|x - X_0|^2}{R^2}\right], \qquad V_{\rm rep} \geq 0, \tag{264}$$

then the net effective well depth is

$$\boxed{V_0^{\rm eff} = \frac{g^2}{M_h^2 - E} - V_{\rm rep}.} \tag{265}$$

Hence the Gaussian variational trapping criterion from the previous step becomes

$$\boxed{\frac{V_0^{\rm eff} R^2}{Z_\ell} > \frac{9\sqrt{3}}{4}.} \tag{266}$$

Equivalently,

$$\boxed{\frac{R^2}{Z_\ell}\left(\frac{g^2}{M_h^2 - E} - V_{\rm rep}\right) > \frac{9\sqrt{3}}{4}.} \tag{267}$$

This is the minimal surrogate criterion for a Schur-complement-induced single-defect carrier seed.

LXXX. 8. CLASS-II-INSPIRED AUXILIARY-FIELD INTERPRETATION

The same mechanism appears in the standard quadratic integration of a heavy auxiliary field. Suppose the UV/Class-II-inspired quadratic sector contains

$$\mathcal{L}_{\rm quad} = \frac{1}{2}\,\psi L_\ell \psi + \frac{1}{2}\,X L_h X + X\,\mathcal{J}[\psi, P_{X_0}], \tag{268}$$

with

$$\mathcal{J}[\psi, P_{X_0}] = g\,f_{X_0}(x)\psi \tag{269}$$

at leading defect-local order. Integrating out X gives

$$\mathcal{L}_{\text{eff}}[\psi] = \frac{1}{2}\,\psi L_\ell \psi - \frac{1}{2}\,\mathcal{J}[\psi, P_{X_0}](L_h)^{-1}\mathcal{J}[\psi, P_{X_0}] + \cdots . \tag{270}$$

Thus

$$\boxed{\Delta L_{\text{eff}} = -\,g^2\, f_{X_0}(x)\,(L_h)^{-1}\, f_{X_0}(x)} \tag{271}$$

is automatically negative semidefinite.

So the carrier-lowering sign is not a model-dependent miracle. It is the generic sign of integrating out a positive heavy quadratic sector.

LXXXI. 9. WHAT IS GENUINELY ESTABLISHED AND WHAT REMAINS OPEN

What is genuinely established at the surrogate/block-operator level is:

$$\boxed{\text{The Schur-complement contribution always lowers the effective light-sector energy if the heavy block is positive.}} \tag{272}$$

$$\boxed{\text{A localized defect-heavy mixing profile induces a negative effective well in the light sector.}} \tag{273}$$

$$\boxed{\text{The single-defect trapping criterion becomes } \frac{R^2}{Z_\ell}\left(\frac{g^2}{M_h^2 - E} - V_{\text{rep}}\right) > \frac{9\sqrt{3}}{4}.} \tag{274}$$

What remains open is:

$$\boxed{\text{whether the actual microscopic Class II Hessian produces the required positive heavy block and localized light-heavy mixing pr}} \tag{275}$$

$$\boxed{\text{whether the induced well is strong enough to overcome any bare repulsive shift in the actual theory.}} \tag{276}$$

LXXXII. 10. STRATEGIC CONSEQUENCE

The correct next bridge is no longer to guess the sign of a bare local term. Instead one should derive, from the actual or quasi-actual Class II quadratic sector:

1. the light block L_ℓ,
2. the heavy block L_h,
3. the defect-local mixing operator C,
4. and the resulting induced well depth

$$V_0^{\text{eff}} = \frac{g^2}{M_h^2 - E} - V_{\text{rep}}.$$

This is the proper route from the microscopic/Class-II program to single-defect carrier localization.

LXXXIII. FINAL SUMMARY

The single-defect carrier-lowering mechanism can arise naturally from a two-sector Hessian

$$\mathbb{H} = \begin{pmatrix} L_\ell & C \\ C^\dagger & L_h \end{pmatrix}, \tag{277}$$

where L_ℓ is the light carrier block, L_h is a positive heavy block, and C is a defect-local light-heavy mixing operator.

After integrating out the heavy sector, the exact reduced light operator is

$$L_{\rm eff}(E) = L_\ell - C(L_h - E)^{-1} C^\dagger. \tag{278}$$

If $L_h - E > 0$, then

$$-C(L_h - E)^{-1} C^\dagger \tag{279}$$

is negative semidefinite. Hence integrating out the heavy sector always lowers the effective light-sector energy.

For a localized mixing profile

$$C = g\, f_{X_0}(x), \qquad f_{X_0}(x) = \exp\left[-\frac{|x - X_0|^2}{2R^2}\right], \tag{280}$$

the induced effective well is

$$V_{\rm ind}(x) = \frac{g^2}{M_h^2 - E} \exp\left[-\frac{|x - X_0|^2}{R^2}\right]. \tag{281}$$

If the bare light sector also contains a repulsive defect shift $V_{\rm rep} \exp(-|x - X_0|^2/R^2)$, then the net effective well depth is

$$V_0^{\rm eff} = \frac{g^2}{M_h^2 - E} - V_{\rm rep}. \tag{282}$$

The Gaussian variational single-defect trapping criterion becomes

$$\boxed{\frac{V_0^{\rm eff} R^2}{Z_\ell} > \frac{9\sqrt{3}}{4}.} \tag{283}$$

Thus the old positive-sign local toy operator fails because it contains no carrier-lowering mechanism, whereas a Schur-complement-induced well provides exactly the missing negative contribution. The remaining open task is to derive the actual Class II quadratic blocks L_ℓ, L_h, and C and to check whether the induced well is present and strong enough in the microscopic theory.

LXXXIV. CLASS-II-INSPIRED QUADRATIC BLOCK ANSATZ FOR SINGLE-DEFECT CARRIER LOCALIZATION

LXXXV. 1. GOAL

The previous variational analysis showed that the naive local surrogate

$$\mathcal{L}_{X_0}^{\rm naive} = -Z_\ell \nabla^2 + \mu^2 + 2\lambda \rho_{X_0}(x) \tag{284}$$

does not localize a carrier seed when $\lambda > 0$. Therefore the real microscopic question is:

$$\boxed{\text{Which part of the Class II quadratic sector can generate an attractive defect-centered light-sector well?}} \tag{285}$$

The key point is that the Class II mediator X_i^a naturally splits into two logically distinct channels:

$$\boxed{U_i^a(P)\text{: mass-like carrier-lowering channel,} \qquad V_{ij}^a(P)\text{: derivative/transport channel.}} \tag{286}$$

The first is the relevant seed-localization channel. The second is the relevant CKM-transport/hierarchy channel.

LXXXVI. 2. MICROSCOPIC COEFFICIENT TEMPLATE

In the explicit minimal microscopic coefficient template already written in the notes, the Ψ-sector appears as

$$\boxed{\mathcal{L}_{\rm mic} = \mathcal{L}_P[P] + \frac{i}{2}\Psi^\dagger Z^0(P)\overleftrightarrow{\partial_t}\Psi + \frac{i}{2}\Psi^\dagger\Big(Z^i(P) + X^a_j V^a_{ji}(P)\Big)\overleftrightarrow{\partial_i}\Psi - \Psi^\dagger\Big(M_0(P) - X^a_i U^a_i(P)\Big)\Psi + \frac{M_X^2}{2}X^a_i X^a_i.} \tag{287}$$

Equivalently,

$$U^a_i(P) = \alpha_X A^a_i(P) + \beta_X C^a_i(P), \qquad V^a_{ij}(P) = \alpha_X B^a_{ij}(P) + \beta_X D^a_{ij}(P). \tag{288}$$

Thus U^a_i enters as a local mass-like coefficient field, whereas V^a_{ij} enters as a derivative coefficient field.

LXXXVII. 3. SINGLE-DEFECT BACKGROUND SPLIT

Let $P_{X_0}(x)$ be a single-defect background centered at X_0. Expand the heavy field around its defect-induced background expectation:

$$X^a_i(x) = \bar{X}^a_i(x) + \delta X^a_i(x). \tag{289}$$

The minimal background source compatible with the Class II notes is a shell/projector current of the form

$$\bar{\mathcal{J}}^a_i(x) = \bar{\mathcal{J}}^a_i[P_{X_0}] \sim m^a(P_{X_0})\,\partial_i\phi_0(x), \tag{290}$$

or more generally a defect-local current built from P_{X_0} and its gradients.

The heavy-field equation is algebraic:

$$M_X^2 X^a_i + \mathcal{J}^{a,\rm tot}_i = 0. \tag{291}$$

Therefore the background part is

$$\boxed{\bar{X}^a_i(x) = -\frac{1}{M_X^2}\,\bar{\mathcal{J}}^a_i(x)} \tag{292}$$

at the minimal tree level.

LXXXVIII. 4. LIGHT-SECTOR LOCALIZATION OPERATOR

Now write the light carrier fluctuation as χ and isolate the quadratic light-sector operator around the defect background:

$$\mathcal{L}^{(2)}_\chi = \chi^\dagger\Big[-Z_\ell\nabla^2 + M_0(P_{X_0}) - \bar{X}^a_i U^a_i(P_{X_0})\Big]\chi + \text{derivative terms from } V^a_{ij} + \cdots. \tag{293}$$

Hence the single-defect light operator is

$$\boxed{L^{\rm seed}_\ell = -Z_\ell\nabla^2 + U_{\rm bare}(x) - \bar{X}^a_i(x)\,U^a_i(P_{X_0}(x)),} \tag{294}$$

where

$$U_{\rm bare}(x) := M_0(P_{X_0}(x)). \tag{295}$$

Substituting the background solution for $\bar{X}^a_i$,

$$\boxed{L^{\rm seed}_\ell = -Z_\ell\nabla^2 + U_{\rm bare}(x) + \frac{1}{M_X^2}\,\bar{\mathcal{J}}^a_i(x)\,U^a_i(P_{X_0}(x)).} \tag{296}$$

Thus the effective local defect shift is

$$\boxed{\delta U_{\rm eff}(x) = \frac{1}{M_X^2}\,\bar{\mathcal{J}}^a_i(x)\,U^a_i(P_{X_0}(x)).} \tag{297}$$

LXXXIX. 5. ATTRACTIVE-WELL CONDITION

The sign of the induced shift is not fixed by M_X^2 alone; it is fixed by the local contraction

$$\bar{\mathcal{J}}_i^a(x)\, U_i^a(P_{X_0}(x)). \tag{298}$$

To obtain a carrier-lowering well, one needs

$$\boxed{\delta U_{\rm eff}(x) < 0 \quad \text{in the defect core region.}} \tag{299}$$

Equivalently, if we write

$$U_{\rm bare}(x) = U_\infty + V_{\rm rep}(x), \qquad V_{\rm rep}(x) \geq 0, \tag{300}$$

and parameterize the induced attractive contribution as

$$\delta U_{\rm eff}(x) \approx -V_{\rm ind}(x), \qquad V_{\rm ind}(x) \geq 0, \tag{301}$$

then

$$\boxed{L_\ell^{\rm seed} = -Z_\ell \nabla^2 + U_\infty + V_{\rm rep}(x) - V_{\rm ind}(x).} \tag{302}$$

So the net effective well depth is

$$\boxed{V_0^{\rm eff}(x) = V_{\rm ind}(x) - V_{\rm rep}(x).} \tag{303}$$

This is the correct Class-II-inspired meaning of the earlier surrogate quantity $V_0^{\rm eff}$.

XC. 6. IDENTIFICATION OF THE PREVIOUS ABSTRACT PARAMETERS

The previous abstract Schur-complement notation

$$L_{\rm eff} = L_\ell - C(L_h - E)^{-1} C^\dagger \tag{304}$$

is now identified concretely as follows:

$$\boxed{L_h \leftrightarrow M_X^2 + \mathcal{L}_{X,\rm kin}^{\rm ref},} \tag{305}$$

$$\boxed{M_h^2 \leftrightarrow M_X^2,} \tag{306}$$

$$\boxed{C \leftrightarrow \text{defect-local Class II mixing channel carried by } X_i^a,} \tag{307}$$

$$\boxed{V_{\rm rep}(x) \leftrightarrow U_{\rm bare}(x) - U_\infty,} \tag{308}$$

$$\boxed{V_{\rm ind}(x) \leftrightarrow -\frac{1}{M_X^2} \bar{\mathcal{J}}_i^a(x) U_i^a(P_{X_0}(x)) \quad \text{(when this is positive).}} \tag{309}$$

Thus the abstract trapping criterion derived previously becomes, in the Class II microscopic language:

$$\boxed{\frac{R^2}{Z_\ell}\left(V_{\rm ind}^{\rm core} - V_{\rm rep}^{\rm core}\right) > \frac{9\sqrt{3}}{4},} \tag{310}$$

where $V_{\rm ind}^{\rm core}$ and $V_{\rm rep}^{\rm core}$ denote the effective core amplitudes of the attractive and repulsive parts.

XCI. 7. MINIMAL GAUSSIAN CORE REDUCTION

Assume that near the defect core

$$V_{\rm rep}(x) \approx V_{\rm rep}\, e^{-|x-X_0|^2/R^2}, \qquad V_{\rm ind}(x) \approx V_{\rm ind}\, e^{-|x-X_0|^2/R^2}. \tag{311}$$

Then

$$\boxed{V_0^{\rm eff} = V_{\rm ind} - V_{\rm rep}.} \tag{312}$$

If the background current and mass-like carrier coefficient are aligned so that

$$\bar{\mathcal{J}}_i^a(x)\, U_i^a(P_{X_0}(x)) \approx -\Gamma_{\rm loc}\, e^{-|x-X_0|^2/R^2}, \qquad \Gamma_{\rm loc} > 0, \tag{313}$$

then

$$\boxed{V_{\rm ind} = \frac{\Gamma_{\rm loc}}{M_X^2}.} \tag{314}$$

Hence the single-defect trapping condition becomes

$$\boxed{\frac{R^2}{Z_\ell}\left(\frac{\Gamma_{\rm loc}}{M_X^2} - V_{\rm rep}\right) > \frac{9\sqrt{3}}{4}.} \tag{315}$$

This is the Class-II-inspired replacement of the earlier toy inequality

$$\frac{V_0 R^2}{Z_\ell} > \frac{9\sqrt{3}}{4}.$$

XCII. 8. ROLE SEPARATION: SEED LOCALIZATION VS TRANSPORT HIERARCHY

At this point the logic becomes very clean.

A. 8.1. Seed localization channel

The quantity

$$\bar{\mathcal{J}}_i^a\, U_i^a(P_{X_0}) \tag{316}$$

controls whether a single defect lowers the carrier energy and creates a localized seed mode.

This is the $K1A1$ channel.

B. 8.2. CKM transport channel

The derivative coefficient

$$V_{ij}^a(P) \tag{317}$$

controls the transport/hierarchy operator once the family subspace already exists.

This is the later $K4$ channel.

Therefore:

$$\boxed{U_i^a(P) \text{ is the seed-localization carrier}, \qquad V_{ij}^a(P) \text{ is the hierarchy/transport carrier.}} \tag{318}$$

This distinction prevents the program from mixing two logically different tasks.

XCIII. 9. RELATION TO THE EXPLICIT CLASS II NOTES

The already-constructed one-mediator Class II truncation

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda\, m^a X_i^a\, \partial_i \phi_0 + g_X\, X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.} \tag{319}$$

is the transport-oriented realization that explicitly generates the enriched tensor

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \tag{320}$$

This is exactly the mechanism needed later for $(\lambda_\parallel, \alpha, \beta)$. However, by itself it is not yet the most transparent localization ansatz, because it emphasizes the derivative witness channel.

For single-defect seed localization, the more relevant piece is the microscopic coefficient template where the mediator enters the mass-like term through

$$-\Psi^\dagger (M_0(P) - X_i^a U_i^a(P))\Psi. \tag{321}$$

Thus the notes already contain both required ingredients; they simply play different roles.

XCIV. 10. WHAT IS ACTUALLY ESTABLISHED AND WHAT REMAINS OPEN

What is now structurally established is:

$$\boxed{\text{The Class II mediator naturally splits into a mass-like channel } U_i^a(P) \text{ and a derivative channel } V_{ij}^a(P).} \tag{322}$$

$$\boxed{\text{The mass-like channel is the correct microscopic source of the single-defect carrier-lowering well.}} \tag{323}$$

$$\boxed{\text{The derivative channel is the correct microscopic source of the later hierarchy/transport operator.}} \tag{324}$$

What remains open is:

$$\boxed{\text{the actual computation of } \bar{\mathcal{J}}_i^a(P_{X_0})\, U_i^a(P_{X_0}) \text{ in the intended Class II branch,}} \tag{325}$$

$$\boxed{\text{and whether its core amplitude exceeds the repulsive bare shift strongly enough to satisfy the trapping bound.}} \tag{326}$$

XCV. 11. IMMEDIATE NEXT STEP

The correct next calculation is therefore:

$$\boxed{\text{compute the core scalar contraction } \Gamma_{\text{loc}}(x) := -\bar{\mathcal{J}}_i^a(x)\, U_i^a(P_{X_0}(x))} \tag{327}$$

in an actual or quasi-actual single-defect background, fit its core profile to

$$\Gamma_{\text{loc}}\, e^{-|x-X_0|^2/R^2},$$

and compare

$$\frac{R^2}{Z_\ell}\left(\frac{\Gamma_{\text{loc}}}{M_X^2} - V_{\text{rep}}\right) \tag{328}$$

against the threshold $9\sqrt{3}/4$.

Only after this step will the single-defect seed problem stop being schematic.

XCVI. FINAL SUMMARY

The Class II mediator X_i^a plays two logically distinct roles in the TECT program.

First, through the mass-like coefficient channel

$$U_i^a(P),$$

it can lower the local carrier energy in a single-defect background and thereby generate a localized carrier seed.

Second, through the derivative coefficient channel

$$V_{ij}^a(P),$$

it can generate the later transport/hierarchy operator relevant for the CKM program.

Using the minimal microscopic coefficient template,

$$\mathcal{L}_{\rm mic} = \cdots - \Psi^\dagger\big(M_0(P) - X_i^a U_i^a(P)\big)\Psi + \frac{i}{2}\Psi^\dagger\big(Z^i(P) + X_j^a V_{ji}^a(P)\big)\overleftrightarrow{\partial_i}\,\Psi + \frac{M_X^2}{2} X_i^a X_i^a,$$

and eliminating the heavy field around a single-defect background P_{X_0}, one obtains the seed-localization light operator

$$L_\ell^{\rm seed} = -Z_\ell \nabla^2 + U_{\rm bare}(x) + \frac{1}{M_X^2}\,\bar{\mathcal{J}}_i^a(x)\,U_i^a(P_{X_0}(x)).$$

Thus the induced attractive well is controlled by the defect-core scalar contraction

$$-\bar{\mathcal{J}}_i^a(x)\,U_i^a(P_{X_0}(x)).$$

If near the core

$$-\bar{\mathcal{J}}_i^a(x)\,U_i^a(P_{X_0}(x)) \approx \Gamma_{\rm loc}\, e^{-|x-X_0|^2/R^2}, \qquad \Gamma_{\rm loc} > 0,$$

then the effective well depth is

$$V_{\rm ind} = \frac{\Gamma_{\rm loc}}{M_X^2},$$

and the single-defect Gaussian trapping criterion becomes

$$\boxed{\frac{R^2}{Z_\ell}\left(\frac{\Gamma_{\rm loc}}{M_X^2} - V_{\rm rep}\right) > \frac{9\sqrt{3}}{4}.}$$

Therefore the correct next microscopic question is no longer the sign of a bare toy potential. It is the actual computation of the core contraction

$$\bar{\mathcal{J}}_i^a(P_{X_0})\,U_i^a(P_{X_0})$$

in the intended Class II branch, and whether it is strong enough to overcome the repulsive bare shift and localize a carrier seed.

XCVII. EXPLICIT GAUSSIAN-CORE EVALUATION OF THE SINGLE-DEFECT LOCALIZATION SCALAR $\Gamma_{\rm loc}$

XCVIII. 1. STARTING POINT FROM THE CLASS II DEFECT OPERATOR

In the locked single-defect Class II coefficient template, the relevant mass-like coefficient is

$$\boxed{m_{\rm eff}(x) := m(\bar\sigma(x)) - 3\chi_X(x)\,u_X(\bar\sigma(x)),} \tag{329}$$

where the locked mediator scalar is

$$\boxed{\chi_X(x) := \frac{1}{3}\,\bar{X}_i^a(x)\,E_i{}^a(x),} \tag{330}$$

and the mass-like channel is

$$\boxed{U_i^a(P) = u_X(\sigma)\, E_i{}^a\, \Gamma^\sharp.} \tag{331}$$

Hence on the locked background one has

$$\boxed{\bar{X}_i^a U_i^a = 3\chi_X\, u_X\, \Gamma^\sharp.} \tag{332}$$

This is exactly the defect-background mass-shift mechanism already extracted in the notes.

The single-defect localization question is therefore reduced to the sign and magnitude of the defect-core lowering of m_{eff}.

XCIX. 2. MINIMAL RADIAL SINGLE-DEFECT ANSATZ

Let

$$r := |x - X_0|. \tag{333}$$

We take the minimal Gaussian-core ansatz

$$\boxed{\bar{\sigma}(r) = \sigma_\infty + \delta\sigma\, e^{-r^2/R^2},} \tag{334}$$

$$\boxed{\chi_X(r) = \chi_0\, e^{-r^2/R^2},} \tag{335}$$

with R the core width scale.

We also expand the constitutive functions around the bulk value σ_∞:

$$\boxed{m(\sigma) = m_\infty + m_1(\sigma - \sigma_\infty) + \frac{1}{2}m_2(\sigma - \sigma_\infty)^2 + \cdots,} \tag{336}$$

$$\boxed{u_X(\sigma) = u_\infty + u_1(\sigma - \sigma_\infty) + \frac{1}{2}u_2(\sigma - \sigma_\infty)^2 + \cdots,} \tag{337}$$

where

$$m_\infty := m(\sigma_\infty), \qquad u_\infty := u_X(\sigma_\infty), \qquad m_1 := m'(\sigma_\infty), \qquad u_1 := u_X'(\sigma_\infty). \tag{338}$$

C. 3. EXPLICIT CORE EXPANSION OF $m_{\text{eff}}(r)$

Substitute the ansatz into

$$m_{\text{eff}}(r) = m(\bar{\sigma}(r)) - 3\chi_X(r)\, u_X(\bar{\sigma}(r)). \tag{339}$$

First,

$$m(\bar{\sigma}(r)) = m_\infty + m_1\delta\sigma\, e^{-r^2/R^2} + \frac{1}{2}m_2(\delta\sigma)^2 e^{-2r^2/R^2} + \cdots. \tag{340}$$

Second,

$$u_X(\bar{\sigma}(r)) = u_\infty + u_1\delta\sigma\, e^{-r^2/R^2} + \frac{1}{2}u_2(\delta\sigma)^2 e^{-2r^2/R^2} + \cdots, \tag{341}$$

$$3\chi_X(r)\, u_X(\bar{\sigma}(r)) = 3\chi_0 u_\infty\, e^{-r^2/R^2} + 3\chi_0 u_1\delta\sigma\, e^{-2r^2/R^2} + \cdots. \tag{342}$$

Therefore

$$m_{\text{eff}}(r) = m_\infty + \big(m_1\delta\sigma - 3\chi_0 u_\infty\big)e^{-r^2/R^2} \tag{343}$$

$$+ \left(\frac{1}{2}m_2(\delta\sigma)^2 - 3\chi_0 u_1 \delta\sigma\right) e^{-2r^2/R^2} + \cdots . \tag{344}$$

Thus the leading Gaussian core coefficient is

$$\boxed{m_{\text{eff}}(r) = m_\infty - \Gamma_{\text{loc}}\, e^{-r^2/R^2} + O(e^{-2r^2/R^2}),} \tag{345}$$

with

$$\boxed{\Gamma_{\text{loc}} := 3\chi_0 u_\infty - m_1 \delta\sigma.} \tag{346}$$

CI. 4. INTERPRETATION OF THE SIGN

The carrier-lowering condition is precisely

$$\boxed{\Gamma_{\text{loc}} > 0.} \tag{347}$$

Equivalently,

$$m_{\text{eff}}(0) < m_\infty \tag{348}$$

at leading core order.

So the single-defect core becomes attractive for the carrier if and only if the mediator-induced lowering term

$$3\chi_0 u_\infty \tag{349}$$

dominates over the bare defect-induced upward shift

$$m_1 \delta\sigma. \tag{350}$$

This is the explicit leading-order answer to the question:

Where does the single-defect carrier-lowering well come from in the Class II branch?

CII. 5. REWRITING χ_0 IN TERMS OF THE MICROSCOPIC SOURCE

The heavy mediator is algebraic:

$$\boxed{M_X^2 X_i^a + \mathcal{J}_i^{a,\text{tot}} = 0, \qquad X_i^a = -\frac{1}{M_X^2}\mathcal{J}_i^{a,\text{tot}}.} \tag{351}$$

In the minimal working branch,

$$\mathcal{J}_i^{a,\text{tot}} = \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P,\Psi] \tag{352}$$

at lowest order.

Project onto the locked carrier direction $E_i{}^a$. Define the aligned scalar source amplitude by

$$\boxed{\frac{1}{3}E_i{}^a\, \mathcal{J}_i^{a,\text{tot}}(r) = -\mathrm{j}_0\, e^{-r^2/R^2}} \tag{353}$$

with $\mathrm{j}_0 > 0$ in the aligned attractive branch.

Then

$$\chi_X(r) = \frac{1}{3}\bar{X}_i^a E_i{}^a = -\frac{1}{3M_X^2}E_i{}^a \mathcal{J}_i^{a,\text{tot}}(r) = \frac{\mathrm{j}_0}{M_X^2}e^{-r^2/R^2}. \tag{354}$$

Hence

$$\boxed{\chi_0 = \frac{\mathrm{j}_0}{M_X^2}.} \tag{355}$$

Substituting into the core lowering amplitude gives

$$\boxed{\Gamma_{\mathrm{loc}} = \frac{3u_\infty}{M_X^2}\,\mathrm{j}_0 - m_1\delta\sigma.} \tag{356}$$

This is the desired explicit Class-II-inspired single-defect scalar criterion.

CIII. 6. SPLITTING THE SOURCE INTO J- AND K-CHANNELS

If one wants the source decomposition more explicitly, write

$$\frac{1}{3}E_i{}^a J_i^a[\bar{\Psi}](r) = -j_J\, e^{-r^2/R^2}, \qquad \frac{1}{3}E_i{}^a K_i^a[\bar{P},\bar{\Psi}](r) = -j_K\, e^{-r^2/R^2}. \tag{357}$$

Then

$$\mathrm{j}_0 = \alpha_X j_J + \beta_X j_K, \tag{358}$$

and therefore

$$\boxed{\Gamma_{\mathrm{loc}} = \frac{3u_\infty}{M_X^2}\big(\alpha_X j_J + \beta_X j_K\big) - m_1\delta\sigma.} \tag{359}$$

This is the cleanest minimal decomposition of the seed-localization amplitude.

CIV. 7. RELATION TO THE PREVIOUS GAUSSIAN TRAPPING THRESHOLD

In the previous Schrödinger-type surrogate, the single-defect trapping criterion was

$$\frac{V_0^{\mathrm{eff}}R^2}{Z_\ell} > \frac{9\sqrt{3}}{4}. \tag{360}$$

At the present first-order defect-operator level, the exact positive conversion factor from the local mass-lowering coefficient Γ_{loc} to the second-order scalar well depth still depends on the detailed reduction of the canonically normalized defect operator

$$\mathbb{O}^{(0)}_{1\mathrm{def}} = i\widetilde{\Gamma}^i_{\mathrm{def}}(\partial_i + \Omega_i^{\mathrm{def}}) + \Phi_{\mathrm{def}}, \tag{361}$$

which is fixed structurally but not yet fully constitutively specified.

Therefore the honest bridge is:

$$\boxed{V_0^{\mathrm{eff}} = c_{\mathrm{seed}}\,\Gamma_{\mathrm{loc}}, \qquad c_{\mathrm{seed}} > 0,} \tag{362}$$

with c_{seed} still to be extracted from the explicit one-defect reduction.

The trapping criterion then becomes

$$\boxed{\frac{c_{\mathrm{seed}}\Gamma_{\mathrm{loc}}R^2}{Z_\ell} > \frac{9\sqrt{3}}{4}.} \tag{363}$$

CV. 8. IMMEDIATE COROLLARIES

a. Corollary 1. If

$$\frac{3u_\infty}{M_X^2}\,\mathrm{j}_0 < m_1\delta\sigma, \tag{364}$$

then the defect core does not lower the carrier mass at leading Gaussian order, and the single-defect seed fails already at the first-order coefficient level.

b. Corollary 2. If

$$\frac{3u_\infty}{M_X^2}\,\mathrm{j}_0 > m_1\delta\sigma, \tag{365}$$

then the defect core lowers $m_{\rm eff}$ at leading Gaussian order, and the existence of a trapped seed is reduced to whether the quantitative threshold

$$\frac{c_{\rm seed}\Gamma_{\rm loc}R^2}{Z_\ell} > \frac{9\sqrt{3}}{4} \tag{366}$$

is met.

c. Corollary 3. The single-defect seed problem is therefore not controlled by the transport coefficients (g_E, g_O, g_c) directly. It is controlled first by the scalar core contraction

$$\boxed{\Gamma_{\rm loc} = \frac{3u_\infty}{M_X^2}\,\mathrm{j}_0 - m_1\delta\sigma.} \tag{367}$$

Only after the seed exists does the derivative channel become the relevant hierarchy generator.

CVI. 9. HONEST STATUS

What is genuinely closed at this step is:

$$\boxed{\text{The leading Gaussian core lowering amplitude is explicitly } \Gamma_{\rm loc} = 3\chi_0 u_\infty - m_1\delta\sigma.} \tag{368}$$

$$\boxed{\text{Using the algebraic Class II equation, this becomes } \Gamma_{\rm loc} = \frac{3u_\infty}{M_X^2}\mathrm{j}_0 - m_1\delta\sigma.} \tag{369}$$

What remains open is:

$$\boxed{\text{the actual constitutive evaluation of } u_\infty,\ m_1,\ \delta\sigma,\ \mathrm{j}_0,\ c_{\rm seed} \text{ in the intended Class II defect branch.}} \tag{370}$$

Thus the single-defect localization problem has now been reduced to a finite core-coefficient audit.

CVII. FINAL SUMMARY

In the locked Class II single-defect coefficient template, the seed-localization mass coefficient is

$$m_{\rm eff}(x) = m(\bar\sigma(x)) - 3\chi_X(x)\,u_X(\bar\sigma(x)). \tag{371}$$

Using the minimal Gaussian core ansatz

$$\bar\sigma(r) = \sigma_\infty + \delta\sigma e^{-r^2/R^2}, \qquad \chi_X(r) = \chi_0 e^{-r^2/R^2}, \tag{372}$$

and expanding

$$m(\sigma) = m_\infty + m_1(\sigma - \sigma_\infty) + \cdots, \qquad u_X(\sigma) = u_\infty + u_1(\sigma - \sigma_\infty) + \cdots, \tag{373}$$

one obtains

$$m_{\rm eff}(r) = m_\infty - \Gamma_{\rm loc}\, e^{-r^2/R^2} + O(e^{-2r^2/R^2}), \tag{374}$$

with the explicit leading core-lowering amplitude

$$\boxed{\Gamma_{\rm loc} = 3\chi_0 u_\infty - m_1 \delta\sigma.} \tag{375}$$

Using the algebraic heavy-field equation

$$M_X^2 X_i^a + \mathcal{J}_i^{a,\rm tot} = 0, \tag{376}$$

and defining the aligned scalar source amplitude by

$$\frac{1}{3} E_i{}^a \mathcal{J}_i^{a,\rm tot}(r) = -\mathrm{j}_0 e^{-r^2/R^2}, \tag{377}$$

one gets

$$\chi_0 = \frac{\mathrm{j}_0}{M_X^2}, \tag{378}$$

so that

$$\boxed{\Gamma_{\rm loc} = \frac{3u_\infty}{M_X^2}\,\mathrm{j}_0 - m_1\delta\sigma.} \tag{379}$$

Thus the single-defect seed problem is reduced to a finite scalar core audit. A necessary leading-order carrier-lowering condition is

$$\Gamma_{\rm loc} > 0. \tag{380}$$

If the positive reduction coefficient from the first-order defect operator to the second-order trapping surrogate is $c_{\rm seed} > 0$, then the quantitative Gaussian trapping criterion becomes

$$\frac{c_{\rm seed}\Gamma_{\rm loc}R^2}{Z_\ell} > \frac{9\sqrt{3}}{4}. \tag{381}$$

Therefore the next microscopic task is to evaluate the constitutive core data

$$u_\infty, \quad m_1, \quad \delta\sigma, \quad \mathrm{j}_0, \quad c_{\rm seed} \tag{382}$$

on the intended Class II single-defect branch.

CVIII. EXPLICIT SINGLE-DEFECT CORE ANSATZ FOR THE ALIGNED SOURCE j_0

CIX. 1. GOAL

The remaining microscopic quantity controlling single-defect seed localization is the aligned scalar source amplitude

$$\boxed{\frac{1}{3} E_i{}^a\, \mathcal{J}_i^{a,\rm tot}(\xi) = -\mathrm{j}_0\, e^{-\xi^2/R^2} + O(e^{-2\xi^2/R^2}),} \tag{383}$$

where

$$\mathcal{J}_i^{a,\rm tot} = \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P,\Psi] \tag{384}$$

in the minimal Class II branch.

The purpose of this step is to write j_0 explicitly in terms of the single-defect core data.

CX. 2. MINIMAL LOCKED SINGLE-DEFECT ANSATZ

Use the defect-normal coordinate ξ, with the defect core located at $\xi = 0$. Take the minimal locked background

$$\Psi_B(\xi) = \phi_0(\xi)\, z_B(\xi), \qquad P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \qquad z_B^\dagger z_B = 1. \tag{385}$$

We choose the minimal embedded $SU(2)_{(13)}$ representative

$$\boxed{z_B(\xi) = \begin{pmatrix} \cos\theta(\xi) \\ 0 \\ \sin\theta(\xi) \end{pmatrix}, \qquad \partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi).} \tag{386}$$

Near the defect core, assume the Gaussian-core approximation

$$\boxed{\phi_0(\xi) \approx \phi_c, \qquad \theta'(\xi) \approx \theta_1\, e^{-\xi^2/R^2}, \qquad \mathcal{O}[\Psi_B](\xi) \approx \mathcal{O}_c,} \tag{387}$$

with $\phi_c, \theta_1, \mathcal{O}_c$ constants at leading core order.

CXI. 3. ALIGNED INTERNAL PROJECTOR

Let the aligned internal/spatial carrier direction be

$$\boxed{E_i{}^a = \hat{n}_i\, m^a,} \tag{388}$$

where $\hat{n}_i$ is the defect-normal unit vector and m^a is the internal aligned direction of the branch. We define the projected scalar source by

$$\frac{1}{3} E_i{}^a \mathcal{J}_i^{a,\mathrm{tot}} = \frac{1}{3} \hat{n}_i m^a \big(\alpha_X J_i^a + \beta_X K_i^a\big). \tag{389}$$

CXII. 4. EXPLICIT ANSATZ FOR THE J-CURRENT CONTRIBUTION

The minimal microscopic current is

$$\boxed{J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \Psi^\dagger T_a (\partial_i \Psi).} \tag{390}$$

On the locked defect background $\Psi_B = \phi_0 z_B$, the derivative along the normal direction is

$$\partial_\xi \Psi_B = (\partial_\xi \phi_0)\, z_B + \phi_0\, \partial_\xi z_B. \tag{391}$$

Hence

$$J_\xi^a[\Psi_B] = 2\phi_0 (\partial_\xi \phi_0)\, z_B^\dagger T_a z_B + \phi_0^2 \Big[(\partial_\xi z_B^\dagger) T_a z_B + z_B^\dagger T_a (\partial_\xi z_B) \Big]. \tag{392}$$

The first term is odd if $\partial_\xi \phi_0$ is odd and is not the most stable core-even contribution. The second term is proportional to $\phi_0^2 \theta'(\xi)$ and provides the natural even Gaussian core source.

Define the branch-dependent internal contraction constant

$$\boxed{C_J := \frac{1}{3} m^a \Big[e_1^\dagger T_a z_B + z_B^\dagger T_a e_1 \Big]_{\xi=0}.} \tag{393}$$

Then at leading core order

$$\boxed{\frac{1}{3} E_i{}^a J_i^a[\Psi_B] = -\, j_J\, e^{-\xi^2/R^2} + O(e^{-2\xi^2/R^2}),} \tag{394}$$

with

$$\boxed{j_J = \phi_c^2\, \theta_1\, C_J.} \tag{395}$$

CXIII. 5. EXPLICIT ANSATZ FOR THE K-CURRENT CONTRIBUTION

The minimal projector/test-mode current is

$$\boxed{K_i^a[P,\Psi] = \mathrm{Tr}(T_a \partial_i P)\, \mathcal{O}[\Psi].} \tag{396}$$

For the defect background,

$$\partial_\xi P_B = (\partial_\xi z_B) z_B^\dagger + z_B(\partial_\xi z_B^\dagger) = \theta'(\xi)\, \mathcal{T}_\xi, \tag{397}$$

with $\mathcal{T}_\xi$ the fixed tangent projector generator.

Define the aligned projector contraction constant

$$\boxed{C_K := \frac{1}{3}\, m^a\, \mathrm{Tr}(T_a \mathcal{T}_\xi)\Big|_{\xi=0}.} \tag{398}$$

Then using $\mathcal{O}[\Psi_B](\xi) \approx \mathcal{O}_c$,

$$\boxed{\frac{1}{3} E_i{}^a K_i^a[P_B, \Psi_B] = -\, j_K\, e^{-\xi^2/R^2} + O(e^{-2\xi^2/R^2}),} \tag{399}$$

with

$$\boxed{j_K = \mathcal{O}_c\, \theta_1\, C_K.} \tag{400}$$

CXIV. 6. THE EXPLICIT ALIGNED SCALAR SOURCE AMPLITUDE

Combining the two channels,

$$\frac{1}{3} E_i{}^a \mathcal{J}_i^{a,\mathrm{tot}} = \frac{1}{3} E_i{}^a \big(\alpha_X J_i^a + \beta_X K_i^a\big) = -\big(\alpha_X j_J + \beta_X j_K\big) e^{-\xi^2/R^2} + O(e^{-2\xi^2/R^2}). \tag{401}$$

Therefore

$$\boxed{\mathrm{j}_0 = \alpha_X j_J + \beta_X j_K.} \tag{402}$$

Substituting the explicit core amplitudes gives

$$\boxed{\mathrm{j}_0 = \theta_1 \Big(\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K\Big).} \tag{403}$$

CXV. 7. CORE-LOWERING AMPLITUDE IN EXPLICIT SINGLE-DEFECT DATA

From the previous step, the single-defect mass-lowering coefficient is

$$\Gamma_{\mathrm{loc}} = \frac{3u_\infty}{M_X^2}\, \mathrm{j}_0 - m_1 \delta\sigma. \tag{404}$$

Thus

$$\boxed{\Gamma_{\mathrm{loc}} = \frac{3u_\infty}{M_X^2}\, \theta_1 \Big(\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K\Big) - m_1 \delta\sigma.} \tag{405}$$

This is the first explicit microscopic single-defect seed formula in terms of constitutive core data.

CXVI. 8. NECESSARY SIGN CONDITION

A necessary leading-order condition for carrier lowering is

$$\boxed{\Gamma_{\rm loc} > 0.} \tag{406}$$

Equivalently,

$$\boxed{\frac{3u_\infty}{M_X^2}\,\theta_1\Big(\alpha_X\phi_c^2 C_J + \beta_X \mathcal{O}_c C_K\Big) > m_1\delta\sigma.} \tag{407}$$

This exhibits the precise competition:

1. mediator-induced lowering through the aligned J- and K-channel source;
2. bare repulsive scalar-core shift through $m_1\delta\sigma$.

CXVII. 9. MINIMAL NORMALIZED BRANCH

In the simplest normalized primitive branch, one may set

$$C_J \sim C_K \sim C_\parallel, \qquad C_\parallel = 1, \tag{408}$$

which yields the simplified estimate

$$\boxed{\mathfrak{j}_0 \approx \theta_1\big(\alpha_X\phi_c^2 + \beta_X\mathcal{O}_c\big),} \tag{409}$$

and therefore

$$\boxed{\Gamma_{\rm loc} \approx \frac{3u_\infty\theta_1}{M_X^2}\big(\alpha_X\phi_c^2 + \beta_X\mathcal{O}_c\big) - m_1\delta\sigma.} \tag{410}$$

This is the most economical working formula for the next microscopic audit.

CXVIII. 10. INTERPRETATION

The seed-localization problem is now reduced to five core quantities:

$$\phi_c, \qquad \theta_1, \qquad \mathcal{O}_c, \qquad u_\infty, \qquad m_1\delta\sigma, \tag{411}$$

together with the branch constants

$$C_J, \qquad C_K, \tag{412}$$

and the microscopic couplings

$$\alpha_X, \qquad \beta_X, \qquad M_X^2. \tag{413}$$

Thus the next concrete task is no longer structural. It is the finite evaluation of these core constants on the intended Class II single-defect branch.

CXIX. FINAL SUMMARY

Using the minimal locked single-defect background

$$\Psi_B(\xi) = \phi_0(\xi) z_B(\xi), \qquad P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \qquad z_B(\xi) = \begin{pmatrix} \cos\theta(\xi) \\ 0 \\ \sin\theta(\xi) \end{pmatrix},$$

and the Gaussian-core approximation

$$\phi_0(\xi) \approx \phi_c, \qquad \theta'(\xi) \approx \theta_1 e^{-\xi^2/R^2}, \qquad \mathcal{O}[\Psi_B](\xi) \approx \mathcal{O}_c,$$

the aligned scalar source amplitude is

$$\mathrm{j}_0 = \alpha_X j_J + \beta_X j_K,$$

with

$$j_J = \phi_c^2 \theta_1 C_J, \qquad j_K = \mathcal{O}_c \theta_1 C_K.$$

Hence

$$\boxed{\mathrm{j}_0 = \theta_1 \big(\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K\big).}$$

Therefore the leading single-defect carrier-lowering amplitude becomes

$$\boxed{\Gamma_{\rm loc} = \frac{3u_\infty}{M_X^2}\,\theta_1 \big(\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K\big) - m_1 \delta\sigma.}$$

A necessary leading-order seed condition is

$$\Gamma_{\rm loc} > 0.$$

In the simplest normalized primitive branch with $C_J \sim C_K \sim 1$, this reduces to

$$\Gamma_{\rm loc} \approx \frac{3u_\infty \theta_1}{M_X^2} \big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\big) - m_1 \delta\sigma.$$

Thus the single-defect seed problem is reduced to a finite core-coefficient audit of the Class II branch.

CXX. EXPLICIT EVALUATION OF C_J AND C_K ON THE EMBEDDED $SU(2)_{(13)}$ SINGLE-DEFECT BRANCH

CXXI. 1. GOAL

We now evaluate the branch constants

$$C_J = \frac{1}{3}\, s^a \Big(e_1^\dagger T_a z_B + z_B^\dagger T_a e_1\Big)_{\xi=0}, \qquad C_K = \frac{1}{3}\, s^a \,\mathrm{Tr}\Big[T_a (e_1 z_B^\dagger + z_B e_1^\dagger)\Big]_{\xi=0}, \tag{414}$$

for the explicit embedded $SU(2)_{(13)}$ defect branch.

Here s^a denotes the internal direction selected by the aligned projector entering the scalar source

$$\frac{1}{3} E_i{}^a \mathcal{J}_i^{a,\mathrm{tot}} = \frac{1}{3} \hat{n}_i s^a \mathcal{J}_i^{a,\mathrm{tot}}. \tag{415}$$

The key issue is whether the physically relevant aligned direction is the background orbit direction m_a, or the defect tangent direction $k_a = A_a$.

CXXII. 2. EXPLICIT EMBEDDED $SU(2)_{(13)}$ BRANCH

Take the standard embedded generators

$$T_1 = \frac{1}{2}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad T_2 = \frac{1}{2}\begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \qquad T_3 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{416}$$

and the real defect representative

$$z_B(\xi) = \begin{pmatrix} \cos\theta(\xi) \\ 0 \\ \sin\theta(\xi) \end{pmatrix}, \qquad e_1(\xi) = \begin{pmatrix} -\sin\theta(\xi) \\ 0 \\ \cos\theta(\xi) \end{pmatrix}. \tag{417}$$

Define

$$m_a(\theta) := z_B^\dagger T_a z_B, \tag{418}$$

$$k_a(\theta) := \mathrm{Tr}\Big[T_a(e_1 z_B^\dagger + z_B e_1^\dagger)\Big], \tag{419}$$

$$A_a(\theta) := e_1^\dagger T_a z_B + z_B^\dagger T_a e_1. \tag{420}$$

Direct evaluation gives

$$\boxed{m_1 = \frac{1}{2}\sin 2\theta, \qquad m_2 = 0, \qquad m_3 = \frac{1}{2}\cos 2\theta,} \tag{421}$$

$$\boxed{k_1 = \cos 2\theta, \qquad k_2 = 0, \qquad k_3 = -\sin 2\theta,} \tag{422}$$

and

$$\boxed{A_1 = \cos 2\theta, \qquad A_2 = 0, \qquad A_3 = -\sin 2\theta.} \tag{423}$$

Therefore

$$\boxed{A_a(\theta) = k_a(\theta).} \tag{424}$$

CXXIII. 3. GEOMETRIC IDENTITIES

The orbit vector m_a and the tangent vector $k_a = A_a$ satisfy

$$m_a m_a = \frac{1}{4}\big(\sin^2 2\theta + \cos^2 2\theta\big) = \frac{1}{4}, \tag{425}$$

$$k_a k_a = \cos^2 2\theta + \sin^2 2\theta = 1, \tag{426}$$

and crucially

$$m_a k_a = \frac{1}{2}\sin 2\theta\cos 2\theta + \frac{1}{2}\cos 2\theta(-\sin 2\theta) = 0. \tag{427}$$

Hence

$$\boxed{m_a(\theta)\, k_a(\theta) = 0.} \tag{428}$$

Thus the background orbit direction and the defect tangent current direction are exactly orthogonal on the embedded $SU(2)_{(13)}$ branch.

CXXIV. 4. GENERAL FORMULA FOR C_J AND C_K

For any chosen aligned internal direction s^a,

$$\boxed{C_J = \frac{1}{3}\, s^a A_a = \frac{1}{3}\, s^a k_a, \qquad C_K = \frac{1}{3}\, s^a k_a.} \tag{429}$$

Therefore

$$\boxed{C_J = C_K = \frac{1}{3}\, s \cdot k.} \tag{430}$$

This immediately implies that the two source channels are automatically coherent on this branch: they carry the same projection factor.

CXXV. 5. ORBIT-ALIGNED BRANCH

Let the aligned internal direction be the normalized orbit vector

$$\hat{m}^a := 2m^a = (\sin 2\theta,\, 0,\, \cos 2\theta). \tag{431}$$

Then

$$\hat{m} \cdot k = \sin 2\theta \cos 2\theta + \cos 2\theta(-\sin 2\theta) = 0. \tag{432}$$

Hence

$$\boxed{C_J = C_K = 0 \qquad \text{(orbit-aligned branch).}} \tag{433}$$

Therefore, if one projects the source along the background orbit direction, the leading single-defect source amplitude vanishes:

$$\mathrm{j}_0 = 0. \tag{434}$$

Consequently

$$\boxed{\Gamma_{\mathrm{loc}} = -m_1 \delta\sigma \qquad \text{(orbit-aligned branch).}} \tag{435}$$

So in the orbit-aligned branch the mediator does not lower the carrier mass at leading order.

CXXVI. 6. TANGENT-ALIGNED BRANCH

Now choose the aligned internal direction to be the unit tangent current direction

$$\hat{k}^a := k^a = (\cos 2\theta,\, 0,\, -\sin 2\theta). \tag{436}$$

Since $k \cdot k = 1$, we obtain

$$\boxed{C_J = C_K = \frac{1}{3} \qquad \text{(tangent-aligned branch).}} \tag{437}$$

If one instead chooses the opposite tangent direction $-\hat{k}$, then

$$C_J = C_K = -\frac{1}{3}. \tag{438}$$

The attractive branch is the one for which the final Γ_{loc} is positive.

CXXVII. 7. EXPLICIT SOURCE AMPLITUDE j_0

From the previous step,

$$\mathrm{j}_0 = \theta_1 \Big(\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K \Big). \tag{439}$$

Therefore:

A. 7.1. Orbit-aligned branch

$$\boxed{\mathrm{j}_0 = 0.} \tag{440}$$

B. 7.2. Tangent-aligned attractive branch

$$\boxed{\mathrm{j}_0 = \frac{\theta_1}{3} \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \Big).} \tag{441}$$

CXXVIII. 8. EXPLICIT CORE-LOWERING AMPLITUDE

The general leading single-defect carrier-lowering amplitude was

$$\Gamma_{\rm loc} = \frac{3u_\infty}{M_X^2}\, \mathrm{j}_0 - m_1 \delta\sigma. \tag{442}$$

Hence:

A. 8.1. Orbit-aligned branch

$$\boxed{\Gamma_{\rm loc} = -\, m_1 \delta\sigma.} \tag{443}$$

B. 8.2. Tangent-aligned attractive branch

$$\boxed{\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2} \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \Big) - m_1 \delta\sigma.} \tag{444}$$

Thus the necessary leading seed condition becomes

$$\boxed{\frac{u_\infty \theta_1}{M_X^2} \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \Big) > m_1 \delta\sigma.} \tag{445}$$

CXXIX. 9. INTERPRETATION

This yields a sharp structural conclusion.

$$\boxed{\text{Single-defect seed localization is not sourced by the orbit direction } m_a.} \tag{446}$$

$$\boxed{\text{It is sourced by the defect tangent current direction } k_a = A_a.} \tag{447}$$

Equivalently:

$$\boxed{\text{the seed-localization channel is tangent-aligned, not orbit-aligned.}} \tag{448}$$

This also explains why the two channels J and K reinforce each other rather than compete at leading order on this branch: both are projected onto the same tangent direction $k_a = A_a$.

CXXX. 10. CONSEQUENCE FOR THE NEXT STEP

The remaining task is now finite and explicit. One must evaluate, on the intended single-defect Class II branch,

$$u_\infty, \qquad \theta_1, \qquad \phi_c, \qquad \mathcal{O}_c, \qquad m_1\delta\sigma, \qquad M_X^2, \tag{449}$$

and then test the inequality

$$\boxed{\frac{u_\infty\theta_1}{M_X^2}\left(\alpha_X\phi_c^2 + \beta_X\mathcal{O}_c\right) > m_1\delta\sigma.} \tag{450}$$

Only if this inequality holds can the single-defect branch support a carrier seed already at leading Gaussian core order.

CXXXI. FINAL SUMMARY

On the explicit embedded $SU(2)_{(13)}$ single-defect branch,

$$m_a(\theta) = \left(\frac{1}{2}\sin 2\theta,\, 0,\, \frac{1}{2}\cos 2\theta\right), \tag{451}$$

$$k_a(\theta) = A_a(\theta) = (\cos 2\theta,\, 0,\, -\sin 2\theta)\,. \tag{452}$$

These satisfy

$$m_a k_a = 0, \qquad k_a k_a = 1. \tag{453}$$

Therefore the branch constants are

$$C_J = C_K = \frac{1}{3}\, s\cdot k, \tag{454}$$

for any chosen aligned internal direction s^a.

If one chooses the orbit-aligned direction $s^a = \hat{m}^a = 2m^a$, then

$$C_J = C_K = 0, \tag{455}$$

so the mediator-induced source vanishes and

$$\Gamma_{\mathrm{loc}} = -m_1\delta\sigma. \tag{456}$$

If instead one chooses the tangent-aligned direction $s^a = \hat{k}^a = k^a$, then

$$C_J = C_K = \frac{1}{3}, \tag{457}$$

and the source amplitude becomes

$$\mathrm{j}_0 = \frac{\theta_1}{3}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big). \tag{458}$$

Hence the leading single-defect carrier-lowering amplitude is

$$\boxed{\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) - m_1 \delta\sigma.} \tag{459}$$

Thus the embedded $SU(2)_{(13)}$ branch yields a sharp structural result: single-defect seed localization is sourced by the tangent current direction $k_a = A_a$, not by the orbit direction m_a.

CXXXII. SIGN AND SIZE AUDIT OF THE SINGLE-DEFECT SEED INEQUALITY

CXXXIII. 1. STARTING POINT

On the tangent-aligned embedded $SU(2)_{(13)}$ single-defect branch, the leading carrier-lowering amplitude is

$$\boxed{\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) - m_1 \delta\sigma.} \tag{460}$$

The seed-localization condition at leading core order is

$$\boxed{\Gamma_{\rm loc} > 0.} \tag{461}$$

Thus the problem is reduced to a competition between the lowering block

$$L_{\rm seed} := \frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) \tag{462}$$

and the repulsive block

$$R_{\rm seed} := m_1 \delta\sigma. \tag{463}$$

CXXXIV. 2. NECESSARY SIGN CONDITION

Since

$$\theta_1 > 0, \qquad \phi_c^2 > 0, \qquad M_X^2 > 0, \tag{464}$$

the sign of the lowering block is controlled by

$$u_\infty \qquad \text{and} \qquad \alpha_X \phi_c^2 + \beta_X \mathcal{O}_c. \tag{465}$$

Therefore a necessary sign condition for seed localization is

$$\boxed{\mathrm{sgn}(u_\infty) = \mathrm{sgn}\big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\big).} \tag{466}$$

Equivalently:

1. if $u_\infty > 0$, then one needs

$$\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c > 0;$$

2. if $u_\infty < 0$, then one needs

$$\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c < 0.$$

CXXXV. 3. ACCIDENTAL CANCELLATION DANGER

Even if the sign condition is satisfied, the source can still fail if the two microscopic channels cancel:

$$\boxed{\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \approx 0.} \tag{467}$$

Thus one must avoid a branch in which the J-current and K-current contributions nearly cancel at the defect core.

CXXXVI. 4. SIZE CONDITION

The quantitative seed criterion is

$$\boxed{\frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) > m_1 \delta\sigma.} \tag{468}$$

This shows four immediate scaling facts:

1. $\Gamma_{\rm loc} \propto M_X^{-2}$: an excessively heavy mediator suppresses seed localization;
2. $\Gamma_{\rm loc} \propto \theta_1$: a sharper defect core strengthens the seed;
3. $\Gamma_{\rm loc} \propto \phi_c^2$ in the J-channel and $\Gamma_{\rm loc} \propto \mathcal{O}_c$ in the K-channel;
4. $m_1 \delta\sigma$ is the competing bare repulsive scalar-core shift.

CXXXVII. 5. NATURAL FAVORABLE WINDOW

The most natural favorable branch is:

$$\boxed{u_\infty > 0, \qquad \alpha_X > 0, \qquad \beta_X \geq 0,} \tag{469}$$

together with the non-cancellation condition

$$\boxed{\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \not\approx 0,} \tag{470}$$

and the quantitative inequality

$$\boxed{\frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) > m_1 \delta\sigma.} \tag{471}$$

CXXXVIII. 6. THREE STRUCTURAL SUBBRANCHES

It is useful to distinguish:

A. J-dominant branch

$$\alpha_X \phi_c^2 \gg \beta_X \mathcal{O}_c. \tag{472}$$

Then

$$\Gamma_{\rm loc} \approx \frac{u_\infty \theta_1}{M_X^2} \alpha_X \phi_c^2 - m_1 \delta\sigma. \tag{473}$$

B. K-dominant branch

$$\beta_X \mathcal{O}_c \gg \alpha_X \phi_c^2. \tag{474}$$

Then

$$\Gamma_{\rm loc} \approx \frac{u_\infty \theta_1}{M_X^2} \beta_X \mathcal{O}_c - m_1 \delta\sigma. \tag{475}$$

C. Balanced branch

$$\alpha_X \phi_c^2 \sim \beta_X \mathcal{O}_c. \tag{476}$$

Then the two channels reinforce each other and jointly produce the seed.

CXXXIX. 7. IMPORTANT DISTINCTION FROM THE TRANSPORT HIERARCHY SECTOR

In the explicit one-mediator truncation, the transport anisotropy ratio obeys

$$\rho_{\rm LO} = -\frac{3}{32} \frac{\alpha_X C_\parallel}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2, \tag{477}$$

so β_X and M_X drop out of the transport ratio.

By contrast, the seed-localization condition depends explicitly on

$$\boxed{\beta_X, \qquad M_X^2.} \tag{478}$$

Therefore:

$$\boxed{\text{transport naturalness and seed-localization naturalness are not the same audit problem.}} \tag{479}$$

CXL. 8. HONEST CURRENT CONCLUSION

What is already structurally fixed is:

$$\boxed{\text{single-defect seed localization requires tangent alignment and a positive net lowering block.}} \tag{480}$$

What remains to be evaluated on the intended Class II branch is:

$$u_\infty, \qquad \phi_c^2, \qquad \mathcal{O}_c, \qquad m_1 \delta\sigma, \qquad \text{and the effective coupling signs of } \alpha_X, \beta_X. \tag{481}$$

Thus the next microscopic task is a finite sign/size audit of these core coefficients.

CXLI. FINAL SUMMARY

On the tangent-aligned single-defect branch, the seed-localization amplitude is

$$\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2} \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \Big) - m_1 \delta\sigma. \tag{482}$$

The necessary leading condition is

$$\Gamma_{\rm loc} > 0. \tag{483}$$

This implies a necessary sign condition:

$$\operatorname{sgn}(u_\infty) = \operatorname{sgn}\big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\big), \tag{484}$$

and a quantitative size condition:

$$\frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) > m_1 \delta\sigma. \tag{485}$$

The most natural favorable window is

$$u_\infty > 0, \qquad \alpha_X > 0, \qquad \beta_X \geq 0, \tag{486}$$

with no accidental cancellation between the J- and K-channel source contributions.

Unlike the transport anisotropy ratio, the seed-localization condition depends explicitly on both β_X and M_X^2. Therefore seed-localization naturalness and transport naturalness are distinct microscopic audit problems.

The next actual task is to evaluate the core quantities

$$u_\infty, \quad \phi_c^2, \quad \mathcal{O}_c, \quad m_1\delta\sigma, \quad \text{and the effective signs of } \alpha_X, \beta_X \tag{487}$$

on the intended Class II branch.

CXLII. CONSTITUTIVE SIGN AUDIT OF u_∞ AND $m_1\delta\sigma$

CXLIII. 1. LOAD-BEARING MASS COEFFICIENT

In the explicit Class II defect-background coefficient template, the actual local mass coefficient is

$$\boxed{m_{\text{eff}}(x) = m(\bar\sigma(x)) - 3\chi_X(x)\, u_X(\bar\sigma(x)).} \tag{488}$$

Thus the one-defect problem is controlled by the same constitutive combination both in the core and in the bulk.

The corresponding normalized radial mass parameter is

$$\mu(r) = \frac{m(\bar\sigma(r)) - 3\chi_X(r) u_X(\bar\sigma(r))}{z_0(\bar\sigma(r))}. \tag{489}$$

CXLIV. 2. CORE-SIDE SEED CONDITION

On the tangent-aligned single-defect branch, the leading Gaussian core lowering amplitude is

$$\boxed{\Gamma_{\text{loc}} = \frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) - m_1 \delta\sigma.} \tag{490}$$

A necessary leading-order seed condition is

$$\boxed{\Gamma_{\text{loc}} > 0.} \tag{491}$$

This yields:

a. Sign condition

$$\boxed{u_\infty \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) > 0.} \tag{492}$$

b. Size condition

$$\boxed{\frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) > m_1 \delta\sigma.} \tag{493}$$

CXLV. 3. BULK-SIDE GAP CONDITION

Assuming the standard normalization $z_0(\sigma_\infty) > 0$, the asymptotic bulk gap condition is

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.} \tag{494}$$

Thus the sign of $u_\infty = u_X(\sigma_\infty)$ enters the bulk audit as well.

A. Case A: $u_\infty \geq 0$, $\chi_\infty \geq 0$

Then the Class II correction reduces the bulk gap, so one needs

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_\infty \geq 0.} \tag{495}$$

B. Case B: $u_\infty \leq 0$, $\chi_\infty \geq 0$

Then the Class II correction helps the bulk gap, so the condition is weaker:

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_\infty.} \tag{496}$$

C. Case C: simplest vacuum asymptotic branch

If

$$\boxed{\chi_\infty = 0,} \tag{497}$$

then

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}, \qquad \mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \tag{498}$$

This is the cleanest branch because the bulk-gap condition no longer constrains u_∞.

CXLVI. 4. SIGN OF $m_1\delta\sigma$

The scalar core term $m_1\delta\sigma$ controls whether the bare defect core is repulsive or attractive for the carrier.

- If
$$m_1\delta\sigma > 0,$$
the bare scalar core is repulsive and the mediator-induced lowering block must overcome it.
- If
$$m_1\delta\sigma = 0,$$
the scalar core is neutral at leading order.
- If
$$m_1\delta\sigma < 0,$$
the bare scalar core itself helps localization.

CXLVII. 5. MOST NATURAL FAVORABLE BRANCH

The structurally cleanest favorable window is

$$\boxed{\chi_\infty = 0, \qquad u_\infty > 0, \qquad \alpha_X \phi_c^2 + \beta_X \mathcal{O}_c > 0, \qquad m(\sigma_\infty) > 0.} \tag{499}$$

Then the bulk-gap condition is automatically decoupled from u_∞, while the core seed is strengthened by the positive mediator-induced lowering block.

CXLVIII. 6. HONEST CONCLUSION

At the present stage, the following is structurally fixed:

$$\boxed{\text{core localization and bulk positivity are controlled by the same constitutive mass combination} \quad m - 3\chi_X u_X.} \tag{500}$$

The remaining open constitutive audit is finite: one must determine the actual signs and relative magnitudes of

$$u_\infty, \qquad m_1 \delta\sigma, \qquad \alpha_X \phi_c^2, \qquad \beta_X \mathcal{O}_c \tag{501}$$

on the intended Class II branch.

CXLIX. FINAL SUMMARY

The single-defect seed condition and the bulk-gap condition are controlled by the same constitutive mass combination

$$m_{\text{eff}}(x) = m(\bar{\sigma}(x)) - 3\chi_X(x)\, u_X(\bar{\sigma}(x)).$$

On the tangent-aligned branch, the leading seed-localization amplitude is

$$\Gamma_{\text{loc}} = \frac{u_\infty \theta_1}{M_X^2} \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \Big) - m_1 \delta\sigma,$$

so a necessary seed condition is

$$\Gamma_{\text{loc}} > 0.$$

In the bulk, assuming $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.$$

Therefore u_∞ can help the defect core while harming the bulk gap if $\chi_\infty \neq 0$. The cleanest favorable window is the simplest vacuum asymptotic branch

$$\chi_\infty = 0,$$

for which the bulk condition reduces to

$$m(\sigma_\infty) > 0,$$

while the core seed still prefers

$$u_\infty > 0, \qquad \alpha_X \phi_c^2 + \beta_X \mathcal{O}_c > 0.$$

Thus the next microscopic task is a finite constitutive sign-and-size audit of

$$u_\infty, \quad m_1 \delta\sigma, \quad \alpha_X \phi_c^2, \quad \beta_X \mathcal{O}_c$$

on the intended Class II single-defect branch.

CL. J-DOMINANT, K-DOMINANT, AND BALANCED SEED AUDIT ON THE VACUUM-ASYMPTOTIC BRANCH

CLI. 1. VACUUM-ASYMPTOTIC WORKING BRANCH

We now specialize to the simplest vacuum-asymptotic branch

$$\boxed{\chi_\infty = 0, \qquad m(\sigma_\infty) > 0,} \tag{502}$$

so that the bulk-gap condition reduces simply to

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \tag{503}$$

On the tangent-aligned single-defect branch, the leading seed-localization amplitude is

$$\boxed{\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}\Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c\Big) - m_1 \delta\sigma.} \tag{504}$$

Define

$$\boxed{A_J := \alpha_X \phi_c^2, \qquad A_K := \beta_X \mathcal{O}_c, \qquad R_* := m_1 \delta\sigma.} \tag{505}$$

Then

$$\boxed{\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}(A_J + A_K) - R_*.} \tag{506}$$

The leading seed condition is

$$\boxed{\Gamma_{\rm loc} > 0.} \tag{507}$$

CLII. 2. DIMENSIONLESS DOMINANCE PARAMETERS

It is useful to introduce the channel ratio

$$\boxed{r_{JK} := \frac{A_K}{A_J} = \frac{\beta_X \mathcal{O}_c}{\alpha_X \phi_c^2},} \tag{508}$$

whenever $A_J \neq 0$, and the dimensionless seed margin

$$\boxed{\Delta_* := \frac{u_\infty \theta_1}{M_X^2}(A_J + A_K) - R_*.} \tag{509}$$

The three regimes are then:

$$\boxed{|r_{JK}| \ll 1 \quad \Rightarrow \quad \text{J-dominant branch,}} \tag{510}$$

$$\boxed{|r_{JK}| \gg 1 \quad \Rightarrow \quad \text{K-dominant branch,}} \tag{511}$$

$$\boxed{|r_{JK}| = O(1) \quad \Rightarrow \quad \text{balanced branch.}} \tag{512}$$

CLIII. 3. J-DOMINANT SEED BRANCH

Assume

$$|A_J| \gg |A_K|. \tag{513}$$

Then

$$\boxed{\Gamma_{\rm loc} \approx \frac{u_\infty \theta_1}{M_X^2} \alpha_X \phi_c^2 - R_*.} \tag{514}$$

The sign condition is

$$\boxed{u_\infty \alpha_X > 0.} \tag{515}$$

If $u_\infty > 0$, the quantitative seed threshold becomes

$$\boxed{\alpha_X > \alpha_{\rm crit}^{(J)} := \frac{M_X^2 R_*}{u_\infty \theta_1 \phi_c^2}.} \tag{516}$$

a. Advantage. This is the simplest analytic branch; the seed is controlled by one dominant microscopic channel.

b. Disadvantage. If one later wants good compatibility with the transport sector, a very large α_X may be costly, because the later longitudinal/transverse anisotropy audit is sensitive to α_X whereas the seed sector is not protected by any such cancellation.

CLIV. 4. K-DOMINANT SEED BRANCH

Assume

$$|A_K| \gg |A_J|. \tag{517}$$

Then

$$\boxed{\Gamma_{\rm loc} \approx \frac{u_\infty \theta_1}{M_X^2} \beta_X \mathcal{O}_c - R_*.} \tag{518}$$

The sign condition is

$$\boxed{u_\infty \beta_X \mathcal{O}_c > 0.} \tag{519}$$

If $u_\infty > 0$ and $\mathcal{O}_c > 0$, the threshold is

$$\boxed{\beta_X > \beta_{\rm crit}^{(K)} := \frac{M_X^2 R_*}{u_\infty \theta_1 \mathcal{O}_c}.} \tag{520}$$

a. Advantage. This branch can generate the seed without requiring a large α_X. Hence it is potentially more compatible with later transport naturalness, where α_X often plays the role of the longitudinal enhancement parameter.

b. Disadvantage. The branch relies strongly on the geometric/projector channel $\mathcal{O}_c$, whose actual constitutive magnitude still has to be audited.

CLV. 5. BALANCED SEED BRANCH

Assume

$$A_J \sim A_K \qquad \text{and} \qquad u_\infty(A_J + A_K) > 0. \tag{521}$$

Then

$$\boxed{\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}(A_J + A_K) - R_*,} \tag{522}$$

with neither channel negligible.

The threshold may be written as

$$\boxed{A_J + A_K > \frac{M_X^2\, R_*}{u_\infty \theta_1}.} \tag{523}$$

Equivalently,

$$\boxed{\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c > \frac{M_X^2\, R_*}{u_\infty \theta_1}.} \tag{524}$$

a. Advantage. This is the most robust branch against moderate fluctuations in either channel: each channel helps the other cross the seed threshold.

b. Second advantage. It allows one to avoid pushing α_X alone too hard. Thus it is structurally the most promising branch if one wishes to keep later transport anisotropy under control.

c. Disadvantage. The branch is more constitutively involved: one must audit both ϕ_c^2 and $\mathcal{O}_c$, and accidental cancellation must be avoided.

CLVI. 6. ACCIDENTAL-CANCELLATION WALL

Regardless of dominance classification, a structurally bad branch is

$$\boxed{A_J + A_K \approx 0.} \tag{525}$$

Then

$$\Gamma_{\rm loc} \approx -R_*, \tag{526}$$

so the mediator-induced seed disappears at leading order.

Thus the forbidden window is not simply negative sign, but also the near-cancellation locus

$$\boxed{\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \approx 0.} \tag{527}$$

CLVII. 7. COMBINED SEED/TRANSPORT NATURALNESS VERDICT

The seed-localization sector and the later transport-hierarchy sector are not controlled by the same effective ratios.

For seed localization, one needs

$$\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}(A_J + A_K) - R_* > 0. \tag{528}$$

By contrast, in the later one-mediator transport audit, the anisotropy ratio depends essentially on α_X relative to transverse couplings, while the common factor β_X / M_X^2 cancels from the leading ratio.

Therefore:

$$\boxed{\text{A pure J-dominant seed branch tends to push directly on } \alpha_X,} \tag{529}$$

whereas

$$\boxed{\text{a K-assisted or balanced seed branch can create localization without demanding an excessively large } \alpha_X.} \tag{530}$$

This makes the balanced branch, and secondarily the K-dominant branch, the most natural candidates for a combined seed-plus-transport program.

CLVIII. 8. PREFERRED ORDERING OF SUBBRANCHES

The structurally preferred ordering is:

$$\boxed{\text{Balanced} \succ \text{K-dominant} \succ \text{J-dominant.}} \tag{531}$$

The reason is not that J-dominant is impossible, but that it is the most likely to create tension with the later transport audit by forcing α_X to do too much work alone.

CLIX. 9. HONEST CURRENT CONCLUSION

At the present stage, what is already fixed is:

$$\boxed{\text{The vacuum-asymptotic branch } (\chi_\infty = 0) \text{ cleanly decouples bulk positivity from the sign of } u_\infty.} \tag{532}$$

$$\boxed{\text{The seed branch is controlled by the single scalar sum } A_J + A_K = \alpha_X \phi_c^2 + \beta_X \mathcal{O}_c.} \tag{533}$$

$$\boxed{\text{Balanced or K-assisted seed localization is structurally more compatible with the later transport sector than a purely J-domina}} \tag{534}$$

What remains open is the actual constitutive size audit of

$$\phi_c^2, \qquad \mathcal{O}_c, \qquad R_* = m_1 \delta\sigma, \tag{535}$$

on the intended Class II branch.

CLX. 10. NEXT FINITE AUDIT TARGETS

The next finite audit should therefore determine:

$$\boxed{\text{(i) whether } \phi_c^2 \gg \mathcal{O}_c, \quad \mathcal{O}_c \gg \phi_c^2, \quad \text{or } \phi_c^2 \sim \mathcal{O}_c;} \tag{536}$$

$$\boxed{\text{(ii) the critical values } \alpha_{\text{crit}}^{(J)},\ \beta_{\text{crit}}^{(K)};} \tag{537}$$

$$\boxed{\text{(iii) whether the actual branch lies in the J-dominant, K-dominant, or balanced window.}} \tag{538}$$

CLXI. FINAL SUMMARY

On the vacuum-asymptotic branch $\chi_\infty = 0$, the single-defect seed condition is

$$\Gamma_{\text{loc}} = \frac{u_\infty \theta_1}{M_X^2} \left(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \right) - m_1 \delta\sigma > 0.$$

Define

$$A_J = \alpha_X \phi_c^2, \qquad A_K = \beta_X \mathcal{O}_c.$$

Then the three seed subbranches are:

1. **J-dominant:**

$$|A_J| \gg |A_K|, \qquad \Gamma_{\rm loc} \approx \frac{u_\infty \theta_1}{M_X^2} A_J - R_*.$$

2. **K-dominant:**

$$|A_K| \gg |A_J|, \qquad \Gamma_{\rm loc} \approx \frac{u_\infty \theta_1}{M_X^2} A_K - R_*.$$

3. **Balanced:**

$$A_J \sim A_K, \qquad \Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2} (A_J + A_K) - R_*.$$

The structurally bad window is the cancellation locus

$$A_J + A_K \approx 0.$$

For the combined seed-plus-transport program, the preferred ordering is

$$\boxed{\text{Balanced} \ \succ \ \text{K-dominant} \ \succ \ \text{J-dominant.}}$$

The reason is that a pure J-dominant seed tends to force α_X to be large, while a K-assisted or balanced seed can create localization without placing all the burden on the longitudinal coupling.

CLXII. CORE SCALING AUDIT OF ϕ_c^2 VERSUS $\mathcal{O}_c$

CLXIII. 1. STARTING POINT

On the vacuum-asymptotic branch $\chi_\infty = 0$, the tangent-aligned single-defect seed amplitude is

$$\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2} \Big(\alpha_X \phi_c^2 + \beta_X \mathcal{O}_c \Big) - R_*, \qquad R_* := m_1 \delta\sigma. \tag{539}$$

Define

$$A_J := \alpha_X \phi_c^2, \qquad A_K := \beta_X \mathcal{O}_c. \tag{540}$$

The question is now:

$$\boxed{\text{What is the natural scaling of } \phi_c^2 \text{ relative to } \mathcal{O}_c \text{ on the intended single-defect branch?}} \tag{541}$$

CLXIV. 2. WHY THE COMPARISON IS DIMENSIONALLY MEANINGFUL

The microscopic defect currents on the locked background have the form

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0)\phi_0 \, m_a + \delta_{i\xi} \phi_0^2 \, \mathcal{A}_\xi^a, \tag{542}$$

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) \, k_a \, \mathcal{O}[\psi]. \tag{543}$$

Moreover, the minimal shell/test operator is

$$\boxed{\mathcal{O}_j[\psi] = D_j \psi.} \tag{544}$$

At the relativistic infrared fixed point,

$$[\partial_\mu] = 1, \qquad [\psi] = 1. \tag{545}$$

Hence

$$[\mathcal{O}_j[\psi]] = [D_j \psi] = 2. \tag{546}$$

If the condensate carrier field has the same canonical scaling class at the working branch, then

$$[\phi_c^2] = 2 \tag{547}$$

as well. Thus ϕ_c^2 and $\mathcal{O}_c$ are directly comparable at the core-scaling level.

CLXV. 3. CORE AMPLITUDE ESTIMATE FOR $\mathcal{O}_c$

Since the minimal shell/test mode is derivative-based,

$$\mathcal{O}_c \sim |D\psi|_{\text{core}}. \tag{548}$$

Introduce a characteristic shell/worldvolume scale ℓ_ψ (or equivalently $q_\psi \sim \ell_\psi^{-1}$) and a core shell amplitude ψ_c. Then

$$\boxed{\mathcal{O}_c \sim \frac{\psi_c}{\ell_\psi} \sim q_\psi \psi_c.} \tag{549}$$

Thus the channel ratio becomes

$$\boxed{r_{JK} := \frac{A_K}{A_J} = \frac{\beta_X \mathcal{O}_c}{\alpha_X \phi_c^2} \sim \frac{\beta_X}{\alpha_X}\,\frac{q_\psi \psi_c}{\phi_c^2}.} \tag{550}$$

CLXVI. 4. A USEFUL DIMENSIONLESS CORE PARAMETER

Define the purely core-geometric ratio

$$\boxed{\xi_* := \frac{\mathcal{O}_c}{\phi_c^2} \sim \frac{q_\psi \psi_c}{\phi_c^2}.} \tag{551}$$

Then

$$\boxed{r_{JK} = \frac{\beta_X}{\alpha_X}\,\xi_*.} \tag{552}$$

Therefore the J-dominant / K-dominant / balanced classification is reduced to the single dimensionless quantity ξ_*, dressed by the coupling ratio β_X/α_X.

CLXVII. 5. DEFECT-CORE SCALE ESTIMATE

If the shell/test mode varies on the same characteristic scale as the defect core, $\ell_\psi \sim R$, then

$$\mathcal{O}_c \sim \frac{\psi_c}{R}. \tag{553}$$

Hence

$$\boxed{\xi_* \sim \frac{\psi_c}{R\,\phi_c^2}.} \tag{554}$$

If, furthermore, the shell amplitude is of the same order as the condensate core amplitude,

$$\psi_c \sim \phi_c, \tag{555}$$

then

$$\boxed{\xi_* \sim \frac{1}{R\phi_c}.} \tag{556}$$

This provides the simplest practical scaling audit:

$$R\phi_c \gg 1 \quad \Rightarrow \quad \xi_* \ll 1 \quad \Rightarrow \quad J\text{-favoring core scaling}, \tag{557}$$

$$R\phi_c \sim 1 \quad \Rightarrow \quad \xi_* = O(1) \quad \Rightarrow \quad balancedcorescaling, \tag{558}$$

$$R\phi_c \ll 1 \quad \Rightarrow \quad \xi_* \gg 1 \quad \Rightarrow \quad K\text{-favoring core scaling}. \tag{559}$$

CLXVIII. 6. GENERICITY VERDICT

The shell/test mode enters through one derivative, whereas the J-channel core source contains the nondifferentiated amplitude factor ϕ_c^2. Therefore the K-channel is not automatically enhanced; on the contrary, it is typically derivative-suppressed unless the shell mode varies on a sufficiently short scale or β_X/α_X is large.

This leads to the genericity ordering:

$$\boxed{\text{balanced} \gtrsim \text{mild J-dominant} \gg \text{pure K-dominant.}} \tag{560}$$

In words:

1. a purely K-dominant branch is possible, but not the most natural generic core-scaling outcome;
2. a balanced branch is the most structurally plausible if $\ell_\psi \sim R$ and the couplings are of comparable size;
3. a mild J-dominant branch is also plausible when $R\phi_c$ is moderately large.

CLXIX. 7. DISTINCTION BETWEEN GENERICITY AND OVERALL DESIRABILITY

There is an important distinction between:

$$\text{(i) generic core scaling naturalness,} \qquad \text{and} \qquad \text{(ii) overall seed-plus-transport desirability.} \tag{561}$$

For genericity alone, a large pure K-dominant branch is somewhat less natural because it requires either

$$q_\psi R \gg 1, \qquad \text{or} \qquad \beta_X/\alpha_X \gg 1, \tag{562}$$

or both.

However, for the combined seed-plus-transport program, one does not want α_X to carry all the burden by itself, because the later transport audit is sensitive to the longitudinal carrier enhancement. Therefore the most desirable branch remains:

$$\boxed{\text{balanced} \succ \text{K-assisted} \succ \text{pure J-dominant.}} \tag{563}$$

These two orderings are not contradictory:

- **genericity:** balanced and mild J-dominant are easiest to obtain by core scaling;
- **desirability:** balanced is best because it helps seed localization without forcing α_X alone to become too large.

CLXX. 8. PRACTICAL BRANCH TEST

The finite audit now reduces to the following quantities:

$$\boxed{R\phi_c, \qquad \frac{\psi_c}{\phi_c}, \qquad \frac{\beta_X}{\alpha_X}.} \tag{564}$$

Indeed,

$$r_{JK} \sim \frac{\beta_X}{\alpha_X}\frac{\psi_c}{R\phi_c^2} = \frac{\beta_X}{\alpha_X}\frac{\psi_c/\phi_c}{R\phi_c}. \tag{565}$$

So:

1. if $R\phi_c \gg 1$ and $\beta_X/\alpha_X = O(1)$, the seed tends to be J-leaning;
2. if $R\phi_c \sim 1$ and $\beta_X/\alpha_X = O(1)$, the seed tends to be balanced;
3. if $R\phi_c \ll 1$ or $\beta_X/\alpha_X \gg 1$, the seed may become K-dominant.

CLXXI. 9. HONEST CURRENT CONCLUSION

What is now structurally established is:

$$\boxed{\text{The relative size of } A_J = \alpha_X \phi_c^2 \text{ and } A_K = \beta_X \mathcal{O}_c \text{ is controlled by the dimensionless core ratio } \xi_* = \mathcal{O}_c/\phi_c^2.} \quad (566)$$

Under the minimal shell ansatz $\mathcal{O}_j[\psi] = D_j\psi$, one has

$$\boxed{\xi_* \sim \frac{q_\psi \psi_c}{\phi_c^2} \sim \frac{\psi_c/\phi_c}{R\phi_c} \quad (\ell_\psi \sim R).} \quad (567)$$

Therefore the most natural current expectation is:

$$\boxed{\text{balanced or mild } J\text{-dominant seed scaling is more generic than pure } K\text{-dominant scaling.}} \quad (568)$$

At the same time, for the full seed-plus-transport program, the most desirable branch remains balanced.

CLXXII. FINAL SUMMARY

The single-defect seed channels are

$$A_J = \alpha_X \phi_c^2, \qquad A_K = \beta_X \mathcal{O}_c.$$

Since the minimal shell/test operator is

$$\mathcal{O}_j[\psi] = D_j\psi,$$

the core shell amplitude scales as

$$\mathcal{O}_c \sim q_\psi \psi_c \sim \frac{\psi_c}{\ell_\psi}.$$

Hence the relative channel ratio is

$$r_{JK} = \frac{A_K}{A_J} \sim \frac{\beta_X}{\alpha_X} \frac{q_\psi \psi_c}{\phi_c^2}.$$

If the shell variation scale is comparable to the defect-core scale, $\ell_\psi \sim R$, then

$$r_{JK} \sim \frac{\beta_X}{\alpha_X} \frac{\psi_c/\phi_c}{R\phi_c}.$$

Therefore:

- $R\phi_c \gg 1$ tends to favor a J-leaning seed;
- $R\phi_c \sim 1$ tends to favor a balanced seed;
- $R\phi_c \ll 1$ or $\beta_X/\alpha_X \gg 1$ can lead to K-dominance.

Thus the most natural generic expectation is

$$\boxed{\text{balanced or mild } J\text{-dominant seed scaling,}}$$

whereas a pure K-dominant seed is possible but less generic.

For the combined seed-plus-transport program, the preferred branch remains

$$\boxed{\text{balanced.}}$$

CLXXIII. FROM CORE-SCALING HEURISTICS TO THE ACTUAL ONE-DEFECT WORKING CRITERION

CLXXIV. 1. THE ROLE OF $R\phi_c$ AND ψ_c/ϕ_c

The heuristic core-scaling audit showed that the relative size of the seed channels

$$A_J = \alpha_X \phi_c^2, \qquad A_K = \beta_X \mathcal{O}_c \tag{569}$$

is controlled by

$$r_{JK} \sim \frac{\beta_X}{\alpha_X} \frac{\psi_c/\phi_c}{R\phi_c}, \tag{570}$$

under the minimal shell/test assumption

$$\mathcal{O}_j[\psi] = D_j \psi, \qquad \ell_\psi \sim R. \tag{571}$$

This remains useful as a pre-screening tool for classifying the branch as:

$$J\text{-leaning}, \qquad \text{balanced}, \qquad K\text{-leaning}. \tag{572}$$

CLXXV. 2. WHY THIS IS NOT THE FINAL ONE-DEFECT CRITERION

However, the actual one-defect/one-mode program is already structurally reduced to a sharper set of variables. The working chain is:

$$\boxed{\text{remote-gap isolation} \Longrightarrow \text{low-slot channel projection} \Longrightarrow \text{radial one-defect criterion in terms of } (g_c, g_E, \mu_\infty).} \tag{573}$$

The core coefficients are

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}, \tag{574}$$

and the bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \tag{575}$$

The sufficient one-defect/one-mode criterion is then

$$\boxed{\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},} \tag{576}$$

with competitor exclusion in the remaining projected channels.

CLXXVI. 3. INTERPRETATION

Therefore $R\phi_c$ and ψ_c/ϕ_c are not the final load-bearing variables of the one-defect theorem. They are only constitutive pre-audit variables which help predict the likely sign and relative size of the projected transport coefficients.

In other words:

$$\boxed{(R\phi_c,\ \psi_c/\phi_c) \text{ guide the branch classification,}} \tag{577}$$

but

$$\boxed{(g_E,\ g_c,\ \mu_\infty,\ \kappa_*) \text{ decide the actual one-defect acceptance.}} \tag{578}$$

CLXXVII. 4. GENERICITY VERSUS THEOREM-LEVEL ACCEPTANCE

At the heuristic core-scaling level:

$$\boxed{\text{balanced or mild } J\text{-leaning branches are more generic than pure } K\text{-dominant branches.}} \tag{579}$$

But theorem-level acceptance is controlled instead by:

$$g_E > 0, \qquad \xi_E g_E + \xi_c g_c > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}. \tag{580}$$

Hence a branch that looks generic at the $(R\phi_c, \psi_c/\phi_c)$ level may still fail the actual one-defect criterion if the projected channel data are wrong.

CLXXVIII. 5. CORRECT NEXT STEP

The mathematically correct next step is therefore not to continue refining only $(R\phi_c, \psi_c/\phi_c)$, but to evaluate directly on the intended single-defect branch:

$$\boxed{g_E, \qquad g_c, \qquad \mu_\infty, \qquad \kappa_*.} \tag{581}$$

The core-scaling variables should then be used only to interpret why the projected coefficients come out in a J-leaning, balanced, or K-leaning manner.

CLXXIX. FINAL SUMMARY

The heuristic ratios $R\phi_c$ and ψ_c/ϕ_c remain useful for pre-classifying the single-defect seed branch as J-leaning, balanced, or K-leaning:

$$r_{JK} \sim \frac{\beta_X}{\alpha_X} \frac{\psi_c/\phi_c}{R\phi_c}. \tag{582}$$

However, the actual one-defect/one-mode working theorem is already reduced to the projected coefficients

$$g_E, \qquad g_c, \qquad \mu_\infty, \qquad \kappa_*, \tag{583}$$

with

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad \mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \tag{584}$$

The sufficient one-defect/one-mode criterion is

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}, \tag{585}$$

together with competitor exclusion.

Therefore:

$$\boxed{(R\phi_c, \psi_c/\phi_c) \text{ are heuristic branch guides,}} \tag{586}$$

while

$$\boxed{(g_E, g_c, \mu_\infty, \kappa_*) \text{ are the actual one-defect acceptance data.}} \tag{587}$$

The correct next step is thus to evaluate the projected one-defect coefficients directly on the intended Class II branch.

CLXXX. MINIMAL PROJECTION FORMULA FOR g_E AND g_c ON THE ONE-DEFECT BRANCH

CLXXXI. 1. GOAL

The heuristic core variables $(R\phi_c, \psi_c/\phi_c)$ are useful only for branch intuition. The actual one-defect acceptance theorem is controlled by the projected transport data

$$g_c, \qquad g_E, \qquad g_O, \qquad \mu_\infty, \qquad \kappa_*. \tag{588}$$

Therefore the correct next task is to write the explicit projection formula that extracts g_c and g_E from the shell-reduced microscopic operator.

CLXXXII. 2. SHELL-PROJECTED DERIVATIVE DATA

Let the shell-projected effective operator be

$$\boxed{H_{\rm shell}(p) = \Pi_{S_*}\, H_{\rm eff}(k_* + p)\, \Pi_{S_*},} \tag{589}$$

and let the low-slot projector be

$$\boxed{P_* = \frac{1}{2\pi i} \oint_\Gamma (z - H_{\rm shell}(0))^{-1}\, dz.} \tag{590}$$

Define the first derivative blocks

$$\boxed{K_i = \partial_{k_i} H_{\rm eff}(k)|_{k=k_*}\,, \qquad M_i = P_* K_i P_*.} \tag{591}$$

In the adapted shell frame $(\hat{n}, e_1, e_2)$, define

$$\boxed{M_\parallel = \hat{n}_i M_i, \qquad M_1 = e_{1i} M_i, \qquad M_2 = e_{2i} M_i.} \tag{592}$$

CLXXXIII. 3. EXACT EXTRACTION OF $\lambda_\parallel, \alpha, \beta$

Under the locked Pauli selection rule,

$$M_\parallel = \lambda_\parallel \sigma_3, \qquad M_1 = \alpha\sigma_1 + \beta\sigma_2, \qquad M_2 = \alpha\sigma_2 - \beta\sigma_1. \tag{593}$$

Therefore the reduced coefficients are extracted exactly by

$$\boxed{\lambda_\parallel = \frac{1}{2}\,\mathrm{Tr}(\sigma_3 M_\parallel),} \tag{594}$$

$$\boxed{\alpha = \frac{1}{4}\,\mathrm{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \qquad \beta = \frac{1}{4}\,\mathrm{Tr}(\sigma_2 M_1 - \sigma_1 M_2).} \tag{595}$$

Equivalently, written directly in terms of the projected derivative blocks,

$$\boxed{\lambda_\parallel = \frac{1}{2}\,\mathrm{Tr}\Big(\sigma_3 P_* K_\parallel P_*\Big), \qquad K_\parallel := \hat{n}_i K_i,} \tag{596}$$

$$\boxed{\alpha = \frac{1}{4}\,\mathrm{Tr}\Big(\sigma_1 P_* K_1 P_* + \sigma_2 P_* K_2 P_*\Big),} \tag{597}$$

$$\boxed{\beta = \frac{1}{4}\,\mathrm{Tr}\Big(\sigma_2 P_* K_1 P_* - \sigma_1 P_* K_2 P_*\Big).} \tag{598}$$

CLXXXIV. 4. EXACT NORMALIZATION LAYER

The exact overlap-normalization layer is

$$\boxed{\lambda_\parallel = C_\parallel g_c, \qquad \alpha = C_E g_E, \qquad \beta = C_O g_O.} \tag{599}$$

Hence the microscopic transport coefficients are

$$\boxed{g_c = \frac{1}{C_\parallel}\,\frac{1}{2}\,\mathrm{Tr}\Big(\sigma_3 P_* K_\parallel P_*\Big),} \tag{600}$$

$$\boxed{g_E = \frac{1}{C_E}\,\frac{1}{4}\,\mathrm{Tr}\Big(\sigma_1 P_* K_1 P_* + \sigma_2 P_* K_2 P_*\Big),} \tag{601}$$

$$\boxed{g_O = \frac{1}{C_O}\,\frac{1}{4}\,\mathrm{Tr}\Big(\sigma_2 P_* K_1 P_* - \sigma_1 P_* K_2 P_*\Big).} \tag{602}$$

These are the minimal projection formulas to be used on the actual or quasi-actual one-defect branch.

CLXXXV. 5. CORE MATCHING TO THE ONE-DEFECT RADIAL COEFFICIENTS

The exact core formulas are

$$\boxed{a_0 = \frac{g_E^{\mathrm{core}} + g_c^{\mathrm{core}}}{z_{0,0}}, \qquad b_0 = \frac{g_E^{\mathrm{core}}}{z_{0,0}}, \qquad c_0 = \frac{g_O^{\mathrm{core}}}{z_{0,0}}.} \tag{603}$$

Under the matching hypothesis

$$\boxed{g_E^{\mathrm{core}} = \xi_E g_E, \qquad g_c^{\mathrm{core}} = \xi_c g_c, \qquad g_O^{\mathrm{core}} = \xi_O g_O,} \tag{604}$$

one gets

$$\boxed{a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}.} \tag{605}$$

The bulk mass parameter is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \tag{606}$$

CLXXXVI. 6. WORKING ONE-DEFECT/ONE-MODE CRITERION

The sufficient one-defect/one-mode criterion is

$$\boxed{\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},} \tag{607}$$

together with competitor exclusion in every other projected channel.

Equivalently, in terms of the projected transport data,

$$\boxed{\mu_\infty > 0, \qquad \xi_E g_E + \xi_c g_c > 0, \qquad \xi_E g_E > 0,} \tag{608}$$

and

$$\boxed{2|\kappa_*|\,\xi_E g_E > \xi_E g_E + \xi_c g_c.} \tag{609}$$

Thus the actual one-defect problem is completely reduced to the finite audit of

$$g_c, \qquad g_E, \qquad \mu_\infty, \qquad \kappa_*, \tag{610}$$

after the normalization constants and matching constants are fixed.

CLXXXVII. 7. CANONICAL EXPLICIT BRANCH

For the explicit canonical branch

$$\boxed{T_\star = T^1, \qquad \gamma_E = 1, \qquad \gamma_O = 0,} \tag{611}$$

the extracted Class II/Bloch coefficients are

$$\boxed{g_c = \frac{3\sqrt{3}\pi}{512}\,\frac{\alpha_X\beta_X}{M_X^2}\,\theta_0^3\kappa^3,} \tag{612}$$

$$\boxed{g_E = \frac{\sqrt{3}\pi}{16}\,\frac{\beta_X}{M_X^2}\,\theta_0\kappa, \qquad g_O = 0.} \tag{613}$$

Hence

$$\boxed{b_0 = \frac{\xi_E}{z_{0,0}}\,\frac{\sqrt{3}\pi}{16}\,\frac{\beta_X}{M_X^2}\,\theta_0\kappa,} \tag{614}$$

and

$$\boxed{a_0 = b_0(1+R_c), \qquad R_c = \frac{\xi_c}{\xi_E}\,\frac{3}{32}\,\alpha_X(\theta_0\kappa)^2.} \tag{615}$$

Therefore in the most relevant lowest-channel case $\kappa_* = -1$, the sufficient criterion reduces to

$$\boxed{-1 < R_c < 1, \qquad \alpha > 0, \qquad \mu_\infty > 0, \qquad \Delta^{\rm full}_{{\rm bench},\rho} > \eta_{{\rm rem},\rho}.} \tag{616}$$

CLXXXVIII. 8. INTERPRETATION

The point of this reorganization is simple:

$$\boxed{(R\phi_c, \psi_c/\phi_c) \text{ remain useful only as heuristic branch guides,}} \tag{617}$$

whereas

$$\boxed{(g_E, g_c, \mu_\infty, \kappa_*) \text{ are the actual one-defect acceptance data.}} \tag{618}$$

Thus the mathematically correct next step is no longer a heuristic core-scaling discussion, but the explicit projected derivative audit on the intended single-defect branch.

CLXXXIX. FINAL SUMMARY

The actual one-defect acceptance theorem is controlled by the projected transport data

$$g_c, \qquad g_E, \qquad g_O, \qquad \mu_\infty, \qquad \kappa_*,$$

not directly by the heuristic core ratios $(R\phi_c, \psi_c/\phi_c)$.

The exact extraction chain is:

$$H_{\rm eff}(k) \to K_i = \partial_{k_i} H_{\rm eff}(k)\big|_{k=k_*} \to M_i = P_* K_i P_* \to (\lambda_\parallel, \alpha, \beta) \to (g_c, g_E, g_O),$$

with

$$\lambda_\parallel = \frac{1}{2}\,\mathrm{Tr}(\sigma_3 M_\parallel), \qquad \alpha = \frac{1}{4}\,\mathrm{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \qquad \beta = \frac{1}{4}\,\mathrm{Tr}(\sigma_2 M_1 - \sigma_1 M_2),$$

and

$$g_c = \frac{\lambda_\parallel}{C_\parallel}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.$$

The one-defect core coefficients are

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}},$$

and the bulk mass is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Hence the sufficient one-defect/one-mode criterion is

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},$$

with competitor exclusion.

Therefore the mathematically correct next task is to compute the projected derivative matrices M_i, extract g_c, g_E, and evaluate the finite one-defect inequalities directly on the intended Class II branch.

CXC. MINIMAL FINITE CONSTRUCTION OF P_* AND K_i IN A SHELL-ADAPTED LOW-SLOT BASIS

CXCI. 1. GOAL

The actual one-defect acceptance theorem is controlled by the data

$$g_E, \qquad g_c, \qquad g_O, \qquad \mu_\infty, \qquad \kappa_*. \tag{619}$$

The purpose of the present step is to give a finite-dimensional construction of

$$P_*, \qquad K_i, \qquad g_E,\ g_c,\ g_O, \tag{620}$$

from a shell-adapted low-slot basis.

CXCII. 2. SHELL-ADAPTED FINITE BASIS

Choose a finite shell-adapted basis

$$\mathcal{B}_* = \{\, b_\mu \,\}_{\mu=1}^{N_*} \tag{621}$$

centered around the selected shell point G_*, or more generally around the selected one-defect low-slot region. Define the overlap matrix

$$\boxed{S_{\mu\nu} := \langle b_\mu, b_\nu \rangle,} \tag{622}$$

and the effective operator matrix

$$\boxed{H_{\mu\nu}(k) := \langle b_\mu,\, H_{\mathrm{eff}}(k)\, b_\nu \rangle.} \tag{623}$$

No orthonormality is assumed. All finite extraction is performed in this generalized basis.

CXCIII. 3. LOW-SLOT GENERALIZED EIGENPROBLEM

At the selected expansion point $k = k_*$, solve the generalized Hermitian eigenproblem

$$\boxed{H(k_*)\, u_a = \varepsilon_a\, S\, u_a, \qquad a = 1, \ldots, N_*.} \tag{624}$$

Select the two-dimensional low slot

$$\mathcal{H}_* = \operatorname{span}\{u_1, u_2\}, \tag{625}$$

and normalize the chosen two columns by

$$\boxed{u_a^\dagger S u_b = \delta_{ab}.} \tag{626}$$

Let

$$\boxed{U = (u_1, u_2)} \tag{627}$$

be the $N_* \times 2$ matrix of low-slot eigenvectors.

Then the coefficient-space projector onto the low slot is

$$\boxed{P_*^{(\mathrm{coef})} = U\, U^\dagger S.} \tag{628}$$

Equivalently, for any matrix X acting in coefficient space, the reduced 2×2 low-slot matrix is simply

$$\boxed{X_* := U^\dagger X U.} \tag{629}$$

CXCIV. 4. FINITE CONSTRUCTION OF THE DERIVATIVE BLOCKS

Define the first momentum derivatives

$$\boxed{K_i := \partial_{k_i} H(k)\big|_{k=k_*}.} \tag{630}$$

If analytic derivatives are unavailable, use the central finite-difference approximation

$$\boxed{K_i \approx \frac{H(k_* + \delta k\, e_i) - H(k_* - \delta k\, e_i)}{2\delta k},} \tag{631}$$

with δk chosen inside the linear response window.

In the shell-adapted frame $(\hat{n}, e_1, e_2)$, define

$$\boxed{K_\parallel := \hat{n}_i K_i, \qquad K_1 := e_{1i} K_i, \qquad K_2 := e_{2i} K_i.} \tag{632}$$

Project them to the low slot:

$$\boxed{M_\parallel := U^\dagger K_\parallel U, \qquad M_1 := U^\dagger K_1 U, \qquad M_2 := U^\dagger K_2 U.} \tag{633}$$

These are 2×2 Hermitian matrices.

CXCV. 5. REMOVING IDENTITY PIECES AND FIXING THE PAULI GAUGE

Each reduced matrix can be decomposed as

$$M_a = \frac{1}{2}\operatorname{Tr}(M_a)\, \mathbf{1}_2 + \widetilde{M}_a, \qquad a \in \{\parallel, 1, 2\}, \tag{634}$$

where $\widetilde{M}_a$ is traceless Hermitian.

Only the traceless parts carry the Weyl data. Thus define

$$\boxed{\widetilde{M}_\parallel = M_\parallel - \frac{1}{2}\operatorname{Tr}(M_\parallel)\mathbf{1}_2,} \tag{635}$$

$$\boxed{\widetilde{M}_1 = M_1 - \frac{1}{2}\operatorname{Tr}(M_1)\mathbf{1}_2, \qquad \widetilde{M}_2 = M_2 - \frac{1}{2}\operatorname{Tr}(M_2)\mathbf{1}_2.} \tag{636}$$

Now choose an $SU(2)$ change of basis W inside the low slot such that

$$\boxed{W^\dagger \widetilde{M}_\parallel W = \lambda_\parallel \sigma_3,} \tag{637}$$

with $\lambda_\parallel \geq 0$. Then rotate the transverse blocks:

$$\boxed{\widehat{M}_1 := W^\dagger \widetilde{M}_1 W, \qquad \widehat{M}_2 := W^\dagger \widetilde{M}_2 W.} \tag{638}$$

This is the finite-matrix version of the locked Pauli gauge.

CXCVI. 6. EXACT FINITE EXTRACTION OF $\lambda_\parallel, \alpha, \beta$

In the locked Pauli gauge, the exact extraction formulas are

$$\boxed{\lambda_\parallel = \frac{1}{2}\operatorname{Tr}(\sigma_3 \widehat{M}_\parallel), \qquad \widehat{M}_\parallel := W^\dagger \widetilde{M}_\parallel W = \lambda_\parallel \sigma_3,} \tag{639}$$

$$\boxed{\alpha = \frac{1}{4}\operatorname{Tr}(\sigma_1 \widehat{M}_1 + \sigma_2 \widehat{M}_2),} \tag{640}$$

$$\boxed{\beta = \frac{1}{4}\operatorname{Tr}(\sigma_2 \widehat{M}_1 - \sigma_1 \widehat{M}_2).} \tag{641}$$

Equivalently, before introducing W, one may say: the low-slot basis must first be rotated so that the longitudinal traceless block becomes proportional to σ_3, and only then should α, β be read from the transverse blocks.

CXCVII. 7. EXTRACTION OF g_c, g_E, g_O

Once $\lambda_\parallel, \alpha, \beta$ are extracted, the exact normalization layer is

$$\boxed{\lambda_\parallel = C_\parallel g_c, \qquad \alpha = C_E g_E, \qquad \beta = C_O g_O.} \tag{642}$$

Therefore

$$\boxed{g_c = \frac{\lambda_\parallel}{C_\parallel}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.} \tag{643}$$

This is the final finite projection formula needed for the one-defect theorem.

CXCVIII. 8. BULK MASS AND CORE MATCHING

The bulk mass parameter is not extracted from the shell derivative blocks; it is an independent constitutive quantity:

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \tag{644}$$

The core coefficients entering the one-defect theorem are

$$\boxed{a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}.} \tag{645}$$

Thus the sufficient one-defect criterion becomes

$$\boxed{\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},} \tag{646}$$

with competitor exclusion in the remaining projected channels.

CXCIX. 9. IMPORTANT SEPARATION: WHAT IS AND IS NOT EXTRACTED HERE

The present finite construction extracts

$$\boxed{P_*, \qquad K_i, \qquad \lambda_\parallel, \alpha, \beta, \qquad g_c, g_E, g_O.} \tag{647}$$

It does *not* by itself determine κ_*. The quantity κ_* belongs to the already-isolated radial/projected-channel problem and must be imported from that sector.

Thus the full one-defect acceptance data are assembled as:

$$\boxed{\text{shell derivative extraction} \oplus \text{bulk constitutive audit} \oplus \text{radial channel label } \kappa_*.} \tag{648}$$

CC. 10. MINIMAL FINITE ALGORITHM

The actual computational algorithm is therefore:

1. Choose a shell-adapted finite basis $\{b_\mu\}$ and build $S_{\mu\nu}$, $H_{\mu\nu}(k)$.
2. Solve the generalized eigenproblem at k_*, isolate the two-dimensional low slot, and form U.
3. Compute $K_i = \partial_{k_i} H(k)|_{k_*}$ analytically or by central differences.
4. Project to the low slot:
$$M_\parallel = U^\dagger K_\parallel U, \qquad M_1 = U^\dagger K_1 U, \qquad M_2 = U^\dagger K_2 U.$$
5. Remove identity parts and perform the $SU(2)$ gauge-fixing rotation W so that the longitudinal block is $\lambda_\parallel \sigma_3$.
6. Extract $\lambda_\parallel, \alpha, \beta$.
7. Divide by the normalization constants $C_\parallel, C_E, C_O$ to obtain g_c, g_E, g_O.
8. Combine with μ_∞ and the imported κ_* to test the one-defect criterion.

CCI. 11. HONEST FINAL STATUS

What is now fully fixed is the finite-matrix route from the shell-reduced microscopic operator to the coefficients g_E, g_c, g_O.

What remains open is not the projection formula, but the actual microscopic evaluation of the matrices $H(k)$ and their derivatives on the intended Class II one-defect branch.

CCII. FINAL SUMMARY

The actual one-defect theorem should be fed not by heuristic core ratios, but by the projected transport data

$$g_c, \qquad g_E, \qquad g_O, \qquad \mu_\infty, \qquad \kappa_*.$$

Given a shell-adapted finite basis $\{b_\mu\}$, one builds

$$S_{\mu\nu} = \langle b_\mu, b_\nu \rangle, \qquad H_{\mu\nu}(k) = \langle b_\mu, H_{\mathrm{eff}}(k) b_\nu \rangle,$$

solves the generalized eigenproblem at k_*, isolates the two-dimensional low slot, and forms the projected derivative blocks

$$M_\parallel = U^\dagger K_\parallel U, \qquad M_1 = U^\dagger K_1 U, \qquad M_2 = U^\dagger K_2 U.$$

After removing identity pieces and fixing the internal $SU(2)$ gauge so that the longitudinal block is proportional to σ_3, one extracts

$$\lambda_\parallel, \qquad \alpha, \qquad \beta,$$

and then

$$g_c = \frac{\lambda_\parallel}{C_\parallel}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.$$

These are then inserted into

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad \mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)},$$

and the sufficient one-defect criterion

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}$$

is tested, with κ_* imported from the separate radial/projected-channel sector.

Thus the projection formula is now fixed; the remaining open work is the actual microscopic evaluation of $H(k)$ and K_i on the intended Class II one-defect branch.

CCIII. CANONICAL TOY EXTRACTION IN A TWO-STATE LOW SLOT

CCIV. 1. SETUP

Assume that the one-defect shell-adapted generalized eigenproblem has already isolated a two-dimensional low slot

$$\mathcal{H}_* = \mathrm{span}\{u_1, u_2\}, \qquad u_a^\dagger S u_b = \delta_{ab}. \tag{649}$$

Let

$$U = (u_1, u_2) \tag{650}$$

and define the projected derivative blocks

$$M_\parallel := U^\dagger K_\parallel U, \qquad M_1 := U^\dagger K_1 U, \qquad M_2 := U^\dagger K_2 U. \tag{651}$$

Each M_a is a 2×2 Hermitian matrix.

After subtracting identity pieces,

$$\widetilde{M}_a := M_a - \frac{1}{2} \mathrm{Tr}(M_a) \mathbf{1}_2, \qquad a \in \{\parallel, 1, 2\}, \tag{652}$$

we work only with the traceless Hermitian blocks.

CCV. 2. GENERAL PAULI DECOMPOSITION

Any traceless Hermitian 2×2 matrix has the form

$$\widetilde{M}_a = v_a^1 \sigma_1 + v_a^2 \sigma_2 + v_a^3 \sigma_3 =: \vec{v}_a \cdot \vec{\sigma}. \tag{653}$$

Thus

$$\widetilde{M}_\parallel = \vec{v}_\parallel \cdot \vec{\sigma}, \qquad \widetilde{M}_1 = \vec{v}_1 \cdot \vec{\sigma}, \qquad \widetilde{M}_2 = \vec{v}_2 \cdot \vec{\sigma}. \tag{654}$$

Choose an $SU(2)$ rotation W such that

$$W^\dagger \widetilde{M}_\parallel W = |\vec{v}_\parallel|\, \sigma_3. \tag{655}$$

Define

$$\lambda_\parallel := |\vec{v}_\parallel| \geq 0. \tag{656}$$

After this rotation, there remains a residual $U(1)$ freedom

$$W \mapsto W e^{i\varphi\sigma_3/2}, \tag{657}$$

which rotates the (σ_1, σ_2) plane. If the branch is admissible in the locked Pauli sense, one may use this residual freedom to reach the canonical form

$$\boxed{\widehat{M}_\parallel = \lambda_\parallel \sigma_3, \qquad \widehat{M}_1 = \alpha\sigma_1 + \beta\sigma_2, \qquad \widehat{M}_2 = \alpha\sigma_2 - \beta\sigma_1.} \tag{658}$$

CCVI. 3. EXPLICIT 2×2 MATRIX FORM

Using

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{659}$$

the canonical reduced blocks become

$$\boxed{\widehat{M}_\parallel = \begin{pmatrix} \lambda_\parallel & 0 \\ 0 & -\lambda_\parallel \end{pmatrix},} \tag{660}$$

$$\boxed{\widehat{M}_1 = \begin{pmatrix} 0 & \alpha - i\beta \\ \alpha + i\beta & 0 \end{pmatrix},} \tag{661}$$

$$\boxed{\widehat{M}_2 = \begin{pmatrix} 0 & -\beta - i\alpha \\ -\beta + i\alpha & 0 \end{pmatrix}.} \tag{662}$$

Thus:

$$\boxed{\lambda_\parallel = (\widehat{M}_\parallel)_{11} = -(\widehat{M}_\parallel)_{22},} \tag{663}$$

$$\boxed{\alpha = \Re(\widehat{M}_1)_{12}, \qquad \beta = -\Im(\widehat{M}_1)_{12}.} \tag{664}$$

Equivalently, from $\widehat{M}_2$,

$$\boxed{\alpha = -\Im(\widehat{M}_2)_{12}, \qquad \beta = -\Re(\widehat{M}_2)_{12}.} \tag{665}$$

CCVII. 4. TRACE EXTRACTION FORMULAS

The same coefficients are extracted invariantly by

$$\boxed{\lambda_\parallel = \frac{1}{2}\,\mathrm{Tr}(\sigma_3 \widehat{M}_\parallel),} \tag{666}$$

$$\boxed{\alpha = \frac{1}{4}\,\mathrm{Tr}(\sigma_1 \widehat{M}_1 + \sigma_2 \widehat{M}_2), \qquad \beta = \frac{1}{4}\,\mathrm{Tr}(\sigma_2 \widehat{M}_1 - \sigma_1 \widehat{M}_2).} \tag{667}$$

CCVIII. 5. TRANSPORT COEFFICIENTS g_c, g_E, g_O

The exact normalization layer is

$$\lambda_\parallel = C_\parallel g_c, \qquad \alpha = C_E g_E, \qquad \beta = C_O g_O. \tag{668}$$

Therefore

$$\boxed{g_c = \frac{\lambda_\parallel}{C_\parallel}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.} \tag{669}$$

Substituting the explicit matrix entries,

$$\boxed{g_c = \frac{1}{C_\parallel}\,(\widehat{M}_\parallel)_{11},} \tag{670}$$

$$\boxed{g_E = \frac{1}{C_E}\,\Re(\widehat{M}_1)_{12}, \qquad g_O = -\frac{1}{C_O}\,\Im(\widehat{M}_1)_{12}.} \tag{671}$$

CCIX. 6. CORE COEFFICIENTS a_0, b_0, c_0

The one-defect theorem uses

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}. \tag{672}$$

Thus

$$\boxed{a_0 = \frac{1}{z_{0,0}}\left(\frac{\xi_E}{C_E}\alpha + \frac{\xi_c}{C_\parallel}\lambda_\parallel\right),} \tag{673}$$

$$\boxed{b_0 = \frac{1}{z_{0,0}}\frac{\xi_E}{C_E}\alpha, \qquad c_0 = \frac{1}{z_{0,0}}\frac{\xi_O}{C_O}\beta.} \tag{674}$$

CCX. 7. CANONICAL ONE-DEFECT CRITERION IN THE TWO-STATE SLOT

The sufficient one-defect/one-mode criterion is

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}. \tag{675}$$

Substituting a_0, b_0,

$$\boxed{\mu_\infty > 0,} \tag{676}$$

$$\boxed{\frac{\xi_E}{C_E}\alpha + \frac{\xi_c}{C_\parallel}\lambda_\parallel > 0,} \tag{677}$$

$$\boxed{\alpha > 0 \quad \text{if } \frac{\xi_E}{C_E} > 0,} \tag{678}$$

and

$$\boxed{2|\kappa_*|\frac{\xi_E}{C_E}\alpha > \frac{\xi_E}{C_E}\alpha + \frac{\xi_c}{C_\parallel}\lambda_\parallel.} \tag{679}$$

In the canonical lowest-channel case

$$|\kappa_*| = 1, \tag{680}$$

this becomes

$$\boxed{\frac{\xi_E}{C_E}\alpha > \frac{\xi_c}{C_\parallel}\lambda_\parallel.} \tag{681}$$

This is the sharp reduced inequality: the transverse coefficient must dominate the longitudinal coefficient after matching/normalization.

CCXI. 8. SIMPLEST TOY EXAMPLE

Consider the minimal canonical toy matrices

$$\boxed{\widehat{M}_\parallel = \begin{pmatrix} \delta_c & 0 \\ 0 & -\delta_c \end{pmatrix}, \qquad \widehat{M}_1 = \begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix}, \qquad \widehat{M}_2 = \begin{pmatrix} 0 & -it \\ it & 0 \end{pmatrix},} \tag{682}$$

with

$$\delta_c > 0, \qquad t > 0. \tag{683}$$

Then

$$\boxed{\lambda_\parallel = \delta_c, \qquad \alpha = t, \qquad \beta = 0.} \tag{684}$$

Hence

$$\boxed{g_c = \frac{\delta_c}{C_\parallel}, \qquad g_E = \frac{t}{C_E}, \qquad g_O = 0.} \tag{685}$$

The core coefficients are

$$\boxed{a_0 = \frac{1}{z_{0,0}}\left(\frac{\xi_E}{C_E}t + \frac{\xi_c}{C_\parallel}\delta_c\right),} \tag{686}$$

$$\boxed{b_0 = \frac{1}{z_{0,0}}\frac{\xi_E}{C_E}t.} \tag{687}$$

For $|\kappa_*| = 1$, the reduced acceptance window is

$$\boxed{\mu_\infty > 0, \qquad t > 0, \qquad \frac{\xi_E}{C_E}t + \frac{\xi_c}{C_\parallel}\delta_c > 0, \qquad \frac{\xi_E}{C_E}t > \frac{\xi_c}{C_\parallel}\delta_c.} \tag{688}$$

So the simplest toy lesson is:

$$\boxed{\text{one-defect acceptance requires the transverse slot splitting } t \text{ to dominate the longitudinal slot splitting } \delta_c.} \tag{689}$$

CCXII. 9. INTERPRETATION

This canonical toy extraction is not yet the microscopic answer. Its role is to make the one-defect theorem completely concrete: once the reduced 2×2 derivative blocks are known, the transport coefficients and the acceptance inequalities are immediate.

Thus the remaining open work is not algebraic. It is the actual microscopic production of the matrices

$$M_\parallel, \qquad M_1, \qquad M_2 \tag{690}$$

on the intended Class II one-defect branch.

CCXIII. FINAL SUMMARY

In a two-state low slot, after fixing the internal $SU(2)$ gauge, the canonical reduced derivative blocks can be written as

$$\widehat{M_\parallel} = \lambda_\parallel \sigma_3, \qquad \widehat{M_1} = \alpha\sigma_1 + \beta\sigma_2, \qquad \widehat{M_2} = \alpha\sigma_2 - \beta\sigma_1.$$

Equivalently,

$$\widehat{M_\parallel} = \begin{pmatrix} \lambda_\parallel & 0 \\ 0 & -\lambda_\parallel \end{pmatrix}, \quad \widehat{M_1} = \begin{pmatrix} 0 & \alpha - i\beta \\ \alpha + i\beta & 0 \end{pmatrix}, \quad \widehat{M_2} = \begin{pmatrix} 0 & -\beta - i\alpha \\ -\beta + i\alpha & 0 \end{pmatrix}.$$

Thus the microscopic transport coefficients are

$$g_c = \frac{\lambda_\parallel}{C_\parallel}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.$$

The core coefficients entering the one-defect theorem are

$$a_0 = \frac{1}{z_{0,0}} \left(\frac{\xi_E}{C_E}\alpha + \frac{\xi_c}{C_\parallel}\lambda_\parallel \right), \qquad b_0 = \frac{1}{z_{0,0}} \frac{\xi_E}{C_E}\alpha.$$

Therefore, in the canonical lowest-channel case $|\kappa_*| = 1$, the one-defect criterion reduces to

$$\mu_\infty > 0, \qquad \frac{\xi_E}{C_E}\alpha + \frac{\xi_c}{C_\parallel}\lambda_\parallel > 0, \qquad \frac{\xi_E}{C_E}\alpha > \frac{\xi_c}{C_\parallel}\lambda_\parallel.$$

In the simplest toy example with

$$\widehat{M_\parallel} = \begin{pmatrix} \delta_c & 0 \\ 0 & -\delta_c \end{pmatrix}, \qquad \widehat{M_1} = \begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix}, \qquad \widehat{M_2} = \begin{pmatrix} 0 & -it \\ it & 0 \end{pmatrix},$$

one gets

$$\lambda_\parallel = \delta_c, \qquad \alpha = t, \qquad \beta = 0,$$

so one-defect acceptance requires the transverse coefficient t to dominate the longitudinal coefficient δ_c after normalization and matching.

CCXIV. C4 STATUS

At the present stage, the work is indeed inside the C4 sector:

$$\boxed{C4 \supset K1: \text{ common family subspace} \supset K1A1: \text{ single-defect seed localization.}} \tag{691}$$

So the current program is not yet the closure of the full C4 sector, but the closure of its first real microscopic gate. The next decisive checkpoint is:

$$\boxed{\text{actual/quasi-actual evaluation of } (g_E, g_c, \mu_\infty, \kappa_*) \text{ and the one-defect acceptance criterion.}} \tag{692}$$

If this passes, then the C4 matter/family program moves from a broad conditional framework to a much stronger microscopic stage. If it fails, the branch choice or the carrier-seed mechanism must be revised.

Thus the honest diagnosis is:

$$\boxed{\text{Yes, we are doing C4, but still in its early-to-middle microscopic closure stage, not at the final closure yet.}} \tag{693}$$

CCXV. MICROSCOPIC MATRIX-ELEMENT ROUTE TO $M_\parallel, M_1, M_2$ ON THE INTENDED CLASS II ONE-DEFECT BRANCH

CCXVI. 1. GOAL

The actual one-defect acceptance theorem is controlled by

$$g_c, \qquad g_E, \qquad g_O, \qquad \mu_\infty, \qquad \kappa_*. \tag{694}$$

The finite low-slot extraction formulas are already fixed:

$$M_i = P_* \, \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} \, P_*, \tag{695}$$

followed by the Pauli decomposition

$$M_\parallel = \lambda_\parallel \sigma_3, \qquad M_1 = \alpha\sigma_1 + \beta\sigma_2, \qquad M_2 = \alpha\sigma_2 - \beta\sigma_1. \tag{696}$$

But this still leaves the real microscopic question:

$$\boxed{\text{Which Hessian block produces } \lambda_c,\ \Lambda_E,\ \Lambda_O \text{ on the intended Class II one-defect branch?}} \tag{697}$$

CCXVII. 2. BACKGROUND-DEPENDENT HESSIAN BLOCK SPLIT

Let the full quadratic operator around the one-defect background $\bar{\Phi}_{\min}$ be written in block form

$$\boxed{\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}.} \tag{698}$$

Expand each block in tangential derivatives:

$$\boxed{\mathcal{H}_{AB} = \mathcal{H}^{(0)}_{AB}(\xi) - i\,\mathcal{V}^\mu_{AB}(\xi)\,\partial_{y^\mu} + \mathcal{W}^{\mu\nu}_{AB}(\xi)\,\partial_{y^\mu}\partial_{y^\nu} + \cdots,} \tag{699}$$

where only $\mathcal{V}^\mu_{AB}$ contributes to the principal first-order transport symbol.

Restrict to the tangent bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp, \qquad \dim_{\mathbb{C}} Q_B = 2, \tag{700}$$

and define the Q_B-endomorphism-valued block kernels

$$\boxed{\mathcal{M}^\mu_{AB}(\xi) := \Pi_Q\, \mathcal{V}^\mu_{AB}(\xi)\, \Pi_Q.} \tag{701}$$

CCXVIII. 3. CANONICAL PROJECTOR FUNCTIONALS

Let Σ_a be the Pauli basis on Q_B, and let $(\hat{n}, e_1, e_2)$ be the shell-adapted frame. Then the invariant one-defect projector functionals are

$$\boxed{\mathfrak{P}_\parallel[\mathcal{M}] := \frac{1}{2}\operatorname{Tr}_{Q_B}\big(\Sigma_3\, n_\mu \mathcal{M}^\mu\big),} \tag{702}$$

$$\boxed{\mathfrak{P}_E[\mathcal{M}] := \frac{1}{4} E^\mu{}_a \operatorname{Tr}_{Q_B}\big(\Sigma_a\, \mathcal{M}_\mu\big),} \tag{703}$$

$$\boxed{\mathfrak{P}_O[\mathcal{M}] := \frac{1}{4} J^\mu{}_a \operatorname{Tr}_{Q_B}\big(\Sigma_a\, \mathcal{M}_\mu\big).} \tag{704}$$

These extract, respectively,

$$\text{longitudinal singlet}, \qquad \text{even transverse doublet}, \qquad \text{odd transverse doublet}. \tag{705}$$

CCXIX. 4. LONGITUDINAL EXTRACTION FROM THE OO-BLOCK

Define the longitudinal candidate block

$$\boxed{V^{\mu}_{OO}(\xi) := \Pi_Q\, \mathcal{V}^{\mu}_{OO}(\xi)\, \Pi_Q.} \tag{706}$$

Its scalar longitudinal projection is

$$\boxed{a_{3;OO}(\xi) = \mathfrak{P}_{\|}[V_{OO}] = \frac{1}{2}\,\mathrm{Tr}_{Q_B}\big(\Sigma_3\, n_\mu V^{\mu}_{OO}(\xi)\big).} \tag{707}$$

Because of the oddness/parity filter, its branch-sensitive expansion has the form

$$a_{3;OO}(\xi) = \ell^{(1)}_{0;OO}\,\theta'(\xi) + \ell^{(3)}_{0;OO}\,\theta'(\xi)^3 + O\big(\theta'(\xi)\theta(\xi)^2, \theta'(\xi)^5\big). \tag{708}$$

Thus the exact coefficient identification is

$$\boxed{\Lambda^{(0)}_{\|;OO} = \ell^{(1)}_{0;OO}, \qquad \frac{\lambda_{c;OO}}{2} = \ell^{(3)}_{0;OO}.} \tag{709}$$

On the intended concrete Class II-compatible branch one has

$$\boxed{\Lambda^{(0)}_{\|} = 0,} \tag{710}$$

so the practical longitudinal coefficient is

$$\boxed{\lambda_c = \lambda_{c;OO} = 2\,\ell^{(3)}_{0;OO} = 2\lim_{\theta'\to 0}\frac{\mathfrak{P}_{\|}[V_{OO}]}{\theta'(\xi)^3}.} \tag{711}$$

Hence the microscopic source of $M_{\|}$ is the OO-block alone at leading order in the current branch.

CCXX. 5. EVEN/ODD TRANSVERSE EXTRACTION FROM THE $TD + DT$-BLOCK

Define the mixed first-order transverse carrier

$$\boxed{\mathcal{M}^{\mu}_{TD+DT} := \Pi_Q\big(\mathcal{V}^{\mu}_{TD} + \mathcal{V}^{\mu}_{DT}\big)\Pi_Q.} \tag{712}$$

Then the leading transverse coefficient point is extracted by

$$\boxed{\Lambda_E = \mathfrak{P}_E[\mathcal{M}^{\mathrm{inv}}_{TD+DT}], \qquad \Lambda_O = \mathfrak{P}_O[\mathcal{M}^{\mathrm{inv}}_{TD+DT}],} \tag{713}$$

at the minimal mixed-block level. More generally, beyond the minimal truncation,

$$\Lambda_E = \sum_{AB\in\{TD,DT,OD,DO,DD\}} e^{(1)}_{0;AB}, \qquad \Lambda_O = \sum_{AB\in\{TD,DT,OD,DO,DD\}} o^{(1)}_{0;AB}, \tag{714}$$

but the sharp leading audit starts from $TD + DT$.

Thus the microscopic source of M_1, M_2 is the $TD + DT$ channel at leading order.

CCXXI. 6. DICTIONARY TO THE REDUCED SHELL COEFFICIENTS

Once $\lambda_c, \Lambda_E, \Lambda_O$ are extracted from the one-defect Hessian blocks, the reduced shell coefficients are

$$\boxed{\lambda_{\|} = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.} \tag{715}$$

Hence

$$\boxed{M_{\parallel} = \lambda_{\parallel}\sigma_3, \qquad M_1 = \alpha\sigma_1 + \beta\sigma_2, \qquad M_2 = \alpha\sigma_2 - \beta\sigma_1.} \tag{716}$$

Using the normalization layer,

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.} \tag{717}$$

Therefore the full microscopic chain is

$$\boxed{V_{OO} \xrightarrow{\mathfrak{P}_{\parallel}} \lambda_c \longrightarrow \lambda_{\parallel} \longrightarrow g_c,} \tag{718}$$

$$\boxed{\mathcal{M}_{TD+DT} \xrightarrow{\mathfrak{P}_E, \mathfrak{P}_O} (\Lambda_E, \Lambda_O) \longrightarrow (\alpha, \beta) \longrightarrow (g_E, g_O).} \tag{719}$$

CCXXII. 7. CONNECTION TO THE ONE-DEFECT THEOREM

The one-defect radial data are

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}, \tag{720}$$

with bulk mass

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \tag{721}$$

So the sufficient one-defect/one-mode acceptance condition is

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}, \tag{722}$$

together with competitor exclusion and the remote-gap condition

$$\Delta^{\mathrm{full}}_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho}. \tag{723}$$

Thus the only truly remaining microscopic work is the explicit evaluation of the blockwise first-order kernels:

$$\boxed{V^{\mu}_{OO}(\xi), \qquad \mathcal{V}^{\mu}_{TD}(\xi), \qquad \mathcal{V}^{\mu}_{DT}(\xi),} \tag{724}$$

and then the projector functionals $\mathfrak{P}_{\parallel}, \mathfrak{P}_E, \mathfrak{P}_O$.

CCXXIII. 8. PRACTICAL FINITE PROTOCOL

The practical extraction protocol is now completely fixed:

1. Construct the one-defect background $\bar{\Phi}_{\min}$.
2. Compute the first-order Hessian kernels $\mathcal{V}^{\mu}_{AB}(\xi)$.
3. Restrict them to Q_B to obtain $\mathcal{M}^{\mu}_{AB}(\xi)$.
4. Evaluate

$$\lambda_c = 2 \lim_{\theta' \to 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}.$$

5. Evaluate

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}^{\mathrm{inv}}_{TD+DT}], \qquad \Lambda_O = \mathfrak{P}_O[\mathcal{M}^{\mathrm{inv}}_{TD+DT}].$$

6. Convert to

$$\lambda_\parallel, \alpha, \beta$$

   using the overlap integrals $I_{\theta'^3}, I_\theta$.

7. Divide by $C_\parallel, C_E, C_O$ to obtain g_c, g_E, g_O.

8. Insert them into $a_0, b_0, c_0, \mu_\infty$, together with κ_*, and test the one-defect inequalities.

CCXXIV. 9. HONEST FINAL DIAGNOSIS

The algebraic dictionary is now closed. The remaining frontier is not conceptual but explicit: evaluate the actual first-order Hessian kernels in the intended Class II one-defect branch and apply the projector functionals.

CCXXV. FINAL SUMMARY

On the intended Class II one-defect branch, the longitudinal and transverse low-slot matrices should be computed from different microscopic Hessian blocks.

The longitudinal coefficient is read from the OO-block:

$$V^\mu_{OO} = \Pi_Q\, \mathcal{V}^\mu_{OO}\, \Pi_Q, \qquad \lambda_c = 2 \lim_{\theta' \to 0} \frac{\mathfrak{P}_\parallel[V_{OO}]}{\theta'(\xi)^3}.$$

The transverse even/odd coefficients are read from the mixed $TD + DT$-block:

$$\mathcal{M}^\mu_{TD+DT} = \Pi_Q(\mathcal{V}^\mu_{TD} + \mathcal{V}^\mu_{DT})\Pi_Q,$$

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}^{\mathrm{inv}}_{TD+DT}], \qquad \Lambda_O = \mathfrak{P}_O[\mathcal{M}^{\mathrm{inv}}_{TD+DT}].$$

These then determine the reduced shell coefficients through

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O,$$

and finally

$$g_c = \frac{\lambda_\parallel}{C_\parallel}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.$$

Thus the full microscopic chain is

$$\mathcal{H}_{OO} \to \lambda_c \to \lambda_\parallel \to g_c, \qquad \mathcal{H}_{TD}, \mathcal{H}_{DT} \to (\Lambda_E, \Lambda_O) \to (\alpha, \beta) \to (g_E, g_O).$$

Therefore the remaining frontier is no longer algebraic. It is the explicit evaluation of the first-order one-defect Hessian kernels

$$\mathcal{V}^\mu_{OO}, \qquad \mathcal{V}^\mu_{TD}, \qquad \mathcal{V}^\mu_{DT}$$

on the intended Class II branch.

CCXXVI. SOURCE-BY-SOURCE MICROSCOPIC DECOMPOSITION OF THE REMAINING ONE-DEFECT COEFFICIENT POINT

CCXXVII. 1. GOAL

The remaining microscopic coefficient point on the intended Class II one-defect branch is

$$\boxed{\lambda_c, \qquad \Lambda_E, \qquad \Lambda_O.} \tag{725}$$

The purpose of this step is to identify, source by source, which microscopic quadratic sector carries each of these coefficients.

CCXXVIII. 2. EXACT CURRENT WORKING-BRANCH SCHUR SPLIT

After integrating out the heavy mediator, the quadratic correction is

$$\boxed{\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2}\,\delta J\delta J^\dagger - \frac{\alpha_X\beta_X}{M_X^2}\left(\delta J\delta K^\dagger + \delta K\delta J^\dagger\right) - \frac{\beta_X^2}{M_X^2}\,\delta K\delta K^\dagger.} \tag{726}$$

This already suggests a source decomposition into the sectors

$$JJ, \qquad JK + KJ, \qquad KK.$$

CCXXIX. 3. LONGITUDINAL SOURCE CLASSIFICATION

The pure JJ-piece is axial only:

$$\boxed{\lambda_\parallel^{(JJ)} \neq 0 \text{ is allowed}, \qquad \alpha^{(JJ)} = \beta^{(JJ)} = 0.} \tag{727}$$

Hence the pure JJ-sector is the unique principal longitudinal source.
Equivalently, in the shell-source classification,

$$\boxed{c_\parallel^{(K)} = 0, \qquad c_\parallel^{(JK)} = 0, \qquad c_\parallel^{(KK)} = 0, \qquad c_\parallel^{(JJ)} \text{ allowed}.} \tag{728}$$

Therefore the primitive odd longitudinal singlet coefficient is localized to the JJ-type sector:

$$\boxed{\lambda_c \sim \lambda_c^{(JJ)}.} \tag{729}$$

CCXXX. 4. TRANSVERSE SOURCE CLASSIFICATION

The mixed $JK + KJ$-sector feeds only the transverse family channels:

$$\boxed{JK + KJ \Longrightarrow (c_E, c_O) \text{ only}, \qquad c_\parallel^{(JK)} = 0 \text{ at principal shell order}.} \tag{730}$$

Hence the leading transverse coefficients are sourced by the mixed term:

$$\boxed{\Lambda_E, \Lambda_O \sim (JK + KJ)\text{-sector}.} \tag{731}$$

At the current refined locked + Class II branch,

$$\boxed{c_E = \frac{\alpha_X\beta_X}{M_X^2}(j\cdot k)\Lambda_E^{(0)}, \qquad c_O = \frac{\alpha_X\beta_X}{M_X^2}(j\cdot k)\Lambda_O^{(0)}.} \tag{732}$$

CCXXXI. 5. ROLE OF THE KK-SECTOR

The pure KK-sector does not generate the primitive longitudinal slot:

$$\boxed{c_{\parallel}^{(KK)} = 0 \quad \text{at principal longitudinal shell order.}} \tag{733}$$

Its role is instead residual and transverse-family:

$$\boxed{KK \to \text{same-shell residual / transverse correction sector.}} \tag{734}$$

Thus the principal source hierarchy is

$$\boxed{JJ \to \lambda_c, \qquad JK + KJ \to (\Lambda_E, \Lambda_O), \qquad KK \to \text{corrections.}} \tag{735}$$

CCXXXII. 6. BLOCKWISE HESSIAN TRANSLATION

In the explicit blockwise extraction protocol, the same classification is written as

$$\boxed{\mathcal{H}_{OO} \to \mathcal{V}_{OO}^{\mu} \to a_{3;OO}(\xi) = \frac{1}{2}\,\mathrm{Tr}_{Q_B}\big(\Sigma_3 n_\mu V_{OO}^{\mu}(\xi)\big) \to \lambda_c,} \tag{736}$$

and

$$\boxed{\mathcal{H}_{TD}, \mathcal{H}_{DT} \to \mathcal{M}_{TD+DT}^{\mu} \to e_0^{(1)}, o_0^{(1)} \to \Lambda_E, \Lambda_O.} \tag{737}$$

Thus the source classification and the block classification are consistent:

$$\boxed{JJ \leftrightarrow OO\text{-longitudinal audit,} \qquad JK + KJ \leftrightarrow TD + DT\text{-transverse audit.}} \tag{738}$$

CCXXXIII. 7. REDUCED COEFFICIENT DICTIONARY

After overlap insertion,

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.} \tag{739}$$

Hence

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.} \tag{740}$$

So the complete current one-defect extraction chain is:

$$\boxed{JJ \to \lambda_c \to \lambda_{\parallel} \to g_c, \qquad JK + KJ \to (\Lambda_E, \Lambda_O) \to (\alpha, \beta) \to (g_E, g_O).} \tag{741}$$

CCXXXIV. 8. HONEST CAVEAT

This source-by-source classification is fully closed on the current refined locked + Class II working branch. What is still not yet unconditional is the final source-by-source differentiation of the complete microscopic TECT Hessian that yields the exact numerical coefficient functions themselves.

Thus the remaining frontier is not conceptual but explicit:

$$\boxed{\text{literally differentiate the full one-defect Hessian block-by-block and evaluate the corresponding projector traces.}} \tag{742}$$

CCXXXV. 9. IMMEDIATE NEXT COMPUTATION

The next explicit microscopic steps are therefore:

1. evaluate the JJ-driven OO-block kernel V^{μ}_{OO} and extract λ_c;
2. evaluate the $JK+KJ$-driven $TD+DT$-block kernel $\mathcal{M}^{\mu}_{TD+DT}$ and extract Λ_E, Λ_O;
3. evaluate the KK-sector only as a same-shell residual / competitor correction, not as the principal carrier source.

CCXXXVI. FINAL SUMMARY

The intended Class II one-defect source classification is:

$$\boxed{JJ \to \lambda_c, \qquad JK+KJ \to (\Lambda_E, \Lambda_O), \qquad KK \to \text{transverse residual corrections.}}$$

Equivalently, in blockwise Hessian form:

$$\boxed{\mathcal{H}_{OO} \to \lambda_c, \qquad \mathcal{H}_{TD}, \mathcal{H}_{DT} \to \Lambda_E, \Lambda_O.}$$

The pure JJ-sector is the unique principal longitudinal source, while the mixed $JK+KJ$-sector is the principal transverse source. The KK-sector does not generate the primitive longitudinal slot and should be treated as a correction/residual sector.

After overlap insertion and normalization,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O,$$

and therefore

$$g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.$$

Thus the remaining microscopic task is explicit but finite: compute the OO-block kernel from the JJ-source, compute the $TD+DT$-block kernel from the $JK+KJ$-source, and treat KK only as a correction sector.

CCXXXVII. EXPLICIT LOWEST-DERIVATIVE LOCAL FORM OF THE PURE JJ LONGITUDINAL SOURCE

CCXXXVIII. 1. GOAL

The remaining longitudinal microscopic task is to write the pure JJ-source contribution to the OO-block transport kernel in an explicit local tensor form, and then reduce it to a finite scalar audit.

CCXXXIX. 2. PURE JJ SOURCE AS A LOCAL LOWEST-DERIVATIVE TENSOR

At principal shell order, the background current has the local form

$$\boxed{J_i^a[\phi_0, P_0] = m^a(P_0)\, \partial_i \phi_0.} \tag{743}$$

The Bloch derivative insertion contributes only the spatial tensor δ_{ij} at leading shell order. Therefore the axial pure-JJ coefficient field must take the lowest-derivative local form

$$\boxed{\mathcal{A}_j^{(JJ)}(x) = \sum_m C^{(JJ)}_{jm}\, \partial_m \phi_0(x).} \tag{744}$$

Since no principal spatial tensor survives the shell restriction except δ_{jm}, the coefficient tensor collapses to

$$\boxed{C^{(JJ)}_{jm} = \Xi_{JJ}\,\delta_{jm},} \tag{745}$$

hence

$$\boxed{\mathcal{A}^{(JJ)}_j(x) = \Xi_{JJ}\,\partial_j\phi_0(x).} \tag{746}$$

CCXL. 3. FIRST-SHELL FOURIER COEFFICIENT

Let A_{G_*} be the first-shell Fourier amplitude of ϕ_0. Then

$$\boxed{\widehat{\mathcal{A}}^{(JJ)}_j(G_*) = i\,C^{(JJ)}_{jm}G^m_*\,A_{G_*} = i\,\Xi_{JJ}\,G_{*j}\,A_{G_*}.} \tag{747}$$

Contracting with the shell axis $\hat{n}$,

$$\boxed{\widehat{\mathcal{A}}_{JJ}(G_*) := \hat{n}^j\widehat{\mathcal{A}}^{(JJ)}_j(G_*) = i\,\Xi_{JJ}\,|G_*|\,A_{G_*}.} \tag{748}$$

Thus the pure JJ noncancellation problem reduces to the finite scalar test

$$\boxed{\Xi_{JJ} \neq 0, \qquad A_{G_*} \neq 0.} \tag{749}$$

CCXLI. 4. EMBEDDING INTO THE OO-BLOCK LONGITUDINAL KERNEL

The OO-block longitudinal projector is

$$\boxed{a_{3;OO}(\xi) = \frac{1}{2}\,\mathrm{Tr}_{Q_B}\big(\Sigma_3\,n_\mu\,V^\mu_{OO}(\xi)\big).} \tag{750}$$

Its branch-sensitive expansion is

$$a_{3;OO}(\xi) = \ell^{(1)}_{0;OO}\theta'(\xi) + \ell^{(3)}_{0;OO}\theta'(\xi)^3 + \cdots, \qquad \ell^{(1)}_{0;OO} = 0, \qquad \ell^{(3)}_{0;OO} = \frac{\lambda_c}{2}. \tag{751}$$

Therefore the minimal executable pure-JJ ansatz for the OO-block longitudinal transport kernel is

$$\boxed{V^\mu_{OO,JJ}(\xi) = \frac{\lambda^{(JJ)}_c}{2}\,\theta'(\xi)^3\,n^\mu\Sigma_3 + V^{\mu,\mathrm{rem}}_{OO,JJ}(\xi),} \tag{752}$$

with

$$\mathfrak{P}_\parallel[V^{\mathrm{rem}}_{OO,JJ}] = 0 \qquad \text{at principal shell order.} \tag{753}$$

Equivalently,

$$\boxed{\lambda^{(JJ)}_c = 2\lim_{\theta'\to 0}\frac{\mathfrak{P}_\parallel[V_{OO,JJ}]}{\theta'(\xi)^3}.} \tag{754}$$

CCXLII. 5. FINITE SCALAR REDUCTION

In the current canonical refined locked + Class II working branch, the pure JJ tensor reduces to a finite scalar package:

$$\boxed{\lambda^{(JJ)}_c \sim \frac{\alpha_X^2}{M_X^2}\,\mathcal{N}_{JJ}\,m_\parallel\,u_\parallel\,|G_*|\,A_{G_*}.} \tag{755}$$

Here

$$m_\parallel \quad \text{is the background current scalar,} \tag{756}$$

$$u_\parallel = \frac{1}{2}\,\mathrm{Tr}(\Sigma_3 U_\parallel), \qquad U_\parallel := \hat{n}^j \mathcal{U}_j(0), \tag{757}$$

$$A_{G_*} \quad \text{is the first-shell Fourier amplitude,} \tag{758}$$

and

$$\mathcal{N}_{JJ} \quad \text{is the remaining projector/internal normalization scalar.} \tag{759}$$

Thus the longitudinal frontier is no longer a tensor-classification problem. It is a finite scalar audit:

$$\boxed{\mathcal{N}_{JJ}, \qquad m_\parallel, \qquad u_\parallel, \qquad A_{G_*}.} \tag{760}$$

CCXLIII. 6. REDUCED LONGITUDINAL COEFFICIENT

Once $\lambda_c = \lambda_c^{(JJ)}$ is extracted, the Branch-B reduced coefficient is

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.} \tag{761}$$

Hence the one-defect longitudinal input g_c is

$$\boxed{g_c = \frac{\lambda_\parallel}{C_\parallel} = \frac{I_{\theta'^3}}{2C_\parallel}\lambda_c^{(JJ)}.} \tag{762}$$

CCXLIV. 7. IMMEDIATE NEXT TASK

Therefore the next actual microscopic computation is:

$$\boxed{m_\parallel, \qquad u_\parallel, \qquad A_{G_*}, \qquad \mathcal{N}_{JJ} \longrightarrow \lambda_c^{(JJ)} \longrightarrow \lambda_\parallel \longrightarrow g_c.} \tag{763}$$

No further structural theorem is needed for the longitudinal side.

CCXLV. FINAL SUMMARY

On the current refined locked + Class II working branch, the pure JJ longitudinal source takes the lowest-derivative local form

$$\mathcal{A}_j^{(JJ)}(x) = \Xi_{JJ}\,\partial_j \phi_0(x),$$

because the coefficient tensor collapses to

$$C_{jm}^{(JJ)} = \Xi_{JJ}\,\delta_{jm}.$$

Therefore the first-shell Fourier coefficient is

$$\widehat{\mathcal{A}}_j^{(JJ)}(G_*) = i\,\Xi_{JJ}\,G_{*j}\,A_{G_*},$$

and the shell-normal contraction is

$$\widehat{\mathcal{A}}_{JJ}(G_*) = i\,\Xi_{JJ}\,|G_*|\,A_{G_*}.$$

The OO-block longitudinal kernel may thus be written as

$$V^{\mu}_{OO,JJ}(\xi) = \frac{\lambda_c^{(JJ)}}{2}\,\theta'(\xi)^3\,n^\mu\Sigma_3 + V^{\mu,\mathrm{rem}}_{OO,JJ}(\xi),$$

with

$$\lambda_c^{(JJ)} = 2\lim_{\theta'\to 0}\frac{\mathfrak{P}_\parallel[V_{OO,JJ}]}{\theta'(\xi)^3}.$$

In the current canonical branch, the pure JJ longitudinal coefficient reduces to a finite scalar package:

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2}\,\mathcal{N}_{JJ}\,m_\parallel\,u_\parallel\,|G_*|\,A_{G_*}.$$

Thus the longitudinal frontier is now a finite scalar audit of

$$\mathcal{N}_{JJ}, \qquad m_\parallel, \qquad u_\parallel, \qquad A_{G_*},$$

after which

$$\lambda_\parallel = \frac{\lambda_c}{2}I_{\theta'^3}$$

and

$$g_c = \frac{\lambda_\parallel}{C_\parallel}$$

follow immediately.

CCXLVI. ACTUAL CANONICAL COMPUTATION OF $m_\parallel$ AND $u_\parallel$

CCXLVII. 1. STARTING POINT

The remaining longitudinal scalar is

$$\boxed{L_* = m_\parallel u_\parallel,} \tag{764}$$

with

$$\boxed{m_\parallel = m^3(P_0), \qquad u_\parallel = \frac{1}{2}\,\mathrm{Tr}(\Sigma_3\,\hat{n}^j\mathcal{U}_j(0)) = \frac{1}{2}\Big[(U_\parallel)_{11} - (U_\parallel)_{22}\Big].} \tag{765}$$

Thus longitudinal survival is reduced to the finite scalar test

$$m_\parallel \neq 0, \qquad u_\parallel \neq 0. \tag{766}$$

CCXLVIII. 2. CANONICAL LOCKED BASIS

Fix the canonical locked basis

$$z_0 = e_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad u_1 = e_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}, \qquad u_2 = e_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}. \tag{767}$$

For the standard Hermitian $SU(3)$ generators

$$T^a = \frac{\lambda^a}{2}, \tag{768}$$

define the shell-tangent couplings

$$t^a_{10} := u_1^\dagger T^a z_0 = (T^a)_{21}, \qquad t^a_{20} := u_2^\dagger T^a z_0 = (T^a)_{31}. \tag{769}$$

CCXLIX. 3. SHELL-NORMAL REDUCED MATRIX

In the minimal shell-diagonal truncation, let $\kappa_J \neq 0$ be the shell-normal complex prefactor from the actual Bragg/harmonic/derivative projection. Then the reduced shell-normal J-matrix is

$$\boxed{U_\parallel = \begin{pmatrix} 2\Re(\kappa_J t_{10}) & 0 \\ 0 & 2\Re(\kappa_J t_{20}) \end{pmatrix}.} \tag{770}$$

Therefore

$$\boxed{u_\parallel = \frac{1}{2}\Big[(U_\parallel)_{11} - (U_\parallel)_{22}\Big] = \Re\big[\kappa_J(t_{10} - t_{20})\big].} \tag{771}$$

CCL. 4. ACTIVE GENERATOR DIRECTIONS

Only T^1, T^2, T^4, T^5 couple the locked direction z_0 to the tangent doublet. Their couplings are:

$$T^1: \quad t_{10} = \frac{1}{2}, \qquad t_{20} = 0, \tag{772}$$

$$T^2: \quad t_{10} = \frac{i}{2}, \qquad t_{20} = 0, \tag{773}$$

$$T^4: \quad t_{10} = 0, \qquad t_{20} = \frac{1}{2}, \tag{774}$$

$$T^5: \quad t_{10} = 0, \qquad t_{20} = \frac{i}{2}. \tag{775}$$

For T^3, T^6, T^7, T^8, one has

$$t_{10} = t_{20} = 0, \tag{776}$$

so these directions do not generate a longitudinal J-split.

CCLI. 5. EXPLICIT SINGLE-GENERATOR FORMULAS

Case A: $T_\star = T^1$

$$\boxed{u_\parallel(T^1) = \frac{1}{2}\,\Re(\kappa_J),} \tag{777}$$

hence

$$\boxed{L_*(T^1) = \frac{m_\parallel}{2}\Re(\kappa_J),} \tag{778}$$

and

$$\boxed{\lambda_\parallel(T^1) = \frac{1}{4}\,\alpha_X\, m_\parallel\, \Re(\kappa_J)\, I_{\theta'^3}.} \tag{779}$$

Case B: $T_\star = T^2$

$$\boxed{u_\parallel(T^2) = -\frac{1}{2}\,\Im(\kappa_J),} \tag{780}$$

hence

$$\boxed{L_*(T^2) = -\frac{m_\parallel}{2}\Im(\kappa_J).} \tag{781}$$

Case C: $T_\star = T^4$

$$\boxed{u_\parallel(T^4) = -\frac{1}{2}\,\Re(\kappa_J),} \tag{782}$$

hence

$$\boxed{L_*(T^4) = -\frac{m_\parallel}{2}\Re(\kappa_J).} \tag{783}$$

Case D: $T_\star = T^5$

$$\boxed{u_\parallel(T^5) = \frac{1}{2}\,\Im(\kappa_J),} \tag{784}$$

hence

$$\boxed{L_*(T^5) = \frac{m_\parallel}{2}\Im(\kappa_J).} \tag{785}$$

CCLII. 6. GENERAL ACTIVE DIRECTION

Take

$$T_\star = c_1 T^1 + c_2 T^2 + c_4 T^4 + c_5 T^5, \qquad c_i \in \mathbb{R}. \tag{786}$$

Then

$$t_{10}(T_\star) = \frac{1}{2}(c_1 + ic_2), \qquad t_{20}(T_\star) = \frac{1}{2}(c_4 + ic_5), \tag{787}$$

so

$$\boxed{u_\parallel(T_\star) = \frac{1}{2}\Re\Big[\kappa_J\big((c_1 + ic_2) - (c_4 + ic_5)\big)\Big].} \tag{788}$$

Therefore the exact canonical vanishing condition is

$$\boxed{u_\parallel = 0 \iff \Re\Big[\kappa_J\big((c_1 + ic_2) - (c_4 + ic_5)\big)\Big] = 0.} \tag{789}$$

CCLIII. 7. PRACTICAL LONGITUDINAL VERDICT

The current longitudinal branch is completely reduced to the finite scalar data

$$m_\parallel, \qquad \kappa_J, \qquad T_\star, \qquad A_{G_*}. \tag{790}$$

Once

$$m_\parallel \neq 0, \qquad u_\parallel(T_\star) \neq 0, \qquad \alpha_X \neq 0, \qquad A_{G_*} \neq 0, \tag{791}$$

the longitudinal witness survives:

$$\boxed{\lambda_\parallel = \frac{1}{2}\,\alpha_X\, L_*\, I_{\theta'^3} \neq 0.} \tag{792}$$

CCLIV. 8. PREFERRED EXPLICIT BRANCH

The cleanest practical choice is

$$\boxed{T_\star = T^1, \qquad \Re(\kappa_J) \neq 0, \qquad m_\parallel \neq 0.} \tag{793}$$

Then

$$u_\parallel(T^1) = \frac{1}{2}\Re(\kappa_J) \tag{794}$$

and the branch survives immediately.

By contrast, cancellation directions such as

$$T_\star = \frac{1}{\sqrt{2}}(T^1 + T^4) \tag{795}$$

yield $t_{10} = t_{20}$ and therefore

$$u_\parallel = 0, \tag{796}$$

so the longitudinal seed dies.

CCLV. FINAL SUMMARY

In the canonical locked basis, the longitudinal scalar is

$$L_* = m_\parallel u_\parallel, \qquad m_\parallel = m^3(P_0), \qquad u_\parallel = \frac{1}{2}\Big[(U_\parallel)_{11} - (U_\parallel)_{22}\Big].$$

With the shell-normal complex prefactor κ_J, the reduced J-matrix is

$$U_\parallel = \begin{pmatrix} 2\Re(\kappa_J t_{10}) & 0 \\ 0 & 2\Re(\kappa_J t_{20}) \end{pmatrix},$$

so

$$\boxed{u_\parallel = \Re\big[\kappa_J(t_{10} - t_{20})\big].}$$

Only T^1, T^2, T^4, T^5 are active:

$$T^1: \; u_\parallel = \frac{1}{2}\Re(\kappa_J), \qquad T^2: \; u_\parallel = -\frac{1}{2}\Im(\kappa_J),$$

$$T^4:\ u_{\parallel}=-\frac{1}{2}\Re(\kappa_J),\qquad T^5:\ u_{\parallel}=\frac{1}{2}\Im(\kappa_J).$$

For a general active direction

$$T_\star=c_1T^1+c_2T^2+c_4T^4+c_5T^5,$$

one gets

$$\boxed{u_{\parallel}(T_\star)=\frac{1}{2}\Re\Big[\kappa_J\big((c_1+ic_2)-(c_4+ic_5)\big)\Big].}$$

Hence the exact canonical vanishing condition is

$$u_{\parallel}=0\iff\Re\Big[\kappa_J\big((c_1+ic_2)-(c_4+ic_5)\big)\Big]=0.$$

Thus the longitudinal frontier is fully reduced to the finite scalar audit of

$$m_{\parallel},\qquad \kappa_J,\qquad T_\star,\qquad A_{G_*}.$$

CCLVI. CANONICAL TRANSVERSE REDUCED MATRIX AUDIT AND THE COMPLETE FINITE ONE-DEFECT ACCEPTANCE DATA

CCLVII. 1. GOAL

The longitudinal side has already been reduced to a finite scalar audit through

$$\lambda_c\longrightarrow\lambda_{\parallel}\longrightarrow g_c. \tag{797}$$

We now perform the same reduction for the transverse sector and assemble the full one-defect acceptance data

$$\boxed{(g_c,\ g_E,\ g_O,\ \mu_\infty,\ \kappa_*).} \tag{798}$$

CCLVIII. 2. BRANCH-LEVEL TRANSVERSE TRANSPORT KERNEL

At the current reduced Class II working level, the leading transverse kernel is

$$\boxed{\mathcal{M}^\mu_{TD+DT}=\theta'(\xi)\Big[\Lambda_E\,E^\mu{}_a\Sigma_a+\Lambda_O\,J^\mu{}_a\Sigma_a\Big]+\mathcal{M}^\mu_{\mathrm{corr}},} \tag{799}$$

where $\mathcal{M}^\mu_{\mathrm{corr}}$ starts at higher order and/or receives contributions from

$$OD,\quad DO,\quad DD.$$

Thus the branch-level reduced transverse coefficient point is

$$\boxed{(\Lambda_E,\Lambda_O).} \tag{800}$$

CCLIX. 3. PROJECTION TO THE LOW SLOT

Let the even/odd low modes be

$$u_a^+(\xi)=f_+(\xi)e_a(\xi),\qquad u_a^-(\xi)=f_-(\xi)e_a(\xi),\qquad a=1,2. \tag{801}$$

The projected mixed symbol is

$$\boxed{(M^\mu)_{ab}=\int d\xi\,f_+(\xi)f_-(\xi)\,e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi).} \tag{802}$$

Inside the exact concrete Class II branch,

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.} \tag{803}$$

So the transverse reduced matrices are

$$\boxed{M_1 = \alpha\sigma_1 + \beta\sigma_2, \qquad M_2 = \alpha\sigma_2 - \beta\sigma_1.} \tag{804}$$

CCLX. 4. EXPLICIT CANONICAL 2×2 FORM OF THE TRANSVERSE BLOCKS

Using the Pauli matrices,

$$\boxed{M_1 = \begin{pmatrix} 0 & \alpha - i\beta \\ \alpha + i\beta & 0 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 0 & -\beta - i\alpha \\ -\beta + i\alpha & 0 \end{pmatrix}.} \tag{805}$$

Thus the transverse finite data are read off directly as

$$\boxed{\alpha = \Re(M_1)_{12}, \qquad \beta = -\Im(M_1)_{12}.} \tag{806}$$

Equivalently,

$$\boxed{\alpha = -\Im(M_2)_{12}, \qquad \beta = -\Re(M_2)_{12}.} \tag{807}$$

CCLXI. 5. NORMALIZED TRANSPORT COEFFICIENTS

The exact overlap normalization layer is

$$\boxed{\lambda_{\parallel} = C_c\, g_c, \qquad \alpha = C_E\, g_E, \qquad \beta = C_O\, g_O.} \tag{808}$$

Therefore

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_c}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.} \tag{809}$$

Combining with the branch-level reduced coefficients,

$$\boxed{g_c = \frac{I_{\theta'^3}}{2C_c}\,\lambda_c, \qquad g_E = \frac{I_\theta}{C_E}\,\Lambda_E, \qquad g_O = \frac{I_\theta}{C_O}\,\Lambda_O.} \tag{810}$$

CCLXII. 6. SOURCE-BY-SOURCE FINITE SCALAR PACKAGE FOR THE TRANSVERSE SECTOR

At the current branch level, the principal transverse source is $JK + KJ$. Therefore the transverse coefficients are reduced to the finite scalar package

$$\boxed{\Lambda_E \sim \mathcal{N}_{JK}\,(j\cdot k)\,\Lambda_E^{(0)}, \qquad \Lambda_O \sim \mathcal{N}_{JK}\,(j\cdot k)\,\Lambda_O^{(0)},} \tag{811}$$

with the overall mixed-source Schur prefactor included in $\mathcal{N}_{JK}$. Equivalently, one may write

$$\boxed{\Lambda_E = \frac{\alpha_X\beta_X}{M_X^2}\,\mathfrak{T}_E, \qquad \Lambda_O = \frac{\alpha_X\beta_X}{M_X^2}\,\mathfrak{T}_O,} \tag{812}$$

where $\mathfrak{T}_E, \mathfrak{T}_O$ are the remaining branch-specific finite scalar contractions.
Hence

$$\boxed{g_E = \frac{I_\theta}{C_E}\,\frac{\alpha_X\beta_X}{M_X^2}\,\mathfrak{T}_E, \qquad g_O = \frac{I_\theta}{C_O}\,\frac{\alpha_X\beta_X}{M_X^2}\,\mathfrak{T}_O.} \tag{813}$$

CCLXIII. 7. FULL FINITE ONE-DEFECT ACCEPTANCE DATA

The bulk mass parameter is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \tag{814}$$

The radial core coefficients are

$$\boxed{a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}.} \tag{815}$$

Thus the full one-defect acceptance data are

$$\boxed{(g_c,\ g_E,\ g_O,\ \mu_\infty,\ \kappa_*).} \tag{816}$$

The sufficient one-defect/one-mode criterion is

$$\boxed{\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},} \tag{817}$$

with competitor exclusion and remote-gap isolation.

CCLXIV. 8. CANONICAL LOWEST-CHANNEL REDUCTION

In the practically most relevant lowest-channel case

$$|\kappa_*| = 1, \tag{818}$$

the reduced inequality becomes

$$\boxed{\xi_E g_E > \xi_c g_c.} \tag{819}$$

Indeed,

$$\frac{|\kappa_*| b_0}{a_0} > \frac{1}{2} \iff 2\xi_E g_E > \xi_E g_E + \xi_c g_c \iff \xi_E g_E > \xi_c g_c. \tag{820}$$

Thus the complete practical verdict is:

$$\boxed{\text{bulk positivity } (\mu_\infty > 0) \quad + \quad \text{transverse dominance } (\xi_E g_E > \xi_c g_c) \quad + \quad \text{competitor exclusion.}} \tag{821}$$

CCLXV. 9. CANONICAL EXPLICIT BRANCH

For the explicit canonical branch

$$\boxed{T_\star = T^1, \qquad \gamma_E = 1, \qquad \gamma_O = 0,} \tag{822}$$

the already-extracted working formulas are

$$\boxed{g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \tag{823}$$

$$\boxed{g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \qquad g_O = 0.} \tag{824}$$

Hence

$$\boxed{b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa,} \tag{825}$$

$$\boxed{a_0 = b_0(1 + R_c), \qquad R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.} \tag{826}$$

Therefore, for $\kappa_* = -1$, the sufficient one-defect criterion reduces to

$$\boxed{-1 < R_c < 1, \qquad \alpha > 0, \qquad \mu_\infty > 0, \qquad \Delta^{\text{full}}_{\text{bench},\rho} > \eta_{\text{rem},\rho}.} \tag{827}$$

CCLXVI. 10. HONEST STATUS

What is now closed is the entire finite dictionary:

$$\lambda_c \to g_c, \qquad (\Lambda_E, \Lambda_O) \to (g_E, g_O), \qquad (g_c, g_E, g_O, \mu_\infty, \kappa_*) \to \text{one-defect criterion}.$$

What remains open is not the algebraic reduction. The remaining frontier is the actual microscopic evaluation of

$$\lambda_c, \qquad \Lambda_E, \qquad \Lambda_O, \qquad \mu_\infty, \tag{828}$$

or, equivalently, of the finite scalar packages

$$m_\parallel, \qquad u_\parallel, \qquad A_{G_*}, \qquad \mathfrak{T}_E, \qquad \mathfrak{T}_O, \qquad \mu_\infty. \tag{829}$$

CCLXVII. FINAL SUMMARY

The one-defect acceptance data are now completely reduced to the finite set

$$(g_c,\ g_E,\ g_O,\ \mu_\infty,\ \kappa_*).$$

The longitudinal coefficient is

$$g_c = \frac{I_{\theta'^3}}{2C_c} \lambda_c,$$

while the transverse coefficients are

$$g_E = \frac{I_\theta}{C_E} \Lambda_E, \qquad g_O = \frac{I_\theta}{C_O} \Lambda_O.$$

In the canonical 2×2 reduced slot,

$$M_1 = \begin{pmatrix} 0 & \alpha - i\beta \\ \alpha + i\beta & 0 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 0 & -\beta - i\alpha \\ -\beta + i\alpha & 0 \end{pmatrix},$$

so

$$\alpha = \Re(M_1)_{12}, \qquad \beta = -\Im(M_1)_{12}.$$

The radial core coefficients are

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}},$$

and the bulk mass is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Thus the sufficient one-defect/one-mode condition is

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},$$

with competitor exclusion.

For the canonical lowest-channel case $|\kappa_*| = 1$, this reduces to the sharp inequality

$$\boxed{\xi_E g_E > \xi_c g_c.}$$

Therefore the entire one-defect problem is now a finite audit of the microscopic scalar packages that determine

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O, \quad \mu_\infty.$$

CCLXVIII. PRIORITY LIST FOR THE FINITE ONE-DEFECT AUDIT

CCLXIX. 1. CURRENT TARGET

The one-defect acceptance theorem is already reduced to the finite data

$$\boxed{(g_c,\ g_E,\ g_O,\ \mu_\infty,\ \kappa_*),} \tag{830}$$

with the sufficient criterion

$$\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}, \tag{831}$$

plus competitor exclusion.

The remaining task is therefore not structural but organizational: which finite scalar packages should be audited first?

CCLXX. 2. PRIORITY ORDERING

The correct computational priority order is

$$\boxed{\mu_\infty \longrightarrow g_E \longrightarrow g_c \longrightarrow \kappa_* \longrightarrow \text{competitor exclusion.}} \tag{832}$$

CCLXXI. 3. PRIORITY 1: BULK POSITIVITY

The first test is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)} > 0.} \tag{833}$$

In the vacuum asymptotic branch $\chi_\infty = 0$, this reduces to

$$\boxed{m(\sigma_\infty) > 0.} \tag{834}$$

This is the first priority because if $\mu_\infty \le 0$, the one-defect mode is dead before any transport audit begins.

CCLXXII. 4. PRIORITY 2: TRANSVERSE SUPPORT

The second test is the sign and size of the transverse coefficient

$$\boxed{g_E = \frac{I_\theta}{C_E}\Lambda_E, \qquad \Lambda_E = \frac{\alpha_X \beta_X}{M_X^2}\mathfrak{T}_E.} \tag{835}$$

At minimum one needs

$$\boxed{g_E > 0.} \tag{836}$$

This comes before the longitudinal audit because, in the lowest-channel branch, one-defect survival is ultimately driven by transverse dominance.

CCLXXIII. 5. PRIORITY 3: LONGITUDINAL BURDEN

The third test is the size of

$$\boxed{g_c = \frac{I_{\theta'^3}}{2C_c}\lambda_c, \qquad \lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2}\mathcal{N}_{JJ} m_\parallel u_\parallel |G_*| A_{G_*}.} \tag{837}$$

The practical working branch is

$$\boxed{T_\star = T^1, \qquad u_\parallel(T^1) = \frac{1}{2}\Re(\kappa_J).} \tag{838}$$

The longitudinal audit comes after g_E, because in the lowest-channel acceptance condition g_c acts primarily as the competitor burden against transverse support.

CCLXXIV. 6. PRIORITY 4: LOWEST PROJECTED CHANNEL

The fourth test is the imported radial/projected-channel label κ_*. The practically relevant branch is

$$\boxed{|\kappa_*| = 1.} \tag{839}$$

In this case the acceptance inequality reduces to

$$\boxed{\xi_E g_E > \xi_c g_c.} \tag{840}$$

CCLXXV. 7. PRIORITY 5: COMPETITOR EXCLUSION

Only after the principal candidate survives should one compute the remaining channel competitors and the remote-gap inequality

$$\boxed{\Delta^{\text{full}}_{\text{bench},\rho} > \eta_{\text{rem},\rho}.} \tag{841}$$

This is placed last because it is the most expensive audit and is irrelevant if the principal candidate already fails.

CCLXXVI. 8. FINITE SCALAR PACKAGES TO COMPUTE

The finite audit packages are therefore:

a. Bulk

$$\boxed{m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty).} \tag{842}$$

b. Transverse

$$\boxed{\mathfrak{T}_E, \quad \mathfrak{T}_O, \quad I_\theta, \quad C_E, \quad C_O.} \tag{843}$$

c. Longitudinal

$$\boxed{\mathcal{N}_{JJ}, \quad m_\parallel, \quad u_\parallel, \quad A_{G_*}, \quad I_{\theta'^3}, \quad C_c.} \tag{844}$$

d. Channel / exclusion

$$\boxed{\kappa_*, \quad \Delta^{\mathrm{full}}_{\mathrm{bench},\rho}, \quad \eta_{\mathrm{rem},\rho}.} \tag{845}$$

CCLXXVII. 9. HONEST FINAL DIAGNOSIS

The one-defect problem is no longer blocked by missing structural formulas. It is blocked only by the absence of these finite scalar evaluations on the intended Class II branch.

Thus the correct next move is:

$$\boxed{\text{bulk positivity first, transverse support second, longitudinal burden third.}} \tag{846}$$

CCLXXVIII. FINAL SUMMARY

The finite one-defect audit should now proceed in the following order:

$$\boxed{\mu_\infty \;\to\; g_E \;\to\; g_c \;\to\; \kappa_* \;\to\; \text{competitor exclusion.}} \tag{847}$$

This ordering is optimal because:

1. $\mu_\infty > 0$ is the earliest kill condition;
2. $g_E > 0$ is the first genuine transport support test;
3. g_c measures the longitudinal burden against transverse survival;
4. κ_* is imported from the radial/projected-channel problem;
5. competitor exclusion is the final and most expensive audit.

The corresponding finite scalar packages are:

$$\text{Bulk: } m(\sigma_\infty),\ u_X(\sigma_\infty),\ \chi_\infty,\ z_0(\sigma_\infty),$$

$$\text{Transverse: } \mathfrak{T}_E,\ \mathfrak{T}_O,\ I_\theta,\ C_E,\ C_O,$$

$$\text{Longitudinal: } \mathcal{N}_{JJ},\ m_\parallel,\ u_\parallel,\ A_{G_*},\ I_{\theta'^3},\ C_c,$$

$$\text{Channel/exclusion: } \kappa_*,\ \Delta^{\mathrm{full}}_{\mathrm{bench},\rho},\ \eta_{\mathrm{rem},\rho}.$$

Thus the next real work is a finite scalar audit, not a new structural derivation.

CCLXXIX. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (C4 / ONE-DEFECT ACCEPTANCE STAGE)

CCLXXX. 1. CURRENT MAIN GOAL

The current main goal is no longer a broad discussion of the full C4 sector. The actual frontier has been sharply reduced to:

$$\boxed{\text{close the one-defect acceptance problem inside the C4 matter/family program.}} \tag{848}$$

More precisely, the present location in the proof tree is

$$\boxed{C4 \supset K1 : \text{ common family subspace} \supset K1A1 : \text{ single-defect seed localization} \supset \text{one-defect acceptance audit.}} \tag{849}$$

Thus the immediate target is not yet full CKM extraction, but the first real microscopic gate:

$$\boxed{\text{Does the intended Class II one-defect branch support an acceptable one-defect mode?}} \tag{850}$$

CCLXXXI. 2. CURRENT STRATEGIC PICTURE

The full C4 program is not yet closed. However, the current one-defect subproblem has already been reduced to a finite set of acceptance data:

$$\boxed{(g_c,\ g_E,\ g_O,\ \mu_\infty,\ \kappa_*).} \tag{851}$$

In the practically most relevant lowest-channel case $|\kappa_*| = 1$, the one-defect criterion reduces to the sharp inequality

$$\boxed{\mu_\infty > 0, \qquad \xi_E g_E > \xi_c g_c,} \tag{852}$$

together with competitor exclusion and remote-gap control.
So the real frontier is no longer conceptual. It is a finite scalar audit.

CCLXXXII. 3. GLOBAL STATUS MAP

A. 3.1. Infrared SM+GR checklist status

The current theorem/lemma checklist for an SM+GR-type infrared limit is:

$$C1 : \text{ continuum IR sector,} \qquad C2 : \text{ Einstein-type gravity sector,} \qquad C3 : \text{ gauge sector emergence,} \tag{853}$$

$$C4 : \text{ chiral matter and family sector,} \qquad C5 : \text{ canonical kinetic normalization,} \qquad C6 : \text{ decoupling of extra states,} \tag{854}$$

$$C7 : \text{ suppression of dangerous higher operators,} \qquad C8 : \text{ consistency conditions.} \tag{855}$$

Current honest status:

$$\boxed{\text{Closed: none.}} \tag{856}$$

$$\boxed{\text{Conditional: } C1,\ C4,\ C5.} \tag{857}$$

$$\boxed{\text{Open: } C2,\ C3,\ C6,\ C7,\ C8.} \tag{858}$$

B. 3.2. C4 internal status

Inside C4, the current CKM-entry/minimal-family tree is:

$$K1: \text{ common left-handed family subspace}, \qquad K2: \text{ canonical kinetic normalization}, \tag{859}$$

$$K3: \text{ nondegenerate splittings}, \qquad K4: \text{ hierarchy seed}, \qquad K5: \text{ irreducible CP seed}. \tag{860}$$

Current honest status:

$$\boxed{\text{Closed: none.}} \tag{861}$$

$$\boxed{\text{Conditional: } K1,\ K2,\ K3,\ K4.} \tag{862}$$

$$\boxed{\text{Open: } K5.} \tag{863}$$

The logical order is

$$\boxed{K1 \to K2 \to (K3, K4) \to K5.} \tag{864}$$

The practical work order is

$$\boxed{K4^{\text{toy}} \to K1 \to K2 \to K3 \to K4^{\text{micro}} \to K5.} \tag{865}$$

CCLXXXIII. 4. K1 DECOMPOSITION AND CURRENT LOCATION

The common-family-space condition was decomposed as

$$\boxed{K1A \to K1B \to K1C \to K1D,} \tag{866}$$

where:

$$K1A: \text{ common carrier sector exists}, \qquad K1B: \text{ a three-dimensional family subspace is isolated}, \tag{867}$$

$$K1C: \text{ sector perturbations preserve the family space effectively}, \qquad K1D: \text{ physical left-handed interpretation}. \tag{868}$$

Current status:

$$\boxed{\text{Conditional: } K1A,\ K1B,\ K1C, \qquad \text{Open: } K1D.} \tag{869}$$

The present frontier is inside $K1A1$, namely:

$$\boxed{\text{single-defect seed localization } \to \text{ one-defect acceptance } \to \text{ triplet isolation later.}} \tag{870}$$

CCLXXXIV. 5. SINGLE-DEFECT SEED LOCALIZATION: WHAT HAS BEEN ESTABLISHED

A. 5.1. Variational no-go for the naive positive-sign local surrogate

For the naive local surrogate

$$\mathcal{L}_{X_0} = -Z\nabla^2 + \mu^2 + 2\lambda\rho_{X_0}(x), \tag{871}$$

with $\lambda > 0$, the Gaussian variational test showed:

$$\boxed{\lambda \geq 0 \quad \Longrightarrow \quad \text{no Gaussian variational localized carrier seed below the continuum threshold.}} \tag{872}$$

Thus the old positive-sign local toy operator does not trap a seed.

B. 5.2. Trapping surrogate and threshold

For the Gaussian well surrogate

$$\mathcal{L}_{X_0} = -Z\nabla^2 + U_\infty - V_0 e^{-|x-X_0|^2/R^2}, \qquad V_0 > 0, \tag{873}$$

the Gaussian variational threshold is

$$\boxed{\frac{V_0 R^2}{Z} > \frac{9\sqrt{3}}{4}.} \tag{874}$$

Equivalently, in the old notation $V_0 = -2\lambda$,

$$\boxed{\lambda < 0, \qquad \frac{|\lambda| R^2}{Z} > \frac{9\sqrt{3}}{8}.} \tag{875}$$

C. 5.3. Schur-complement mechanism for the attractive well

A positive heavy block generates a negative semidefinite effective correction:

$$\boxed{L_{\text{eff}}(E) = L_\ell - C(L_h - E)^{-1}C^\dagger,} \tag{876}$$

with

$$\boxed{-C(L_h - E)^{-1}C^\dagger \leq 0.} \tag{877}$$

Hence integrating out the heavy sector automatically lowers the light-sector energy. This provides the correct microscopic carrier-lowering mechanism.

CCLXXXV. 6. CLASS II SINGLE-DEFECT SEED MECHANISM

A. 6.1. Mass-like and derivative-like channels

The Class II mediator X_i^a splits into:

$$\boxed{U_i^a(P): \text{ mass-like seed-localization channel}, \qquad V_{ij}^a(P): \text{ derivative/transport channel.}} \tag{878}$$

The mass-like channel is responsible for seed localization. The derivative channel is responsible later for transport/hierarchy.

B. 6.2. Effective mass coefficient

The load-bearing one-defect mass coefficient is

$$\boxed{m_{\text{eff}}(x) = m(\bar{\sigma}(x)) - 3\chi_X(x)\, u_X(\bar{\sigma}(x)).} \tag{879}$$

Using the Gaussian core ansatz

$$\bar{\sigma}(r) = \sigma_\infty + \delta\sigma e^{-r^2/R^2}, \qquad \chi_X(r) = \chi_0 e^{-r^2/R^2}, \tag{880}$$

one obtains

$$m_{\text{eff}}(r) = m_\infty - \Gamma_{\text{loc}} e^{-r^2/R^2} + O(e^{-2r^2/R^2}), \tag{881}$$

with

$$\boxed{\Gamma_{\text{loc}} = 3\chi_0 u_\infty - m_1 \delta\sigma.} \tag{882}$$

Using the algebraic heavy-field equation

$$M_X^2 X_i^a + \mathcal{J}_i^{a,\text{tot}} = 0, \tag{883}$$

and defining the aligned scalar source amplitude

$$\frac{1}{3} E_i{}^a \mathcal{J}_i^{a,\text{tot}}(r) = -\mathrm{j}_0 e^{-r^2/R^2}, \tag{884}$$

one gets

$$\chi_0 = \frac{\mathrm{j}_0}{M_X^2}, \tag{885}$$

hence

$$\boxed{\Gamma_{\text{loc}} = \frac{3u_\infty}{M_X^2}\, \mathrm{j}_0 - m_1 \delta\sigma.} \tag{886}$$

C. 6.3. J- and K-channel source decomposition

With

$$\mathrm{j}_0 = \alpha_X j_J + \beta_X j_K, \tag{887}$$

and the explicit core amplitudes

$$j_J = \phi_c^2 \theta_1 C_J, \qquad j_K = \mathcal{O}_c \theta_1 C_K, \tag{888}$$

one gets

$$\boxed{\mathrm{j}_0 = \theta_1 (\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K),} \tag{889}$$

hence

$$\boxed{\Gamma_{\text{loc}} = \frac{3u_\infty \theta_1}{M_X^2} (\alpha_X \phi_c^2 C_J + \beta_X \mathcal{O}_c C_K) - m_1 \delta\sigma.} \tag{890}$$

CCLXXXVI. 7. EMBEDDED $SU(2)_{(13)}$ SINGLE-DEFECT BRANCH

For the explicit embedded $SU(2)_{(13)}$ branch, one has

$$m_a(\theta) = \left(\frac{1}{2}\sin 2\theta,\ 0,\ \frac{1}{2}\cos 2\theta\right), \tag{891}$$

$$k_a(\theta) = A_a(\theta) = (\cos 2\theta,\ 0,\ -\sin 2\theta)\,, \tag{892}$$

with

$$\boxed{m_a k_a = 0.} \tag{893}$$

Therefore:

A. Orbit-aligned branch

If the projection direction is orbit-aligned,

$$\boxed{C_J = C_K = 0,} \tag{894}$$

so the mediator-induced seed vanishes at leading order and

$$\Gamma_{\mathrm{loc}} = -m_1\delta\sigma. \tag{895}$$

B. Tangent-aligned branch

If the projection direction is tangent-aligned,

$$\boxed{C_J = C_K = \frac{1}{3}.} \tag{896}$$

Then

$$\boxed{\Gamma_{\mathrm{loc}} = \frac{u_\infty\theta_1}{M_X^2}(\alpha_X\phi_c^2 + \beta_X\mathcal{O}_c) - m_1\delta\sigma.} \tag{897}$$

Hence:

$$\boxed{\text{single-defect seed localization survives only on the tangent-aligned branch at leading order.}} \tag{898}$$

CCLXXXVII. 8. BULK POSITIVITY AND CONSTITUTIVE SIGN AUDIT

The asymptotic mass parameter is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \tag{899}$$

The cleanest favorable branch is the vacuum asymptotic branch:

$$\boxed{\chi_\infty = 0.} \tag{900}$$

Then

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \tag{901}$$

This decouples bulk positivity from the sign of u_∞.
The current favorable sign window is:

$$\boxed{\chi_\infty = 0, \qquad u_\infty > 0, \qquad \alpha_X \phi_c^2 + \beta_X \mathcal{O}_c > 0, \qquad m(\sigma_\infty) > 0.} \tag{902}$$

CCLXXXVIII. 9. HEURISTIC SEED BRANCH CLASSIFICATION

Define

$$A_J = \alpha_X \phi_c^2, \qquad A_K = \beta_X \mathcal{O}_c. \tag{903}$$

Then the seed amplitude is

$$\Gamma_{\rm loc} = \frac{u_\infty \theta_1}{M_X^2}(A_J + A_K) - m_1 \delta\sigma. \tag{904}$$

The three heuristic subbranches are:

$$|A_J| \gg |A_K| \quad \Rightarrow \quad J\text{-dominant}, \tag{905}$$

$$|A_K| \gg |A_J| \quad \Rightarrow \quad K\text{-dominant}, \tag{906}$$

$$A_J \sim A_K \quad \Rightarrow \quad \text{balanced}. \tag{907}$$

The structurally bad locus is

$$\boxed{A_J + A_K \approx 0.} \tag{908}$$

Genericity verdict:

$$\boxed{\text{balanced or mild } J\text{-leaning is more generic than pure } K\text{-dominant.}} \tag{909}$$

Combined seed+transport desirability verdict:

$$\boxed{\text{balanced} \succ \text{K-assisted} \succ \text{pure J-dominant.}} \tag{910}$$

CCLXXXIX. 10. TRANSITION TO THE ACTUAL ONE-DEFECT THEOREM

Heuristic variables $(R\phi_c, \psi_c/\phi_c)$ are no longer the final load-bearing quantities. The actual one-defect theorem is controlled by

$$\boxed{g_c, \qquad g_E, \qquad g_O, \qquad \mu_\infty, \qquad \kappa_*.} \tag{911}$$

The current sufficient one-defect/one-mode criterion is

$$\boxed{\mu_\infty > 0, \qquad a_0 > 0, \qquad b_0 > 0, \qquad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},} \tag{912}$$

with

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \qquad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \qquad c_0 = \frac{\xi_O g_O}{z_{0,0}}. \tag{913}$$

In the lowest-channel case $|\kappa_*| = 1$, this reduces to

$$\boxed{\xi_E g_E > \xi_c g_c.} \tag{914}$$

CCXC. 11. SOURCE-BY-SOURCE MICROSCOPIC CLASSIFICATION

Current working-branch source classification:

$$\boxed{JJ \to \lambda_c, \qquad JK + KJ \to (\Lambda_E, \Lambda_O), \qquad KK \to \text{transverse residual corrections.}} \tag{915}$$

Equivalently, in blockwise Hessian form:

$$\boxed{\mathcal{H}_{OO} \to \lambda_c, \qquad \mathcal{H}_{TD}, \mathcal{H}_{DT} \to \Lambda_E, \Lambda_O.} \tag{916}$$

Thus:

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O,} \tag{917}$$

and therefore

$$\boxed{g_c = \frac{\lambda_\parallel}{C_c}, \qquad g_E = \frac{\alpha}{C_E}, \qquad g_O = \frac{\beta}{C_O}.} \tag{918}$$

CCXCI. 12. LONGITUDINAL FINITE SCALAR REDUCTION

The longitudinal scalar package has already been reduced to

$$\boxed{\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ}\, m_\parallel\, u_\parallel\, |G_*|\, A_{G_*}.} \tag{919}$$

Here

$$m_\parallel = m^3(P_0), \qquad u_\parallel = \Re[\kappa_J(t_{10} - t_{20})]. \tag{920}$$

In the canonical locked basis

$$z_0 = e_1, \qquad u_1 = e_2, \qquad u_2 = e_3,$$

the active generators are T^1, T^2, T^4, T^5, and the cleanest branch is

$$\boxed{T_\star = T^1, \qquad u_\parallel(T^1) = \frac{1}{2}\Re(\kappa_J).} \tag{921}$$

So the longitudinal frontier has been reduced to a finite audit of

$$\boxed{\mathcal{N}_{JJ}, \qquad m_\parallel, \qquad u_\parallel, \qquad A_{G_*}.} \tag{922}$$

CCXCII. 13. TRANSVERSE FINITE SCALAR REDUCTION

The transverse branch-level reduced summary is

$$\boxed{\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \qquad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O.} \tag{923}$$

Hence

$$\boxed{g_E = \frac{I_\theta}{C_E} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \qquad g_O = \frac{I_\theta}{C_O} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O.} \tag{924}$$

So the transverse frontier has been reduced to the finite audit of

$$\boxed{\mathfrak{T}_E, \qquad \mathfrak{T}_O, \qquad I_\theta, \qquad C_E, \qquad C_O.} \tag{925}$$

CCXCIII. 14. FULL FINITE ONE-DEFECT ACCEPTANCE DATA

The full one-defect acceptance data are now:

$$\boxed{(g_c,\ g_E,\ g_O,\ \mu_\infty,\ \kappa_*).} \tag{926}$$

Equivalently, the finite scalar audit packages are:
a. Bulk

$$\boxed{m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty).} \tag{927}$$

b. Transverse

$$\boxed{\mathfrak{T}_E, \quad \mathfrak{T}_O, \quad I_\theta, \quad C_E, \quad C_O.} \tag{928}$$

c. Longitudinal

$$\boxed{\mathcal{N}_{JJ}, \quad m_\parallel, \quad u_\parallel, \quad A_{G_*}, \quad I_{\theta'^3}, \quad C_c.} \tag{929}$$

d. Channel/exclusion

$$\boxed{\kappa_*, \quad \Delta^{\text{full}}_{\text{bench},\rho}, \quad \eta_{\text{rem},\rho}.} \tag{930}$$

CCXCIV. 15. PRIORITY ORDER FOR THE NEXT COMPUTATION

The correct finite audit order is

$$\boxed{\mu_\infty \longrightarrow g_E \longrightarrow g_c \longrightarrow \kappa_* \longrightarrow \text{competitor exclusion.}} \tag{931}$$

Reason:

1. $\mu_\infty > 0$ is the earliest kill condition;
2. $g_E > 0$ is the first transverse support test;
3. g_c is the longitudinal burden against transverse survival;
4. κ_* is imported from the radial/projected-channel sector;
5. competitor exclusion is the final and most expensive audit.

CCXCV. 16. WHAT IS ALREADY CLOSED VS WHAT REMAINS OPEN

A. Already closed structurally

$$\boxed{\text{(i) single-defect seed localization is tangent-aligned at leading order;}} \tag{932}$$

$$\boxed{\text{(ii) bulk positivity is controlled by } \mu_\infty;} \tag{933}$$

$$\boxed{\text{(iii) longitudinal } g_c \text{ is sourced by } JJ;} \tag{934}$$

$$\boxed{\text{(iv) transverse } (g_E, g_O) \text{ are sourced by } JK + KJ;} \tag{935}$$

$$\boxed{\text{(v) the full one-defect theorem is reduced to a finite scalar audit.}} \tag{936}$$

B. Still open explicitly

$$\boxed{\text{actual/quasi-actual evaluation of } \mu_\infty,\ \mathfrak{T}_E,\ \mathcal{N}_{JJ},\ m_\parallel,\ u_\parallel,\ A_{G_*},\ \kappa_*.} \tag{937}$$

So the remaining frontier is not conceptual. It is explicit finite evaluation on the intended Class II branch.

CCXCVI. 17. IMMEDIATE NEXT STEP IN THE NEXT CHAT

The next chat should begin with:

$$\boxed{\text{Priority 1: evaluate } \mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)} \text{ on the chosen vacuum-asymptotic branch.}} \tag{938}$$

Then immediately proceed to:

$$\boxed{\text{Priority 2: evaluate } g_E = \frac{I_\theta}{C_E}\frac{\alpha_X \beta_X}{M_X^2}\mathfrak{T}_E.} \tag{939}$$

Only after that should the longitudinal burden be inserted via

$$\boxed{g_c = \frac{I_{\theta'^3}}{2C_c}\lambda_c^{(JJ)}, \qquad \lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2}\mathcal{N}_{JJ} m_\parallel u_\parallel |G_*| A_{G_*}.} \tag{940}$$

This is the correct continuation point.